FINAL Unit4 MCQ - Topic order

1. A software company wants to develop new features to attract new users to download a popular application. The company created a survey to ask users how they feel about several potential new features. The company sent the survey to all users who spent more than 500 hours using the application during the previous year. A programmer for the company wants to determine which new features should be prioritized by performing an analysis of the survey data. Which of the following best explains why the programmer should be careful about the conclusions drawn from the analysis?

(A) The survey data may be biased in favor of a specific group of users. ✓

(B) The survey data may include responses from users who dislike the application.

(C) The survey data may raise intellectual property concerns for the company.

(D) The survey data may require data encapsulation before they can be analyzed.

Answer A

Correct. The group of users who spent more than 500 hours using the application last year may not be representative of all potential users of the application. As a result, the survey data analysis may lead to outcomes that are biased in favor of users with more experience with the application.

FINAL Unit4 MCQ - Topic order

2. The following class is used to store information about a customer.

```
public class Customer
{
  public String cName;        // The customer's name
  public String cPaymentInfo; // The customer's payment information

  /**
   * Returns the customer's name
   */
  public String getName()
  { /* implementation not shown */ }

  /**
   *  Returns the customer's payment information
   *  when a correct password is entered
   */
  public String getPaymentInfo(String password)
  { /* implementation not shown */ }

  /* There may be other methods that are not shown. */
}
```

Which of the following statements best describes the privacy of the customer's payment information?

(A) It is considered private because retrieving it requires a password.

(B) It is considered private because the value of the instance variable `cName` is not easy to guess.

(C) It is not considered private because the instance variables are declared public. ✓

(D) It is not considered private because the `getName` method has no password parameter.

Answer C

Correct. Instance variables declared `public` are accessible to any object outside the class.

FINAL Unit4 MCQ - Topic order

3. A school district wants to know the average number of hours students spend reading for pleasure each week. The district sends an online survey to students who are members of the district's reading club and asks them to respond.

Which of the following factors is most important for the district to consider before making generalizations from the data obtained from the survey?

(A) The survey results may be unreliable because the data might contain personal information.

(B) The survey results may be inaccurate if some students rounded their reported time reading for pleasure.

(C) The survey results may be inconsistent because students completed the survey at different times during the week.

(D) The survey results may not represent all students because only those who are members of the reading club and choose to respond are included. ✓

Answer D

Correct. The sample is biased toward students who already enjoy reading, so the results cannot be generalized to all students in the district.

4. A company develops an application that helps users find rental housing. It recommends apartments and homes based on income, past search history, and stated preferences. As the application is being used by clients, the company finds that the application produces different recommendations even when the housing needs are similar.

Which of the following is the most likely reason for this behavior?

(A) The application contains algorithmic bias that is creating unfair outcomes for specific groups of users. ✓

(B) The application does not protect the users' data, allowing it to be viewed by other users.

(C) The application encapsulates the data as it is being entered, resulting in invalid data.

(D) The application uses data that has not been sorted, causing some housing options to be overlooked.

Answer A

Correct. This behavior suggests that there may be systematic errors in the application that create unfair outcomes for certain groups or individuals.

FINAL Unit4 MCQ - Topic order

5. The following SocialMediaAccount class is used to represent a user of a social media site. It includes attributes for the user's username, real name, email address, and ID number.

```
public class SocialMediaAccount
{
  public String username;  // The user's username
  private String realName; // The user's real name
  private String email;    // The user's email address
  private int idNumber;    // The user's ID number

  public String getInfo(String str)
  {
    if (str.equals(username))
    {
      return realName + " " + email + " " + idNumber;
    }
    else
    {
      return "username does not match";
    }
  }

  // Constructor not shown.
  // There are no other methods.
}
```

What is the greatest personal privacy risk posed to users by this class design?

(A) The class contains no mutator methods.

(B) The class utilizes data encapsulation.

(C) Personal information can be determined from outside the class using information that has public visibility. ✓

(D) If the ID numbers are assigned consecutively, the total number of users can be determined.

Answer C

Correct. The instance variable username has public visibility, which means it can be accessed outside the class. If the value of this variable is known, it can be passed to the getInfo method to obtain personal information, including the user's real name and email address.

FINAL Unit4 MCQ - Topic order

6. A research team is analyzing survey data to determine the average number of hours high school students spend on homework each night. The survey was conducted online, and students volunteered to participate.

Which of the following factors is most important for the team to consider before drawing conclusions from the data?

(A) The survey data are considered reliable because the survey used self-reported responses from students.

(B) The survey results may not represent all high school students because only those with Internet access and interest in the topic participated. ✓

(C) The survey ensures accuracy because students were able to choose whether or not to participate.

(D) The method of survey collection does not matter as long as there are a large number of responses in the data set.

Answer B

Correct. The method of collecting data online and relying on voluntary participation introduces potential biases. These factors can make the sample unrepresentative of the entire population, which is an important consideration when using the data to draw conclusions.

7. A widely used translation application collects and stores users' voice input to improve its speech-recognition algorithms. Which of the following is the most significant privacy concern raised by this practice?

(A) The application may perform less accurately for speakers with regional accents.

(B) Storing large amounts of voice data increases the risk that the data will become inaccurate.

(C) The quality of the voice collection is dependent on a user's device, which may cause inconsistent results.

(D) Users might assume their conversations are private, even though their voice data is being stored and used to train future models. ✓

Answer D

Correct. This is the most significant privacy concern because it involves a lack of informed consent. If users do not realize their data is being stored and reused, their privacy is at risk. This reflects a key issue in data ethics.

FINAL Unit4 MCQ - Topic order

8. A city transportation office has a data set that contains information about bus ridership in the city. Each row in the data set corresponds to one rider's information for a single week. A sample portion of the data set is shown.

Rider ID	Age Group	Primary Route Used	Has Discount? (Yes/No)
20038	Teen (13 to 19)	Route A	Yes
31932	Adult (20 to 64)	Route B	No
29938	Senior (65 and older)	Route C	Yes
25571	Adult (20 to 64)	Route A	No
38860	Teen (13 to 19)	Route B	No

Which of the following questions can be answered using the data set?

(A) Which rider is entitled to a discount based on age?

(B) Which riders without a discount most frequently use Route A? ✓

(C) What time of day are teens most likely to use Route B?

(D) Which age group prefers Route C because of shorter travel times?

Answer B

Correct. This question can be answered using the data set. An algorithm could examine each row to identify riders whose discount status is "No" and whose primary route is "Route A." The results can then be counted or listed to determine frequency.

FINAL Unit4 MCQ - Topic order

9. A researcher wants to investigate which age group (under 18 years old, or 18 years old and over) is more likely to have downloaded a certain application from an online store. Each customer of the store has a unique ID number. The researcher has the following data sets available.

- Data set 1 contains an entry for each application available from the store. Each entry includes the name of the application and the total number of customers who have downloaded it.
- Data set 2 contains an entry for each application available from the store. Each entry includes the ID number and age of the customer who most recently downloaded the application.
- Data set 3 contains an entry for each customer of the store. Each entry includes the customer's ID number and the customer's age.
- Data set 4 contains an entry for each customer of the store. Each entry includes the customer's ID number and the names of all the applications downloaded by the customer.

Which two data sets can be combined and analyzed to determine the desired information?

(A) Data sets 1 and 2

(B) Data sets 2 and 3

(C) Data sets 2 and 4

(D) Data sets 3 and 4 ✓

Answer D

Correct. Data set 3 can be used to identify the customers in each age group. Data set 4 can be used to identify which customers have downloaded a particular application. These two data sets can be combined using the unique ID number for each customer to determine which age group downloaded the application more often.

FINAL Unit4 MCQ - Topic order

10. A school offers optional courses and clubs to students. A programmer wants to investigate whether students at the school who took a computer science course were more or less likely to participate in the school's photography club. The programmer has the following data sets available.

- Data set 1 contains an entry for each club offered by the school. Each entry includes the name of the club and a list of the names of the club members.
- Data set 2 contains an entry for each member of the photography club. Each entry contains the name of the member, the number of photographs taken by the member, and the member's favorite course.
- Data set 3 contains an entry for each student at the school. Each entry contains the name of the student, a list of the courses the student took, and a list of the clubs the student participated in.
- Data set 4 contains an entry for each student at the school. Each entry contains the name of the student, the student's favorite course, and the student's favorite club.

Which of the data sets is most appropriate for the programmer's investigation?

(A) Data set 1

(B) Data set 2

(C) Data set 3 ✓

(D) Data set 4

Answer C

Correct. For this investigation, the programmer needs to know the proportion of computer science students who joined the photography club and the proportion of all other students who joined the photography club. Since data set 3 indicates the courses and clubs for all students in the school, it includes all of the necessary information to perform an analysis.

FINAL Unit4 MCQ - Topic order

11. A programmer wants to investigate whether players of an online game who are younger than 30 years old tend to spend more time playing the game than players who are 30 years old or older. Each player of the game must create an account with a unique username. The programmer has the following data sets available.

- Data set 1 contains an entry for each player. Each entry contains the player's birth date (month, day, and year) and username.
- Data set 2 contains an entry for each player. Each entry contains the player's username and the average number of hours the player spends playing the game each day.
- Data set 3 contains an entry for each year since the game was released. Each entry includes the number of players who were born in that year.
- Data set 4 contains an entry for each year since the game was released. Each entry includes a list of usernames of the players who created an account in that year.

Which two data sets can be combined and analyzed to determine the desired information?

(A) Data sets 1 and 2 ✓

(B) Data sets 1 and 3

(C) Data sets 2 and 3

(D) Data sets 2 and 4

Answer A

Correct. For this investigation, the programmer needs to know the age of each player and the amount of time each player spends playing the game. Data set 1 can be used to determine the age of each player, and data set 2 contains the average amount of time each player spent playing the game each day. These two data sets can be combined using the unique username.

FINAL Unit4 MCQ - Topic order

12. A retail store has a data set that contains the following information for each sales transaction.

- Store location
- Transaction date
- Time of purchase
- Sales amount, in dollars
- Product category (electronics, clothing, home goods, grocery, etc.)

Which of the following is most easily determined from the data in the data set?

(A) Customer purchasing preferences by category for each store location ✓

(B) Customer satisfaction levels by category for each date

(C) The name of the most popular product at each location

(D) The quantity available for each product at each store location

Answer A

Correct. The time of purchase and product category can be used to analyze the sales for each store location.

FINAL Unit4 MCQ - Topic order

13. A software developer wants to determine which programming language is most commonly used in different types of software projects. They have access to the following data sets.

- Data set 1 contains entries for each registered developer, including their unique developer ID and their favorite programming language.
- Data set 2 contains entries for each software project, including project name, type of project, and the developer ID of the project creator.
- Data set 3 contains entries for each completed project, including project name and user ratings.
- Data set 4 contains entries for each software project, including project name and all programming languages used in the project.

Which two data sets can be combined and analyzed to determine the desired information?

(A) Data sets 1 and 2

(B) Data sets 2 and 4 ✓

(C) Data sets 3 and 4

(D) Data sets 1 and 3

Answer B

Correct. Data set 2 provides information about the type of each project and data set 4 lists all programming languages used in each project. Together they allow analysis of which languages are used across different project types.

FINAL Unit4 MCQ - Topic order

14. A teacher has a data set about the average daily number of hours that students spend on homework, extracurricular activities, and sleep for a given week. Each row in the data set corresponds to one student's data. A sample portion of the data set is shown.

Student	Homework	Extracurricular Activities	Sleep
Elena	2.5	1.5	9.0
Marcus	2.0	2.0	8.5
Jayla	3.5	2.0	8.0
Theo	4.0	1.0	8.0
Aisha	2.5	3.5	7.5
Priya	1.8	2.5	10.0

Which of the following questions can be answered using this data set?

(A) Does the amount of sleep affect a student's homework grade?

(B) How many students spent a total of at least 20 hours per week on homework? ✓

(C) What is the maximum number of hours spent on extracurricular activities in a single day?

(D) Which day of the week are students assigned the most homework?

Answer B

Correct. The data set includes the average number of daily homework hours. Multiplying this value by 7 days would give the total number of homework hours for the week.

FINAL Unit4 MCQ - Topic order

15. A data set contains information for all high schools in a school district. The following information from the previous year is available for each school in the district.

- The name of the high school
- The number of students who arrived at school late 1 or more times during the year
- The number of students who arrived at school late 10 or more times during the year
- The number of students who submitted medical excuse documentation for 1 or more of their late arrivals
- The percentage of students who voted to delay the school start time by one hour

The following shows a portion of the data set for a district with 13 high schools.

High School Name	Number of Students Late at Least 1 Time	Number of Students Late at Least 10 Times	Number of Students Who Submitted Medical Excuse Documentation	Percent of Student Population Who Voted for Later Start Time
School A	402	59	179	62%
School B	63	2	48	29%
⋮	⋮	⋮	⋮	⋮
School M	163	37	44	84%

Which of the following information **cannot** be determined using only the information in the data set?

(A) The district-wide total number of students who were late to school at least 10 times

(B) The district-wide total number of students who were late to school between 1 and 9 times, inclusive

(C) The district-wide percentage of students who voted for a later school start time ✓

(D) The district-wide percentage of the students who were late at least 1 time who submitted medical excuse documentation for at least 1 late arrival

Answer C

Correct. The table does not provide information about the total student population at each school. There is no way to calculate the total student population for the district, nor the total number of students voting to delay start time.

FINAL Unit4 MCQ - Topic order

16. A university wants to determine which major has the highest average GPA among students graduating in a given year. The university has the following data sets available.

- Data set 1 contains an entry for each student. Each entry includes the student's ID, declared major, and graduation year.
- Data set 2 contains an entry for each student. Each entry includes the student's ID, declared major, and assigned faculty advisor.
- Data set 3 contains an entry for each student. Each entry includes the student's ID, course name, and grade received.
- Data set 4 contains an entry for each student. Each entry includes the student's ID and final GPA.

Which two data sets can be combined and analyzed to determine the information the university seeks?

GPA: grade point average

(A) Data sets 1 and 2

(B) Data sets 1 and 4 ✓

(C) Data sets 2 and 3

(D) Data sets 3 and 4

Answer B

Correct. Data set 1 provides major and graduation year for each student ID, and data set 4 provides final GPA for each student ID. By joining the two through student ID, the average GPA by major can be calculated.

FINAL Unit4 MCQ - Topic order

17. Consider the following method.

```
public static int[] addNum(int[] array, int first,
                           int second, int num)
{
  int[] newArray = new int[array.length];
  newArray[first] = array[first] + num;
  newArray[second] = array[second] + num;
  return newArray;
}
```

The following code segment appears in a method in the same class as `addNum`. The code segment is intended to set the contents of `array2` to `{0, 13, 9, 0}`.

```
int[] array1 = /* missing initialization */;
int[] array2 = addNum(array1, 1, 2, 5);
```

Which of the following can replace `/* missing initialization */` so that the code segment works as expected?

(A)

{8, 4}

(B)

{18, 14}

(C) ✓

{5, 8, 4, 5}

(D)

{5, 18, 14, 5}

Answer C

Correct. When `newArray` is declared in the `addNum` method, all its elements are initialized to `0` by default. The method then takes the value passed in `first`, which is `1`, and sets `newArray[1]` to `array[1] + 5`, which is `13`. Then the method takes the value passed in `second`, which is `2`, and sets `newArray[2]` to `array[2] + 5`, which is `9`.

FINAL Unit4 MCQ - Topic order

18. Consider the following method.

```
public int[] transform(int[] a)
{
 a[0]++;
 a[2]++;
 return a;
}
```

The following code segment appears in a method in the same class as `transform`.

```
/* missing code */
arr = transform(arr);
```

After executing the code segment, the array `arr` should contain `{1, 0, 1, 0}`. Which of the following can be used to replace `/* missing code */` so that the code segment works as intended?

I. `int[] arr = {0, 0, 0, 0};`
II. `int[] arr = new int[0];`
III. `int[] arr = new int[4];`

(A) I only

(B) III only

(C) I and II

(D) I and III ✓

Answer D

Correct. The `transform` method increases the first and third elements of the array by `1`. In order to produce the intended result, the array passed to the method should contain `{0, 0, 0, 0}`. Code segment I works as intended. This statement uses an initializer list to create an array with four elements, each equal to `0`. Code segment II does not work as intended. The statement creates a new array of length 0 (an array with no elements). Code segment III works as intended. The statement creates a new array of length 4, whose elements are each assigned the default value `0`.

FINAL Unit4 MCQ - Topic order

19. Consider the following method, which is intended to return the number of strings of length greater than or equal to 3 in an array of `String` objects.

```
public static int checkString(String[] arr)
{
  int count = 0;
  for (int k = 0; k < arr.length; k++)
  {
    if (arr[k].length() >= 3)
    {
      count++;
    }
  }
  return count;
}
```

Which of the following code segments compile without error?

I. `checkString(new String[]);`
II. `checkString(new String[0]);`
III.
```
String[] str = {"cat", "dog"};
checkString(str);
```

(A) III only

(B) I and III only

(C) II and III only ✓

(D) I, II, and III

Answer C

Correct. The size of an array must be established when the array is created, either explicitly by specifying the size within the brackets or implicitly by using an initializer list. Statement I does not compile because the array size is missing. Statement II compiles because it correctly creates an array of size 0. Statement III compiles because it creates an array of size 2, using an initializer list.

FINAL Unit4 MCQ - Topic order

20. The following code segment is intended to create an `int` array named `nums` and set its contents to `{3, 1, 0, 2}`. The code segment does not work as intended.

```
int[] nums = new int[3];

nums[1] = 0;
nums[1] = 1;
nums[0] = 3;

nums.length++;

nums[3] = 2;
```

Which of the following best explains why the code segment does not work as intended?

(A) The assignment statements for the array elements are in the wrong order.

(B) The size of the array cannot be changed once the array is created. ✓

(C) The element `nums[1]` is assigned two different values.

(D) The element `nums[2]` is never set to `0`.

Answer B

Correct. The first statement correctly creates an `int` array, but the array contains only three elements. Each of these elements is assigned the default value `0` at the time the array is created. The statement `nums.length++;` is not valid, and the code segment does not compile. The length of an array is established at the time of creation and cannot be changed.

FINAL Unit4 MCQ - Topic order

21. The method `countTarget` below is intended to return the number of times the value `target` appears in the array `arr`. The method may not work as intended.

```
public int countTarget(int[] arr, int target)
{
  int count = 0;
  for (int j = 0; j <= arr.length; j++)  // line 4
  {
    if (arr[j] == target)
    {
      count++;
    }
  }
  return count;
}
```

Which of the following changes, if any, can be made to line 4 so that the method will work as intended?

(A) Changing `j <= arr.length;` to `j < arr.length;` ✓

(B) Changing `j <= arr.length;` to `j < arr.length - 1;`

(C) Changing `j <= arr.length;` to `j < arr.length + 1;`

(D) No change is necessary; the method works correctly as is.

Answer A

Correct. The last array element is at index `arr.length - 1`, so the loop should only iterate while `j < arr.length`.

FINAL Unit4 MCQ - Topic order

22. Consider the following code segment.

```
/* missing code */
sequence[1] = 1;
sequence[2] = 2;
```

Which of the following code segments can be used replace /* missing code */ so that the code segment sets the contents of the array to {0, 1, 2}?

(A)

```
int[] sequence;
```

(B)

```
int[] sequence = new int[1];
```

(C)

```
int[] sequence = new int[2];
```

(D) ✓

```
int[] sequence = new int[3];
```

Answer D

Correct. This statement creates an array of length 3. It is initialized with default values {0, 0, 0}. The code segment then modifies the elements at indices 1 and 2, resulting in {0, 1, 2}. The first element (at index 0) remains unchanged from its default value.

FINAL Unit4 MCQ - Topic order

23. Consider the following method.

```
public static void updateArr(int[] arr, int val)
{
  arr[val] = val;
}
```

The following code segment appears in a method in the same class as updateArr.

```
int[] numArr = new int[4];
updateArr(numArr, 1);
updateArr(numArr, 3);
```

What are the contents of numArr after executing this code segment?

(A)

{0, 0, 0, 0}

(B) ✓

{0, 1, 0, 3}

(C)

{1, 0, 3, 0}

(D)

{1, 3, 0, 0}

Answer B

Correct. When the array numArr is created with the keyword new, all of its elements are initialized to the default value 0. Arrays are mutable, so the updateArr method modifies the contents of numArr. The first call to updateArr changes the element at index 1 to 1 and the second call changes the element at index 3 to 3.

FINAL Unit4 MCQ - Topic order

24. Assume that an array of integer values has been declared as follows and has been initialized.

```
int[] arr = new int[10];
```

Which of the following code segments correctly interchanges the value of `arr[0]` and `arr[5]` ?

(A)
```
arr[0] = 5;
arr[5] = 0;
```

(B)
```
arr[0] = arr[5];
arr[5] = arr[0];
```

(C)
```
int k = arr[5];
arr[0] = arr[5];
arr[5] = k;
```

(D) ✓
```
int k = arr[0];
arr[0] = arr[5];
arr[5] = k;
```

25. The following code segment is intended to set the contents of the array names to {"Ana", "Billy", "Cathy", "Darius"}. The code segment does not work as intended.

```
String[] names = new String[3]; // Line 1
names[0] = "Ana";               // Line 2
names[1] = "Billy";             // Line 3
names[2] = "Cathy";             // Line 4
names[3] = "Darius";            // Line 5
```

Which of the following changes can be made so that the code segment works as intended?

(A) Changing line 1 to String[] names = new String[4]; ✓

(B) Changing line 1 to String[] names = new names[3];

(C) Changing line 5 to names[2] = "Darius";

(D) Removing line 5

Answer A

Correct. The given code segment creates an array of length 3. Attempting to assign a value to the element at index 3 causes an ArrayIndexOutOfBoundsException because the array is indexed from 0 to 2. By increasing the array length to 4, it allows all four strings to be stored as intended.

FINAL Unit4 MCQ - Topic order

26. Consider the following method.

```
public static int getValue(int[] data, int j, int k)
{
  return data[j] + data[k];
}
```

Which of the following code segments, when appearing in another method in the same class as getValue, will print the value 70?

(A)

```
int[] arr = {40, 30, 20, 10, 0};

System.out.println(getValue(arr, 1, 2));
```

(B) ✓

```
int[] arr = {50, 40, 30, 20, 10};

System.out.println(getValue(arr, 1, 2));
```

(C)

```
int arr = {40, 30, 20, 10, 0};

System.out.println(getValue(arr, 2, 1));
```

(D)

```
int arr = {50, 40, 30, 20, 10};

System.out.println(getValue(arr, 2, 1));
```

Answer B

Correct. The array arr is properly created and is initialized with the initializer list {50, 40, 30, 20, 10}. The call to getValue returns the sum of the values at indices 1 and 2. The sum will be 40 + 30, which evaluates to 70.

FINAL Unit4 MCQ - Topic order

27. Consider the following code segment.

```
int[] arr = {1, 2, 3, 4, 5};
arr[1] = 4;
arr[3] = 2;
```

Which of the following represents the contents of `arr` after the code segment has been executed?

(A)

{1, 2, 3, 4, 1}

(B)

{1, 3, 3, 1, 5}

(C) ✓

{1, 4, 3, 2, 5}

(D)

{4, 2, 2, 4, 5}

Answer C

Correct. The statement arr[1] = 4; sets the element with index 1 (the second element) to 4. The statement arr[3] = 2; sets the element with index 3 (the fourth element) to 2.

FINAL Unit4 MCQ - Topic order

28. Consider the following code segment.

```
int[] scores = {85, 92, 78, 90, 88};
scores[1] *= 2;            // Line 2
scores[2] += 10;           // Line 3
```

Which of the following best describes the behavior of lines 2 and 3 in the code segment?

(A) Line 2 doubles the first element in the array and line 3 adds 10 to the second element.

(B) Line 2 doubles the second element in the array and line 3 adds 10 to the third element. ✓

(C) Line 2 doubles the third element in the array and line 3 adds 10 to the fourth element.

(D) Lines 2 and 3 do not change the contents of the array.

Answer B

Correct. The statement scores[1] *= 2 doubles the second element (index 1), and the statement scores[2] += 10 adds 10 to the third element (index 2).

29. In the following code segment, assume that the string `str` has been properly declared and initialized. The code segment is intended to print the number of strings in the array `animals` that have `str` as a substring.

```
String[] animals = {"horse", "cow", "goat", "dog", "cat", "mouse"};
int count = 0;
for (int i = 0; i <= animals.length; i++)
{
 if (animals[i].indexOf(str) >= 0)
 {
   count++;
 }
}
System.out.println(count);
```

The code segment does not work as intended. Which of the following changes should be made so the code segment works as intended?

(A) The Boolean expression in the `for` loop header should be changed to `i < animals.length`. ✓

(B) The Boolean expression in the `for` loop header should be changed to `i < animals.length - 1`.

(C) The Boolean expression in the `for` loop header should be changed to `i < animals[i].length`.

(D) The condition in the `if` statement should be changed to `animals[i].equals(str)`.

Answer A

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30. Consider the following two code segments.

I.

```
int[] arr = {1, 2, 3, 4, 5};
for (int x = 0; x < arr.length; x++)
{
 System.out.print(arr[x + 3]);
}
```

II.

```
int[] arr = {1, 2, 3, 4, 5};
for (int x : arr)
{
 System.out.print(x + 3);
}
```

Which of the following best describes the behavior of code segment I and code segment II ?

(A) Both code segment I and code segment II will print `45678`.

(B) Code segment I will cause an `ArrayIndexOutOfBoundsException` and code segment II will print `45`.

(C) Code segment I will cause an `ArrayIndexOutOfBoundsException` and code segment II will print `45678`. ✓

(D) Both code segment I and code segment II will cause an `ArrayIndexOutOfBoundsException`.

Answer C

Correct. The first iteration of the `for` loop will print `arr[0 + 3]`, or `4`. The second iteration will print `arr[1 + 3]`, or `5`. The third iteration will attempt to access `arr[2 + 3]`. Since `arr` has five elements, the valid array indices are from `0` to `4`, inclusive. Attempting to access `arr[5]` will cause an `ArrayIndexOutOfBoundsException`. Code segment II uses the enhanced `for` loop. In the first iteration of the loop, `x` is `1` and `1 + 3`, or `4` is printed. During the next iteration of the loop, `x` is `2` and `2 + 3`, or `5` is printed. As `x` takes on the values `3`, `4`, and `5` for the remaining loop iterations, `6`, `7`, and `8` will be printed.

FINAL Unit4 MCQ - Topic order

31. Consider the following method.

```
// precondition: arr contains no duplicates;
//               the elements in arr are in sorted order;
//               0 ≤ low ≤ arr.length; low - 1 ≤ high < arr.length
public static int mystery(int[] arr, int low, int high, int num)
{
  int mid = (low + high) / 2;

  if (low > high)
  {
    return low;
  }
  else if (arr[mid] < num)
  {
    return mystery(arr, mid + 1, high, num);
  }
  else if (arr[mid] > num)
  {
    return mystery(arr, low, mid - 1, num);
  }
  else // arr{mid] == num
  {
    return mid;
   }
}
```

How many calls to `mystery` (including the initial call) are made as a result of the call `mystery(arr, 0, arr.length - 1, 14)` if `arr` is the following array?

	0	1	2	3	4	5	6	7
arr	11	13	25	26	29	30	31	32

(A) 2

(B) 4 ✓

(C) 7

(D) 8

FINAL Unit4 MCQ - Topic order

32. Consider the following recursive method.

```
/** Precondition: 0 <= numVals <= nums.length */
public static int mystery(int[] nums, int v, int numVals)
{
 if (numVals == 0)
 {
   return 0;
 }
 else if (v == nums[numVals - 1])
 {
   return 1 + mystery(nums, v, numVals - 1);
 }
 else
 {
   return mystery(nums, v, numVals - 1);
 }
}
```

Which of the following best describes the value returned by the call `mystery(nums, v, nums.length)` ?

(A) The value `1` is returned.

(B) The number of times that `v` occurs in `nums` is returned. ✓

(C) The number of times that `numVals` occurs in `nums` is returned.

(D) Nothing is returned. A runtime error occurs because of infinite recursion.

Answer B

Correct. A call to the `mystery` method returns the number of occurrences of `v` at indices between `0` and `numVals - 1`, inclusive, in the `nums` array. The `mystery` method calculates this value recursively. The number of occurrences in an array with `numVals` elements is equal to the number of occurrences in the `numVals - 1` leftmost elements of the array plus either zero (if the last element is not equal to `v`) or one (if the last element is equal to `v`). Similarly, the number of occurrences in the `numVals - 1` leftmost elements is equal to the number of occurrences in the `numVals - 2` leftmost elements plus either zero or one. The recursive calls continue until the method is called with a part of the array containing no elements; in this base case, zero is returned.

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33. The following question refer to the following information.

Consider the following data field and method. Method `maxHelper` is intended to return the largest value among the first `numVals` values in an array; however, `maxHelper` does not work as intended.

```
private int[] nums;


// precondition: 0 < numVals <= nums.length
private int maxHelper(int numVals)
{
Line 1:   int max = maxHelper(numVals - 1);


Line 2:   if (max > nums[numVals - 1])
            return max;
          else
            return nums[numVals - 1];
 }
```

Which of the following best describes the conditions under which `maxHelper` does not work as intended?

(A) When `numVals` is 1

(B) When the elements of `nums` are in nonincreasing order

(C) When the elements of `nums` are in nondecreasing order

(D) Method `maxHelper` never works as intended. ✓

FINAL Unit4 MCQ - Topic order

34. The following question refer to the following information.

Consider the following data field and method. Method `maxHelper` is intended to return the largest value among the first `numVals` values in an array; however, `maxHelper` does not work as intended.

```
private int[] nums;


// precondition: 0 < numVals <= nums.length
private int maxHelper(int numVals)
{
Line 1:   int max = maxHelper(numVals - 1);


Line 2:   if (max > nums[numVals - 1])
            return max;
          else
            return nums[numVals - 1];
        }
```

Which of the following corrects the method `maxHelper` so that it works as intended?

(A) Insert the following statement before Line 1.
```
if (numVals == 0)
  return numVals;
```

(B) Insert the following statement before Line 1. ✓
```
if (numVals == 1
  return nums[0];
```

(C) Insert the following statement between Line 1 and Line 2.
```
if (numVals == 0)
  return numVals;
```

(D) Insert the following statement between Line 1 and Line 2.
```
if (numVals == 1)
  return nums[0];
```

FINAL Unit4 MCQ - Topic order

35. Consider the following code segment.

```
int[] arr = {4, 3, 2, 1, 0};
int total = 0;
for (int k = 0; k <= total; k++)
{
  if (arr[k] % 2 == 0)
  {
    total += arr[k];
  }
  else
  {
    total -= arr[k];
  }
}
System.out.print(total);
```

What, if anything, is printed as a result of executing the code segment?

(A) `2`

(B) `1` ✓

(C) `0`

(D) Nothing is printed because the code segment causes a runtime error.

Answer B

Correct. During the first iteration of the `for` loop, the Boolean expression `arr[0] % 2 == 0` or `4 % 2 == 0` evaluates to `true`, so `total` is incremented by `4`, becoming `4`. During the second iteration of the `for` loop, the Boolean expression `arr[1] % 2 == 0` or `3 % 2 == 0` evaluates to `false`, so `total` is decremented by `3`, becoming `1`. At this point, the Boolean expression `k <= total` or `2 <= 1` evaluates to `false`, and the `for` loop terminates. The current value of `total`, `1`, is printed.

FINAL Unit4 MCQ - Topic order

36. Consider the following class definition.

```
public class Toy
{
 private int yearFirstSold;
 public int getYearFirstSold()
 {
   return yearFirstSold;
 }
 /* There may be instance variables, constructors, and other methods not
 shown. */
}
```

The following code segment, which appears in a class other than `Toy`, prints the year each `Toy` object in `toyArray` was first sold by its manufacturer. Assume that `toyArray` is a properly declared and initialized array of `Toy` objects.

```
for (Toy k : toyArray)
{
  System.out.println(k.getYearFirstSold());
}
```

Which of the following could be used in place of the given code segment to produce the same output?

I.

```
for (int k = 0; k < toyArray.length; k++)
{
  System.out.println(getYearFirstSold(k));
}
```

II.

```
for (int k = 0; k < toyArray.length; k++)
{
  System.out.println(k.getYearFirstSold());
}
```

III.

```
for (int k = 0; k < toyArray.length; k++)
{
  System.out.println(toyArray[k].getYearFirstSold());
}
```

(A) II only

(B) III only ✓

(C) I and II

(D) II and III

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Answer B

Correct. Code segment I uses the loop control variable `k` improperly, as a parameter for the `getYearFirstSold` method rather than as an array index. Code segment II also uses the loop control variable `k` improperly, as an object reference rather than as an array index. Code segment III uses the loop control variable `k` as an index into `toyArray` and the expression `toyArray[k]` as the required `Toy` object reference for the `getYearFirstSold` method call.

37. Consider the following code segment.

```
int[] arr = {10, 20, 30, 40, 50};
for (int x = 1; x < arr.length - 1; x++)
{
  arr[x + 1] = arr[x] + arr[x + 1];
}
```

Which of the following represents the contents of `arr` after the code segment has been executed?

(A)

{10, 20, 50, 90, 50}

(B) ✓

{10, 20, 50, 90, 140}

(C)

{10, 30, 60, 100, 50}

(D)

{10, 30, 60, 100, 150}

Answer B

Correct. The code starts with the second element of the array (i.e., the element at index 1) and adds the current element to the next element of the array. The code will set the third element to `20 + 30`, which is `50`. It will then set the fourth element to `50 + 40`, which is `90`. Finally, it will set the last element to `90 + 50`, which is `140`.

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38. Consider the following code segment.

```
int[] arr = {10, 20, 30, 40, 50, 60};
for (int j = arr.length - 1; j >= 0; j -= 2)
{
  arr[j]++;
}
```

What are the contents of arr after executing this code segment?

(A)

{10, 20, 31, 40, 51, 60}

(B) ✓

{10, 21, 30, 41, 50, 61}

(C)

{11, 20, 31, 40, 51, 60}

(D)

{11, 21, 31, 41, 51, 61}

Answer B

Correct. The code segment traverses the array from right to left starting at the last element and decrementing the index by 2 for each iteration of the loop. The value of each accessed element is incremented by 1.

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39. The Fibonacci numbers are a sequence of numbers. The first two numbers are 1 and 1. Each subsequent number is equal to the sum of the previous two numbers. For example, the first seven Fibonacci numbers are 1, 1, 2, 3, 5, 8, and 13.

The following code segment is intended to fill the `fibs` array with the first ten Fibonacci numbers. The code segment does not work as intended.

```
int[] fibs = new int[10];
fibs[0] = 1;
fibs[1] = 1;
for (int j = 1; j < fibs.length; j++)
{
  fibs[j] = fibs[j - 2] + fibs[j - 1];
}
```

Which of the following best explains why the code segment does not work as intended?

(A) In the `for` loop header, the initial value of `j` should be `0`.

(B) In the `for` loop header, the initial value of `j` should be `2`. ✓

(C) The `for` loop condition should be `j < fibs.length - 1`.

(D) The `for` loop condition should be `j < fibs.length + 1`.

Answer B

Correct. If `j` starts at `1`, the attempt to access `fibs[j - 2]` during the first loop iteration will throw an `ArrayIndexOutOfBoundsException`. Starting the loop at `j = 2` avoids this error.

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40. Consider the following code segment.

```
int[] arr1 = {2, 4, 6, 8, 10};
int num = 0;
for (int val : arr1)
{
   num += 10;
}
```

Which of the following best describes the behavior of the code segment?

(A) The loop adds 10 to each element in `arr1`.

(B) The loop adds 10 to `num` once for each element in `arr1`. ✓

(C) The loop adds the value of each element in `arr1` to `num`.

(D) The loop does not change the value of `num` because it uses an enhanced `for` loop.

Answer B

Correct. The enhanced `for` loop iterates for each element in `arr1`. For each iteration, the value 10 is added to `num`.

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41. In the following code segment, `arr` is a one-dimensional array of integers.

```
int sum = 0;
for (int n : arr)
{
  sum = sum + 2 * n;
}
System.out.print(sum);
```

Which of the following code segments will produce the same output as the given code segment?

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(A)

```
int sum = 0;
for (int k = 0; k < arr.length; k++)
{
  sum = sum + 2 * k;
}
System.out.print(sum);
```

(B)

```
int sum = 0;
for (int k = 0; k <= arr.length; k++)
{
  sum = sum + 2 * k;
}
System.out.print(sum);
```

(C) ✓

```
int sum = 0;
for (int k = 0; k < arr.length; k++)
{
  sum = sum + 2 * arr[k];
}
System.out.print(sum);
```

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(D)

```
int sum = arr[0];

for (int k = 1; k <= arr.length; k++)

{

  sum = sum + 2 * arr[k];

}

System.out.print(sum);
```

Answer C

Correct. The original code segment uses an enhanced `for` loop to iterate through the elements of array `arr`, uses the variable `sum` to accumulate the sum of twice the value of the elements in the array, and prints the value of `sum`. This code segment produces the same output using a regular `for` loop. As the index `k` varies from `0` to `arr.length - 1`, twice the value of `arr[k]` accumulates in the variable `sum`.

42. Consider the following code segment.

```
int[] arr = {1, 2, 3, 4, 5, 6, 7};
for (int i = 1; i < arr.length; i += 2)
{
  arr[i] = arr[i - 1];
}
```

Which of the following represents the contents of the array `arr` after the code segment is executed?

(A) `{0, 1, 2, 3, 4, 5, 6}`

(B) `{1, 1, 1, 1, 1, 1, 1}`

(C) `{1, 1, 3, 3, 5, 5, 7}` ✓

(D) `{1, 2, 3, 4, 5, 6, 7}`

Answer C

Correct. The loop control variable `i` is initialized to `1`. During the first iteration of the `for` loop, `arr[1]` is set to `arr[0]` and `i` is incremented by `2` to become `3`. During the second iteration of the `for` loop, `arr[3]` is set to `arr[2]` and `i` is incremented by `2` to become `5`. During the third iteration of the `for` loop, `arr[5]` is set to `arr[4]` and `i` is incremented by `2` to become `7`. At this point, the Boolean expression `i < arr.length` evaluates to `false` and the `for` loop terminates, leaving `{1, 1, 3, 3, 5, 5, 7}` as the contents of `arr`.

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43. Consider the following method.

```
public static int mystery(int[] arr)
{
  int x = 0;
  for (int k = 0; k < arr.length; k = k + 2)
  {
    x = x + arr[k];
  }
  return x;
}
```

Assume that the array nums has been declared and initialized as follows.

```
int[] nums = {3, 6, 1, 0, 1, 4, 2};
```

What value is returned as a result of the call mystery(nums)?

(A)
5

(B) ✓
7

(C)
10

(D)
17

Answer B

Correct. The loop control variable k is initially set to 0 and then incremented by 2 until k is greater than or equal to the length of the array. During the loop, the values of k are 0, 2, 4, and 6. When k has the value 8, the loop terminates since the length of arr is 7. Prior to the loop, x is assigned 0 and then in the body of the loop, the value of arr at index k is added to x. In the first iteration of the loop, when k has the value 0, arr[0] (which is 3) is added to the value of x and x is assigned 3. In the second iteration of the loop, when k has the value 2, arr[2] (which is 1) is added to the value of x and x is assigned 4. In the third iteration of the loop, when k has the value 4, arr[4] (which is 1) is added to the value of x and x is assigned 5. In the fourth and final iteration of the loop, when k has the value 6, arr[6] (which is 2) is added to the value of x and x is assigned 7.

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44. The array `fruits` is declared below.

```
String [] fruits = {"apples", "bananas", "cherries", "dates"};
```

Which of the following code segments will cause an `ArrayIndexOutOfBoundsException` ?

I.

```
for (int i = 0; i <= fruits.length; i++)
{
  System.out.println(fruits[i]);
 }
```

II.

```
for (int i = 0; i <= fruits.length - 1; i++)
{
  System.out.println(fruits[i]);
}
```

III.

```
for (int i = 1; i <= fruits.length; i++)
{
  System.out.println(fruits[i - 1]);
}
```

(A) I only ✓

(B) II only

(C) I and III only

(D) II and III only

Answer A

Correct. Code segment I iterates `i` from `0` to `4`. The last array element is at index `3`, so an `ArrayIndexOutOfBoundsException` occurs when attempting to access the element at index `4`. Code segment II iterates `i` from `0` to `3`. It accesses each element of the array without error. Code segment III iterates `i` from `1` to `4`. Since it accesses array elements at index `i - 1`, it accesses the array elements at indexes `0` to `3` without error.

FINAL Unit4 MCQ - Topic order

45. The following code segment is intended to set the array numbers to {10, 20, 30, 40}.

```
int[] numbers = new int[4];
/* missing code */
```

Which of the following can replace /* missing code */ so that the code segment will work as intended?

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(A)

```
int x = 10;

for (int value : numbers)

{

  value = x;

  x += 10;

}
```

(B)

```
for (int j = 0; j < numbers.length; j++)

{

  numbers[j] = j * 10;

}
```

(C)

```
numbers[0] = 10;

for (int j = 0; j <numbers.length; j++)

{

  numbers[j] = numbers[j] + 10;

}
```

(D) ✓

```
numbers[3] = 40;

for (int j = numbers.length - 1; j > 0; j--)

{

  numbers[j - 1] = numbers[j] – 10;

}
```

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Answer D

Correct. This code segment starts at the last element and sets it to 40. It then traverses the array from right to left, assigning each element a value that is 10 less than the element to its right. The code segment correctly sets numbers to {10, 20, 30, 40}.

46. Consider the following code segment.

```
boolean[] oldVals = {true, false, true, true};
boolean[] newVals = new boolean[4];
for (int j = oldVals.length - 1; j >= 0; j--)
{
  newVals[j] = !(oldVals[j]);
}
```

What, if anything, will be the contents of `newVals` after executing the code segment?

(A) {true, false, true, true}

(B) {false, true, false, false} ✓

(C) {false, false, true, false}

(D) The array `newVals` will not contain any values because the code segment does not compile.

Answer B

Correct. The code segment traverses `oldVals` from right to left and assigns the negation (!) of each element of `oldVals` to the corresponding element in `newVals`.

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47. Consider the following code segment.

```
int[] nums = {5, 10, 15, 20, 25, 30};
for (int k = 0; k < nums.length - 1; k += 2)
{
   nums[k] = nums[k + 1];
}
```

What are the contents of nums after executing the code segment?

(A)

{5, 5, 15, 15, 25, 25}

(B)

{10, 10, 20, 20, 25, 30}

(C) ✓

{10, 10, 20, 20, 30, 30}

(D)

{10, 15, 20, 25, 30, 30}

Answer C

Correct. The elements at indices 0, 2, and 4 are assigned the values of the elements at the next index. Since k increments by 2, the values at indices 1, 3, and 5 remain the same.

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48. Consider the following code segment.

```
int[] arr = {3, 1, 0, 4, 2};
for(int j = 0; j < arr.length; j++)
 {
  System.out.print(arr[j] + j + " ");
 }
```

What, if anything, is printed as a result of executing the code segment?

(A) `3 1 0 4 2`

(B) `3 2 2 7 6` ✓

(C) `6 2 0 8 4`

(D) Nothing is printed, because an `ArrayIndexOutOfBoundsException` is thrown.

Answer B

Correct. The loop iterates from `j = 0` to `j = 4`. For each array element `arr[j]`, the sum of the element and its index `j` is printed.

FINAL Unit4 MCQ - Topic order

49. Consider the following method, which is intended to return the maximum integer value contained in ranges.

```
/** Precondition: ranges contains at least one element. */
public int findMaxValue(int[] ranges)
{
  int max = 0;              // line 4
  for (int value : ranges)
  {
    if (value > max)        // line 7
    {
      max = value;          // line 9
    }
  }
  return max;
}
```

This method does not always work as intended. Which of the following changes can be made so that the code segment works as intended?

(A) Changing line 4 to int max = ranges[0]; ✓

(B) Changing line 4 to int max = Integer.MAX_VALUE;

(C) Changing line 7 to if (value < max)

(D) Changing line 9 to value = max;

Answer A

Correct. The given method only works as intended when the maximum value in ranges is not negative. Assigning max the value of the first element in ranges will ensure that each element in ranges is compared to another element in ranges.

FINAL Unit4 MCQ - Topic order

50. Consider the following method.

```
public static int mystery(int value)
{
 int sum = 0;
 int[] arr = {1, 4, 2, 5, 10, 3, 6, 4};

 for (int item : arr)
 {
   if (item > value)
   {
     sum += item;
   }
 }
 return sum;
}
```

What value is returned as a result of the call `mystery(4)` ?

(A) `6`

(B) `15`

(C) `21` ✓

(D) `29`

Answer C

Correct. The mystery method calculates the sum of those elements of `arr` that exceed `value`. There are three array elements that exceed the parameter value of `4`: `5`, `10`, and `6`. The method returns the sum `5 + 10 + 6 = 21`.

FINAL Unit4 MCQ - Topic order

51. Consider the following method.

```
public static void sort(String[] arr)
{
  for (int pass = arr.length - 1; pass >= 1; pass--)
  {
    String large = arr[0];
    int index = 0;
    for (int k = 0; k <= pass; k++)
    {
      if ((arr[k].compareTo(large)) > 0)
      {
        large = arr[k];
        index = k;
      }
    }
    arr[index] = arr[pass];
    arr[pass] = large;
  }
}
```

Assume `arr` is the following array.

"Ann"	"Mike"	"Walt"	"Lisa"	"Shari"	"Jose"	"Mary"	“Bill”

What is the intermediate value of `arr` after two iterations of the outer `for` loop in the call `sort(arr)`?

(A)

"Ann"	"Mike"	"Walt"	"Lisa"	"Shari"	"Jose"	"Mary"	“Bill”

(B)

"Ann"	"Mike"	"Lisa"	"Shari"	"Jose"	"Mary"	“Bill”	"Walt"

(C)

"Ann"	“Bill”	"Jose"	"Lisa"	"Mary"	"Mike"	"Shari"	"Walt"

(D) ✓

"Ann"	"Mike"	“Bill”	"Lisa"	"Mary"	"Jose"	"Shari"	"Walt"

FINAL Unit4 MCQ - Topic order

52. Consider the static method `selectSort` shown below. Method `selectSort` is intended to sort an array into increasing order; however, it does not always work as intended.

```
// precondition:  numbers.length > 0
// postcondition: numbers is sorted in increasing order
public static void selectSort(int[] numbers)
{
  int temp;
Line 1:  for (int j = 0; j < numbers.length - 1; j++)
  {
Line 2:    int pos = 0;
Line 3:    for (int k = j + 1; k < numbers.length; k++)
     {
Line 4:      if (numbers[k] < numbers[pos])
        {
Line 5:        pos = k;
        }
     }
   temp = numbers[j];
   numbers[j] = numbers[pos];
   numbers[pos] = temp;
  }
}
```

Which of the following changes should be made so that `selectSort` will work as intended?

(A) Line 1 should be changed to
```
for (int j = 0; j < numbers.length - 2; j++)
```

(B) Line 2 should be changed to ✓
```
int pos = j;
```

(C) Line 3 should be changed to
```
for (int k = 0; k < numbers.length; k++)
```

(D) Line 4 should be changed to
```
if (numbers[k] > numbers[pos])
```

FINAL Unit4 MCQ - Topic order

53. The following incomplete method is intended to sort its array parameter `arr` in increasing order.

```
// postcondition: arr is sorted in increasing order
public static void sortArray(int[] arr)
{
  int j, k;

  for (j = arr.length - 1; j > 0; j--)
  {
    int pos = j;

    for ( /* missing code */ )
    {
      if (arr[k] > arr[pos])
      {
        pos = k;
      }
   }
   swap(arr, j, pos);
 }
 }
```

Assume that `swap(arr, j, pos)` exchanges the values of `arr[j]` and `arr[pos]`. Which of the following could be used to replace `/* missing code */` so that executing the code segment sorts the values in array `arr`?

(A) `k = j - 1; k > 0; k--`

(B) `k = j - 1; k >= 0; k--` ✓

(C) `k = 1; k < arr.length; k++`

(D) `k = 0; k <= arr.length; k++`

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54. Consider the following method. Method `allEven` is intended to return `true` if all elements in array `arr` are even numbers; otherwise, it should return `false`.

```
public boolean allEven(int[] arr)
{
  boolean isEven = /* expression */ ;

  for (int k = 0; k < arr.length; k++)
  {
    /* loop body */
  }

  return isEven;
}
```

Which of the following replacements for `/* expression */` and `/* loop body */` should be used so that method `allEven` will work as intended?

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(A)

/* expression */	/* loop body */
false	`if ((arr[k] % 2) == 0)` `  isEven = true;`

(B)

/* expression */	/* loop body */

false	`if ((arr[k] % 2) != 0)`
	`  isEven = false;`
	`else`
	`  isEven = true;`

(C) ✓

/* expression */	/* loop body */
true	`if ((arr[k] % 2) != 0)` `  isEven = false;`

(D)

/* expression */	/* loop body */

true	`if ((arr[k] % 2) != 0)`
	`  isEven = false;`
	`else`
	`  isEven = true;`

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55. Consider the following instance variable and incomplete method. The method is intended to return a string from the array `words` that would be last alphabetically.

```
private String[] words;

public String findLastWord()
{
 /* missing implementation */
}
```

Assume that `words` has been initialized with one or more strings containing only lowercase letters. Which of the following code segments can be used to replace `/* missing implementation */` so that `findLastWord` will work as intended?

(A)
```
int maxIndex = 0;
for (int k = 0; k < words.length; k++)
{
 if (words[k].compareTo(maxIndex) > 0)
 {
   maxIndex = k;
 }
}
return words[maxIndex];
```

(B)
```
int maxIndex = 0;
for (int k = 1; k <= words.length; k++)
{
 if (words[k].compareTo(words[maxIndex]) > 0)
 {
   maxIndex = k;
 }
}
return words[maxIndex];
```

(C)
```
int maxIndex = 0;
for (int k = 1; k < words.length; k++)
{
 if (words[k].compareTo(words[maxIndex]) > 0)
 {
   maxIndex = k;
 }
}
return maxIndex;
```

(D) ✓
```
String maxWord = words[0];
for (int k = 1; k < words.length; k++)
{
 if (words[k].compareTo(maxWord) > 0)
 {
   maxWord = words[k];
 }
}
return maxWord;
```

Answer D

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Correct. The first statement initializes `maxWord`, the alphabetically greatest word, with the first element of the array. The `for` loop then iterates over values of `k` from `1` to the index of the last element in the array, inclusive. Each element of the array is compared to the current value of `maxWord`. If an element of the array is alphabetically greater than `maxWord`, then `maxWord` is updated to contain that array element. When the loop terminates, `maxWord` contains the element of `words` that is last alphabetically.

56. Consider the following instance variable and method.

```
private int[] array;

/** Precondition: array.length > 0
 */
public int checkArray()
{
  int loc = array.length / 2;
  for (int k = 0; k < array.length; k++)
  {
    if (array[k] > array[loc])
    {
      loc = k;
    }
  }
  return loc;
}
```

Which of the following is the best postcondition for checkArray ?

(A) Returns the index of the last element in array array whose value is greater than array [loc]

(B) Returns the largest value in array array

(C) Returns the index of the largest value in array array ✓

(D) Returns the index of the largest value in the second half of array array

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57. Consider an int array named nums, which has been properly declared and initialized with one or more values. Which of the following code segments counts the number of negative values found in nums and stores the count in counter?

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(A)

```
int counter = 0;
int i = 0;
while (i < nums.length)
{
  if (nums[i] < 0)
  {
    counter++;
    i++;
  }
}
```

(B) ✓

```
int counter = 0;
int i = 0;
while (i <= nums.length - 1)
{
  if (nums[i] < 0)
  {
    counter++;
  }
  i++;
}
```

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(C)

```
int counter = 0;
for (int i = 1; i < nums.length; i++)
{
  if (nums[i] < 0)
  {
    counter++;
  }
}
```

(D)

```
int counter = 0;
for (int i : nums)
{
  if (nums[i] < 0)
  {
    counter++;
  }
}
```

Answer B

Correct. In this code segment, i takes on the values 0 through nums.length - 1, inclusive, in the loop. The array elements nums[0] through nums[nums.length - 1] are each compared to 0, and counter is incremented only for negative values.

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58. Consider the following method, which is intended to return the number of elements in `titles` with more than 15 characters. The method does not always work as intended.

```
public static int findLongTitles(String[] titles)
{
  int count = 0;
  for (int x = 0; x < titles.length; x++)     // Line 4
  {
    if (titles[x].length() > 15)              // Line 6
    {
      count++;
      return count;                           // Line 9
    }
  }                                           // Line 11
  return count;                               // Line 12
}
```

Which of the following changes can be made so that the method works as intended?

(A) Changing the Boolean expression in line 4 to `x <= titles[x].length()`

(B) Changing the Boolean expression in line 6 to `titles[x].length() >= 15`

(C) Removing line 9 ✓

(D) Interchanging lines 11 and 12

Answer C

Correct. The return statement in line 9 causes the `for` loop to terminate after the first element in `titles` with a length greater than 15 characters is accessed. Removing the `return` statement allows the `for` loop to traverse the entire array before returning a value.

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59. In the following code segment, data is an int array with at least two elements.

```
int answer = 0;
for (int val : data)
{
   if (data[0] < val)
   {
      answer++;
   }
}
System.out.println(answer);
```

Which of the following best describes the value printed by the code segment?

(A) The number of negative elements in data

(B) The number of positive elements in data

(C) The number of elements in data that are greater than the first element in data ✓

(D) The number of elements in data that are less than than the first element in data

Answer C

Correct. The enhanced for loop compares each array value with the first element of the array (data[0]) and counts the values larger than the first element.

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Consider the following instance variable nums and method findLongest with line numbers added for reference. Method findLongest is intended to find the longest consecutive block of the value target occurring in the array nums; however, findLongest does not work as intended.

For example, if the array nums contains the values [7, 10, 10, 15, 15, 15, 15, 10,10, 10, 15, 10, 10], the call findLongest (10) should return 3, the length of the longest consecutive block of 10s.

```
private int[] nums;

public int findLongest(int target)
{
    int lenCount = 0;
    int maxLen = 0;

Line 1:  for (int val : nums)
Line 2:  {
Line 3:    if (val == target)
Line 4:    {
Line 5:      lenCount++;
Line 6:    }
Line 7:    else
Line 8:    {
Line 9:      if (lenCount > maxLen)
Line 10:     {
Line 11:       maxLen = lenCount;
Line 12:     }
Line 13:   }
Line 14: }
Line 15: if (lenCount > maxLen)
Line 16: {
Line 17:   maxLen = lenCount;
Line 18: }
Line 19: return maxLen;
    }
```

60. The method findLongest does not work as intended. Which of the following best describes the value returned by a call to findLongest ?

(A) It is the length of the array nums.

(B) It is the number of occurrences of the value target in nums. ✓

(C) It is the length of the first consecutive block of the value target in nums.

(D) It is the length of the last consecutive block of the value target in nums.

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61. Which of the following changes should be made so that method findLongest will work as intended?

(A) Insert the statement lenCount = 0;
between lines 2 and 3.

(B) Insert the statement lenCount = 0;
between lines 8 and 9.

(C) Insert the statement lenCount = 0;
between lines 11 and 12.

(D) Insert the statement lenCount = 0;
between lines 12 and 13. ✓

---

62. The following incomplete method is intended to return the largest integer in the array `numbers`.

```
// precondition:  numbers.length > 0

public static int findMax(int[]numbers)
{
  int posOfMax = 0;

  for (int index = 1; index < numbers.length; index++)

  {
    if ( /*condition*/ )
    {
    /* statement */

    }
  }
  return numbers[posOfMax];
}
```

Which of the following can be used to replace `/* condition */` and `/* statement */` so that `findMax` will work as intended?

(A)

/* condition */	/* statement */
numbers[index] > numbers[posOfMax]	posOfMax = numbers[index];

(B) ✓

/* condition */	/* statement */
numbers[index] > numbers[posOfMax]	posOfMax = index;

(C)

/* condition */	/* statement */
numbers[index] > posOfMax	posOfMax = numbers[index];

(D)

/* condition */	/* statement */
numbers[index] < posOfMax	posOfMax = numbers[index];

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63. The following method is intended to return the maximum value that appears in the int array numbers.

```
public static int findMaximum(int[] numbers)
{
  int max = 0;
  for (int num : numbers)
  {
    if (num > max)
    {
      max = num;
    }
  }
  return max;
}
```

The method works as intended for some, but not all, int arrays. Which of the following best describes the conditions in which the method does **not** return the intended result?

(A) When the maximum value in numbers is the first element in the array

(B) When the maximum value in numbers appears multiple times in the array

(C) When the maximum value in numbers is negative ✓

(D) When the maximum value in numbers is Integer.MAX_VALUE

Answer C

Correct. The code segment initially sets the variable max to 0. If the maximum value in the array numbers is negative, the value of max is never changed and the incorrect value 0 is returned.

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64. Consider the following incomplete method. Method `findNext` is intended to return the index of the first occurrence of the value `val` beyond the position `start` in array `arr`.

```
// returns index of first occurrence of val in arr
// after position start;
// returns arr.length if val is not found
public int findNext(int[] arr, int val, int start)
{
  int pos = start + 1;

  while ( /* condition */ )
    pos++;

  return pos;
}
```

For example, consider the following code segment.

```
int[] arr = {11, 22, 100, 33, 100, 11, 44, 100};

System.out.println(findNext(arr, 100, 2));
```

The execution of the code segment should result in the value 4 being printed.

Which of the following expressions could be used to replace `/* condition */` so that `findNext` will work as intended?

(A) `(pos < arr.length) && (arr[pos] != val)` ✓

(B) `(arr[pos] != val) && (pos < arr.length)`

(C) `(pos < arr.length) || (arr[pos] != val)`

(D) `(arr[pos] == val) && (pos < arr.length)`

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65. The code segment below is intended to print the length of the shortest string in the array `wordArray`. Assume that `wordArray` contains at least one element.

```
int shortest = /* missing value */;
for (String word : wordArray)
{
  if (word.length() < shortest)
  {
    shortest = word.length();
  }
}
System.out.println(shortest);
```

Which of the following should be used as the initial value assigned to `shortest` so that the code segment works as intended?

(A) `Integer.MAX_VALUE` ✓

(B) `Integer.MIN_VALUE`

(C) `0`

(D) `wordArray.length`

Answer A

Correct. The code segment compares the length of each string in the array to `shortest`. If a string is the shortest found so far, `shortest` is assigned the length of that string. By initializing `shortest` to `Integer.MAX_VALUE`, the first word of the string will be the shortest found so far (as the length of the string will be less than or equal to `Integer.MAX_VALUE`). Any subsequent word will be compared to the shortest word found so far, and the code segment will work as intended.

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66. Consider the following method, which is intended to return the index `i` for which the value of `predicted[i]` is furthest from the value of `actual[i]`.

For example, if `predicted` contains the values `{60, 65, 62, 68, 70, 72, 68}` and `actual` contains the values `{64, 68, 60, 67, 76, 72, 70}`, then `difference` should return `4`, because the corresponding elements with values that are furthest from each other are `predicted[4]` and `actual[4]`.

```
/**
 *  Precondition: predicted and actual have the same
 *  nonzero length.
 */
public static int difference(int[] predicted, int[] actual)
{
  int num = Integer.MIN_VALUE;
  int idx = -1;
  for (int i = 0; i < predicted.length; i++)
  {
    /* missing code */
  }
  return idx;
}
```

Which of the following can be used to replace `/* missing code */` so that the `difference` method works as intended?

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(A)

```
if (predicted[i] - actual[i] < num)
{
  num = predicted[i] - actual[i];
  idx = i;
}
```

(B)

```
if (predicted[i] - actual[i] > num)
{
  num = predicted[i] - actual[i];
  idx = i;
}
```

(C)

```
if (Math.abs(predicted[i] - actual[i]) < num)
{
  num = Math.abs(predicted[i] - actual[i]);
  idx = i;
}
```

(D) ✓

```
if (Math.abs(predicted[i] - actual[i]) > num)
{
  num = Math.abs(predicted[i] - actual[i]);
  idx = i;
}
```

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Answer D

Correct. The nonnegative difference between corresponding array elements indicates how far apart they are from each other. The variable `num` is used to store the greatest nonnegative difference that has been found while traversing the two arrays and is initialized to the smallest possible `int` value that can be stored in 4 bytes. The variable `idx` is used to store the array index where the greatest nonnegative difference is found and is initialized to `-1`. Each time through the `for` loop, the `Math.abs` method is used to find the absolute value of the difference between the values at the same position in each array. Using absolute value guarantees a nonnegative result. If that value is greater than the currently stored value in `num`, then it becomes the new maximum value and `idx` is updated to the new index.

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67. Consider the following mode method, which is intended to return the most frequently occurring value (mode) in its int[] parameter arr. If there is more than one mode, the method is intended to return the first mode that appears in the array. For example, if the parameter of the method has the contents {6, 5, 1, 5, 2, 6, 5}, then the method is intended to return 5. As another example, if the parameter has the contents {3, 8, 4, 8, 4, 6, 5}, then the method is intended to return 8.

```
/** Precondition: arr.length >= 1 */
public static int mode(int[] arr)
{
  int modeCount = 1;
  int mode = arr[0];

  for (int j = 0; j < arr.length; j++)
  {
    int valCount = 0;
    for (int k = 0; k < arr.length; k++)
    {
      if ( /* missing condition 1 */ )
      {
        valCount++;
      }
    }
    if ( /* missing condition 2 */ )
    {
      modeCount = valCount;
      mode = arr[j];
    }
  }
  return mode;
}
```

Which of the following can replace /* missing condition 1 */ and /* missing condition 2 */ so the method works as intended?

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(A)

/* missing condition 1 */	/* missing condition 2 */
arr[j] == arr[k]	valCount > modeCount

(B)

/* missing condition 1 */	/* missing condition 2 */
arr[j] == arr[k]	modeCount > valCount

(C)

/* missing condition 1 */	/* missing condition 2 */
arr[j] != arr[k]	valCount > modeCount

(D)

/* missing condition 1 */	/* missing condition 2 */
arr[j] != arr[k]	modeCount > valCount

Answer A

Correct. For each element arr[j], the inner loop counts the number of times that arr[j] appears in arr and stores the result in valCount. It does this by evaluating the condition arr[j] == arr[k] and incrementing valCount when the condition evaluates to

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true. After the inner loop completes, the method evaluates the condition valCount > modeCount. If the condition evaluates to true, a new mode has been found and mode and modeCount are updated.

68. Consider the following incomplete method, which is intended to return the longest string in the string array `words`. Assume that the array contains at least one element.

```
public static String longestWord(String[] words)
{
  /* missing declaration and initialization */
  for (int k = 1; k < words.length; k++)
  {
    if (words[k].length() > longest.length())
    {
      longest = words[k];
    }
  }
  return longest;
}
```

Which of the following can replace `/* missing declaration and initialization */` so that the method will work as intended?

(A) `int longest = 0;`

(B) `int longest = words[0].length();`

(C) `String longest = "";`

(D) `String longest = words[0];` ✓

Answer D

Correct. The code segment compares the length of each string in the array to the length of `longest`. If a longer string is found in the array, `longest` is assigned that string. By initializing `longest` to the first element in the array, `longest` will be the longest string found so far. Any subsequent string in the array will be compared to the longest string found so far, and the code segment will work as intended.

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69. Consider the following instance variable and method.

```
private int[] numbers;

/** Precondition: numbers contains int values in no particular order.
 */
public int mystery(int num)
{
  for (int k = numbers.length - 1; k >= 0; k--)
  {
    if (numbers[k] < num)
    {
      return k;
    }
  }
  return -1;
}
```

Which of the following best describes the contents of numbers after the following statement has been executed?

int m = mystery(n);

(A) All values in positions 0 through m are less than n.

(B) All values in positions m+1 through numbers.length-1 are less than n.

(C) All values in positions m+1 through numbers.length-1 are greater than or equal to n. ✓

(D) The largest value that is smaller than n is at position m.

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70. The question refer to the following data field and method.

```
private int[] arr;

// precondition: arr.length > 0
public void mystery()

{

  int s1 = 0;

  int s2 = 0;

  for (int k = 0; k < arr.length; k++)

  {

    int num = arr[k];

    if ((num > 0) && (num % 2 == 0))

      s1 += num;

    else if (num < 0)

      s2 += num;

  }

  System.out.println(s1);

  System.out.println(s2);

}
```

Which of the following best describes the value of `s1` output by the method `mystery` ?

(A) The sum of all positive values in `arr`

(B) The sum of all positive even values in `arr` ✓

(C) The sum of all positive odd values in `arr`

(D) The sum of all values greater than 2 in `arr`

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71. The question refer to the following data field and method.

```
private int[] arr;

// precondition: arr.length > 0
public void mystery()

{

  int s1 = 0;

  int s2 = 0;

  for (int k = 0; k < arr.length; k++)

  {

    int num = arr[k];

    if ((num > 0) && (num % 2 == 0))

      s1 += num;

    else if (num < 0)

      s2 += num;

  }

  System.out.println(s1);

  System.out.println(s2);

}
```

Which of the following best describes the value of `s2` output by the method `mystery` ?

(A) The sum of all positive even values in `arr`

(B) The sum of all negative values in `arr` ✓

(C) The sum of all negative even values in `arr`

(D) The sum of all negative odd values in `arr`

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72. Consider the following instance variable, arr, and incomplete method, partialSum. The method is intended to return an integer array sum such that for all k, sum [k] is equal to arr[0] + arr[1] + ... + arr[k]. For instance, if arr contains the values { 1, 4, 1, 3 }, the array sum will contain the values { 1, 5, 6, 9 }.

```
private int[] arr;
public int[] partialSum()
{
  int[] sum = new int[arr.length];
  for (int j = 0; j < sum.length; j++)
  {
    sum[j] = 0;
  }
  /* missing code */
  return sum;
}
```

The following two implementations of / * *missing code* * / are proposed so that partialSum will work as intended.

Implementation 1

```
for (int j = 0; j < arr.length; j++)
{
  sum[j] = sum[j - 1] + arr[j];
}
```

Implementation 2

```
for (int j = 0; j < arr.length; j++)
{
  for (int k = 0; k <= j; k++)
  {
    sum[j] = sum[j] + arr[k];
  }
}
```

Which of the following statements is true?

(A) Both implementations work as intended, but implementation 1 is faster than implementation 2.

(B) Both implementations work as intended, but implementation 2 is faster than implementation 1.

(C) Implementation 1 does not work as intended, because it will cause anArrayIndexOutOfBoundsException. ✓

(D) Implementation 2 does not work as intended, because it will cause anArrayIndexOutOfBoundsException.

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73. Consider the following method, which is intended to return the average (arithmetic mean) of the values in an integer array. Assume the array contains at least one element.

```
public static double findAvg(double[] values)
{
  double sum = 0.0;
  for (double val : values)
  {
    sum += val;
  }
  return sum / values.length;
}
```

Which of the following preconditions, if any, must be true about the array `values` so that the method works as intended?

(A) The array `values` must be sorted in ascending order.

(B) The array `values` must be sorted in descending order.

(C) The array `values` must not contain values whose sum is not `0`.

(D) No precondition is necessary; the method will always work as intended. ✓

Answer D

Correct. The method works as intended for any `double` array that contains at least one element. After iterating through the array, the sum of the array elements is stored in `sum`. This value is divided by the length of the array, giving the correct average.

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74. Consider the following method, which is intended to return an array of integers that contains the elements of the parameter `arr` arranged in reverse order. For example, if `arr` contains `{7, 2, 3, -5}`, then a new array containing `{-5, 3, 2, 7}` should be returned and the parameter `arr` should be left unchanged.

```
public static int[] reverse(int[] arr)
{
  int[] newArray = new int[arr.length];
  for (int k = 0; k < arr.length; k++)
  {
    /* missing statement */
  }
  return newArray;
}
```

Which of the following statements can be used to replace `/* missing statement */` so that the method works as intended?

(A)

```
newArray[k] = arr[k - arr.length];
```

(B)

```
newArray[k] = arr[k - arr.length - 1];
```

(C)

```
newArray[k] = arr[arr.length - k];
```

(D) ✓

```
newArray[k] = arr[arr.length - k - 1];
```

Answer D

Correct. The first element of `newArray`, at index `0`, is assigned the last element of `arr`, at index `arr.length - 1`, when `k` has the value `0`. The second element of `newArray`, at index `1`, is assigned the next-to-last element of `arr`, at index `arr.length - 2`, when `k` has the value `1`, and so on. In general, `newArray[k]` is assigned the value `arr[arr.length - k - 1]`.

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75. In the following code segment, assume that the int array numArr has been properly declared and initialized. The code segment is intended to reverse the order of the elements in numArr. For example, if numArr initially contains {1, 3, 5, 7, 9}, it should contain {9, 7, 5, 3, 1} after the code segment executes.

```
/* missing loop header */
{
  int temp = numArr[k];
  numArr[k] = numArr[numArr.length - k - 1];
  numArr[numArr.length - k - 1] = temp;
}
```

Which of the following can be used to replace /* missing loop header */ so that the code segment works as intended?

(A) ✓

```
for (int k = 0; k < numArr.length / 2; k++)
```

(B)

```
for (int k = 0; k < numArr.length; k++)
```

(C)

```
for (int k = 0; k < numArr.length / 2; k--)
```

(D)

```
for (int k = numArr.length - 1; k >= 0; k--)
```

Answer A

Correct. With this loop header, the code segment will iterate through the array from left to right, swapping the element k positions from the front of the array with the element k positions from the end of the array. The loop terminates when k has reached the middle of the array, leaving the array elements in reverse order.

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76. The following method is intended to return an array with the values of the parameter `letters` in reverse order. The method does not work as intended.

```
public static String[] reverseArray(String[] letters)
{
  String[] revLetters = new String[letters.length];  // Line 3
  for (int x = 0; x < letters.length; x++)          // Line 4
  {
    revLetters[x] = letters[letters.length - x];    // Line 6
  }
  return revLetters;
}
```

Which of the following changes can be made so the method returns the intended value?

(A) Changing line 3 to `String[] revLetters = new String[letters.length - 1];`

(B) Changing line 4 to `for (int x = 0; x < letters.length - 1; x++)`

(C) Changing line 6 to `revLetters[x] = letters[letters.length - x - 1];` ✓

(D) Changing line 6 to `letters[x] = revLetters[letters.length - x];`

Answer C

Correct. The original line of code would be off by one and cause an `ArrayIndexOutofBoundsException` when attempting to access the element at index `letters.length`. Subtracting 1 from the index will ensure that each element of `revLetters` is assigned an element of `letters`. As `revLetters` is traversed from left to right, it is assigned a value from `letters` from right to left.

FINAL Unit4 MCQ - Topic order

77. Consider a shuffle method that is intended to return a new array that contains all the elements from nums, but in a different order. Let n be the number of elements in nums. The shuffle method should alternate the elements from nums [0] … nums[n / 2 – 1] with the elements from nums[n / 2] …nums[n – 1], as illustrated in the following examples.

Example 1

	0	1	2	3	4	5	6	7
nums	10	20	30	40	50	60	70	80

	0	1	2	3	4	5	6	7
result	10	50	20	60	30	70	40	80

Example 2

	0	1	2	3	4	5	6
nums	10	20	30	40	50	60	70

	0	1	2	3	4	5	6
result	10	40	20	50	30	60	70

The following implementation of the shuffle method does not work as intended.

```
public static int[] shuffle(int[] nums)
{
  int n = nums.length;
  int[] result = new int[n];

  for (int j = 0; j < n / 2; j++)
  {
    result[j * 2] = nums[j];
    result[j * 2 + 1] = nums[j + n / 2];
  }

  return result;
}
```

Which of the following best describes the problem with the given implementation of the shuffle method?

(A) Executing shuffle may cause an ArrayIndexOutOfBoundsException.

(B) The last element of the returned array (result [result.length − 1]) may not have the correct value. ✓

(C) One or more of nums [0] … nums [nums.length / 2 − 1] may have been copied to the wrong position(s) in the returned array.

(D) One or more of nums [nums.length / 2] … nums[nums.length − 1] may have been copied to the wrong position(s) in the returned array.

FINAL Unit4 MCQ - Topic order

78. The `twoInARow` method below is intended to return `true` if any two consecutive elements of the parameter `arr` are equal in value and return `false` otherwise.

```
public boolean twoInARow(int[] arr)
 {
  /* missing loop header */
  {
    if (arr[k] == arr[k + 1])
    {
      return true;
    }
  }
  return false;
 }
```

Which of the following can be used to replace `/* missing loop header */` so that the method will work as intended?

(A) `for (int k = 0; k < arr.length - 1; k++)` ✓

(B) `for (int k = 0; k < arr.length; k++)`

(C) `for (int k = 1; k < arr.length; k++)`

(D) `for (int k = arr.length - 1; k >= 0; k--)`

Answer A

Correct. The loop compares each array element with an array element one position higher. With this header, the loop starts with the element at index `0` and ends with the next-to-last element, ensuring that all consecutive pairs are compared without a bounds error.

FINAL Unit4 MCQ - Topic order

79. A text file named sourceText.txt has the following contents. Assume that the file contains no spaces.

finest_artist
biggest_gamer
nicest_teacher
best_athlete

In the following code segment, a valid Scanner object named sc has been created to read from the text file.

```
File myText = new File("sourceText.txt");
Scanner sc = new Scanner(myText);

ArrayList<String> firstList = new ArrayList<String>();
ArrayList<String> secondList = new ArrayList<String>();

/* missing code */
```

This code segment is intended to assign ["finest", "biggest", "nicest", "best"] to firstList and to assign ["artist", "gamer", "teacher", "athlete"] to secondList.

Which of the following can replace /* missing code */ so that this code segment works as intended?

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(A)

```
while (sc.hasNext())
{
  firstList.add(sc.next());
  secondList.add(sc.next());
}
```

(B)

```
while (sc.hasNext())
{
  firstList.add(sc.next());
}
while (sc.hasNext())
{
  secondList.add(sc.next());
}
```

(C) ✓

```
while (sc.hasNext())
{
  String[] temp = sc.next().split("_");
  firstList.add(temp[0]);
  secondList.add(temp[1]);
}
```

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(D)

```
while (sc.hasNext())
{
  String[] temp = sc.next().split("_");
  firstList.add(temp[0]);
  firstList.add(temp[1]);
}
while (sc.hasNext())
{
  String[] temp = sc.next().split("_");
  secondList.add(temp[0]);
  secondList.add(temp[1]);
}
```

Answer C

Correct. For every string encountered in the file, the code segment will use the split method to place the text before the underscore in temp[0] and the text after the underscore in temp[1]. The value of temp[0] is added to firstList and the value of temp[1] is added to secondList.

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80. In the following code segment, a valid `Scanner` object named `scan` has been created to read from a text file named `"data.txt"`.

```
File inputFile = new File("data.txt");
Scanner scan = new Scanner(inputFile);

String name = scan.next();
int score1 = scan.nextInt();
int score2 = scan.nextInt();
```

Which of the following could be the contents of `"data.txt"` so that the variables `name`, `score1`, and `score2` are each assigned a valid value by the code segment?

(A)

Brody 85 Carmen 90

(B) ✓

Brody 85 90

(C)

Brody 85.7 Carmen 90.6

(D)

Brody 85.7 90.6

Answer B

Correct. The call to `scan.next()` will read the text up to the first whitespace character. The two `scan.nextInt()` calls will correctly read and assign `85` into `score1` and `90` into `score2`. The file provides three valid values in the exact order that the code expects, so the code segment will run without error.

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81. A text file named `colorFile.txt` has the following contents. Assume that the file contains no spaces.

```
blue
yellow
red
green
orange
brown
```

In the following code segment, a valid `Scanner` object named `scan` is created to read from the text file.

```
File colors = new File("colorFile.txt");
Scanner scan = new Scanner(colors);
int value = 0;

while (scan.hasNext())
{
  if (scan.next().length() > 4)
  {
    value++;
  }
}
System.out.println(value);
```

What is printed as a result of executing the code segment?

(A)

1

(B) ✓

4

(C)

5

(D)

6

Answer B

FINAL Unit4 MCQ - Topic order

Correct. The `while` loop executes until all lines have been read from the file. The `scan.next()` method returns the next `String` value read from the file. If the length of the `String` is greater than `4`, then `value` is incremented.

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82. A text file named data.txt has the following contents. Assume that the file contains no spaces.

```
123
456
789
```

In the following code segment, a valid Scanner object named source is created to read from the text file.

```
int x = 0;

File text = new File("data.txt");
Scanner source = new Scanner(text);

while (source.hasNext())
{
  x = source.nextInt();
}
System.out.println(x);
```

What is printed as a result of executing the code segment?

(A)

0

(B)

123

(C) ✓

789

(D)

123456789

Answer C

Correct. The code segment uses nextInt() to obtain each int value from the file. Each value obtained is assigned to the variable x. The value 789 is the last value assigned to the variable before the loop is terminated and the value is printed.

FINAL Unit4 MCQ - Topic order

83. The following code segment is intended to create and initialize the `int` array `numArray` so that it contains 10 integers read from a text file named `"numbers.txt"`. A valid `Scanner` object named `numScan` is created to read from the text file. Assume that the text file contains exactly 10 integer values, each separated by a space.

```
File numberFile = new File("numbers.txt");
Scanner numScan = new Scanner(numberFile);
int[] numArray = new int[10];
for (int j = 0; j < 10; j++)
{
  /* missing code */
}
```

Which of the following can replace `/* missing code */` so that the code segment works as intended?

(A)

```
numArray[j] = numScan.next();
```

(B)

```
numArray[j] = numScan.nextDouble();
```

(C) ✓

```
numArray[j] = numScan.nextInt();
```

(D)

```
numArray[j] = numScan.nextLine();
```

Answer C

Correct. The `nextInt` method returns the next `int` read from the file `numbers.txt`. The statement reads the next `int` from the file and assigns it to the element at index `j` in the array.

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84. In the following code segment, a valid Scanner object named scan is created to read from the text file input.txt.

```
File readFile = new File("input.txt");
Scanner scan = new Scanner(readFile);

int firstIn = scan.nextInt();
int secondIn = scan.nextInt();
int thirdIn = scan.nextInt();
```

Which of the following could be the contents of input.txt so that the variables firstIn, secondIn, and thirdIn are each assigned a valid value by the code segment?

(A)

```
123
```

(B) ✓

```
1
2
3
```

(C)

```
1, 2, 3
```

(D)

```
firstIn: 1 secondIn: 2 thirdIn: 3
```

Answer B

Correct. Despite being on separate lines, scan.nextInt() will read a separate int for each of the three values in the file.

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85. In the following code segment, a valid `Scanner` object named `fileInput` is created to read from the text file `numbers.txt`. Assume that the text file contains only numerical data and spaces.

```
File numData = new File("numbers.txt");
Scanner fileInput = new Scanner(numData);

double sum = 0.0;
while (fileInput.hasNext())
{
  /* missing code */
}

System.out.println(sum);
```

Which of the following can be used to replace `/* missing code */` so that the code segment prints the sum of the positive numbers in the file?

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(A)

```
if (fileInput > 0)
{
  sum = sum + fileInput;
}
```

(B)

```
if (fileInput.nextDouble() > 0)
{
  sum = sum + fileInput.nextDouble();
}
```

(C) ✓

```
double temp = fileInput.nextDouble();
if (temp > 0)
{
  sum = sum + temp;
}
```

(D)

```
double temp = fileInput.nextDouble();
if (temp > 0)
{
  sum = sum + fileInput.nextDouble();
}
```

Answer C

Correct. Each iteration of the loop checks whether there is another value to be read from the file. If so, this value is obtained by the

call `nextDouble()` and stored in the variable `temp`. If this value is positive, it is added `sum`.

86. Consider the following calculate method.

```
public double calculate(double x)
{
  return x + 1.5;
}
```

The following code segment calls the method calculate in the same class.

```
Double d1 = new Double(7.5);
System.out.println(calculate(d1));
```

What, if anything, is printed when the code segment is executed?

(A) 8.5

(B) 9

(C) 9.0 ✓

(D) Nothing is printed because the code does not compile. The actual parameter d1 passed to calculate is a Double, but the formal parameter x is a double.

Answer C

Correct. The variable d1, a Double with the value 7.5, is passed to the calculate method. Since the formal parameter of calculate is a double, Java automatically unboxes the actual parameter. The variable x is incremented by 1.5, resulting in 9.0, and that value is returned and printed.

FINAL Unit4 MCQ - Topic order

87. Consider the following code segment.

```
double val1 = 10.0;
Double val2 = 20.0;
Double val3 = 30.0;
System.out.println(val1 + val2 + val3);
```

What, if anything, is printed when the code segment is executed?

(A) ✓
60.0

(B)
10.050.0

(C)
10.020.030.0

(D) Nothing is printed due to a syntax error.

Answer A

Correct. The code segment prints `60.0`. The expression `val1 + val2 + val3` is computed before it is printed. The Java compiler automatically converts between primitive types and their corresponding wrapper classes using autoboxing and unboxing. In the second statement, the Java compiler applies autoboxing to assign the `double` value `20.0` to the `Double` variable `val2`. In the computation, the Java compiler automatically applies unboxing to the `Double` variables `val2` and `val3` to add the `double` values `20.0` and `30.0`, respectively. The value of the expression `val1 + val2 + val3` is `60.0`.

FINAL Unit4 MCQ - Topic order

88. A text file named grades.txt has the following contents.

89 92 90 88 79 85 77

In the following code segment, a valid Scanner object named scan is created to read from the text file. The code segment is intended to store the values from grades.txt into an int array.

```
File gradeFile = new File("grades.txt");
Scanner scan = new Scanner(gradeFile);

int[] reportCard = new int[7];
for (int i = 0; i < reportCard.length; i++)
{
  int currentGrade = /* missing expression */ ;
  reportCard[i] = currentGrade;
}
```

Which of the following can replace /* missing expression */ so that the code segment works as intended?

(A)

```
Double.parseDouble(scan.next())
```

(B) ✓

```
Integer.parseInt(scan.next())
```

(C)

```
Integer.parseInt(scan.nextInt())
```

(D)

```
Integer.parseInt(scan.nextLine())
```

Answer B

Correct. Each call to scan.next() returns a String from the text file. For example, the first call returns "89". These String values are passed to the parseInt method, which returns an Integer value that can be unboxed into an int.

FINAL Unit4 MCQ - Topic order

89. Consider the following method.

```
public static int timesTwo(int n)
{
  return n * 2;
}
```

The following code segment appears in a method in the same class as the `timesTwo` method.

```
Integer val = 10;
int result1 = timesTwo(val);
Integer result2 = result1;
System.out.print(result2);
```

What, if anything, is printed as a result of executing the code segment?

(A) 10

(B) 20 ✓

(C) Nothing is printed; the code segment will not compile because `timesTwo` cannot accept an `Integer` argument.

(D) Nothing is printed; the code segment will not compile because the `int` variable `result1` cannot be assigned to the `Integer` variable `result2`.

Answer B

Correct. The Java compiler automatically handles conversions between `Integer` and `int`. This allows the `Integer` variable `val` to be used as an argument in the `timesTwo` method call in the second line of the code segment, and the `Integer` variable `result2` to be assigned the value of the `int` variable `result1` in the third line.

FINAL Unit4 MCQ - Topic order

90. Which of the following statements will store the value 34.0 in the double variable value?

(A)

```
double value = (double) "34";
```

(B) ✓

```
double value = Double.parseDouble("34");
```

(C)

```
double value = Integer.parseInt("34") + ".0";
```

(D)

```
double value = Double.parseDouble("34") + ".0";
```

Answer B

Correct. The method parseDouble takes a string argument and returns the argument as a double value.

FINAL Unit4 MCQ - Topic order

91. Consider the following code segment.

```
ArrayList<Integer> numList = new ArrayList<Integer>();

numList.add(3);
numList.add(2);
numList.add(1);
numList.add(1, 0);
numList.set(0, 2);
```

Which of the following represents the contents of numList after executing the code segment?

(A)

[1, 3, 0, 1]

(B) ✓

[2, 0, 2, 1]

(C)

[2, 3, 2, 1]

(D)

[3, 0, 0, 1]

Answer B

Correct. The three single-parameter add method calls create an ArrayList with the contents [3, 2, 1]. The two-parameter add method call inserts an element with the value 0 at position 1, so the ArrayList contains [3, 0, 2, 1]. The set method call sets the value of the element at position 0 to 2, and the ArrayList contains [2, 0, 2, 1] at the end of the code segment.

FINAL Unit4 MCQ - Topic order

92. Consider the following method.

```
public void reviseList(ArrayList<Integer> list)
{
  for (int i = 0; i < list.size(); i++)
  {
    int temp = list.get(i);
    if (temp % 2 == 1)
    {
      list.set(i, temp + 1);
    }
  }
}
```

The reviseList method is called with an ArrayList containing the values [1, 2, 3, 3, 2, 1]. What will be the contents of the ArrayList after executing this method?

(A)

[1, 3, 3, 3, 3, 1]

(B)

[1, 3, 3, 4, 2, 2]

(C) ✓

[2, 2, 4, 4, 2, 2]

(D)

[2, 3, 4, 4, 3, 2]

Answer C

Correct. The code segment traverses list and uses the expression temp % 2 == 1 to determine whether the value of each element is odd. Each odd element is increased by 1 using a call to the set method, overriding the element at index i with the value temp + 1.

FINAL Unit4 MCQ - Topic order

93. Consider the following code segment.

```
ArrayList<String> items = new ArrayList<String>();
items.add("A");
items.add("B");
items.add("C");
items.add(0, "D");
items.remove(3);
items.add(0, "E");
System.out.println(items);
```

What is printed as a result of executing the code segment?

(A) `[A, B, C, E]`

(B) `[A, B, D, E]`

(C) `[E, D, A, B]` ✓

(D) `[E, D, A, C]`

94. Consider the following code segment.

```
ArrayList<String> animals = new ArrayList<>();
animals.add("fox");
animals.add(0, "squirrel");
animals.add("deer");
animals.set(2, "groundhog");
animals.add(1, "mouse");
System.out.println(animals.get(2) + " and " + animals.get(3));
```

What is printed as a result of executing the code segment?

(A) `mouse and fox`

(B) `fox and groundhog` ✓

(C) `groundhog and deer`

(D) `squirrel and groundhog`

Answer B

Correct. After the call `animals.add(0)`, animals contains `["fox"]`. The call `animals.add(0, "squirrel")` inserts `"squirrel"` at index `0`, changing the contents to `["squirrel", "fox"]`. The call `animals.add("deer")` appends `"deer"` to the end of the list, changing the contents to `["squirrel", "fox", "deer"]`. The call `animals.set(2, "groundhog")` sets the element at index `2` to `"groundhog"`, changing the contents to `["squirrel", "fox", "groundhog"]`. The call `animals.add(1, "mouse")` inserts `"mouse"` at index `1`, changing the contents to `["squirrel", "mouse", "fox", "groundhog"]`. Lastly, the code segment prints the elements at indices `2` and `3`, printing `"fox and groundhog"`.

FINAL Unit4 MCQ - Topic order

95. Consider the following code segment.

```
ArrayList<String> words = new ArrayList<String>();
words.add("jump");
words.add("right");
words.add("plan");
/* missing statement */
```

Which of the following can replace `/* missing statement */` so that `words` contains `["jump", "right", "laugh", "plan"]` after the code segment is executed?

(A) ✓

```
words.add(2, "laugh");
```

(B)

```
words.add(3, "laugh");
```

(C)

```
words.set(2, "laugh");
```

(D)

```
words.set(3, "laugh");
```

Answer A

Correct. This statement inserts `"laugh"` at index 2 in `words`. The current element at index 2, `"plan"`, moves one position to the right.

FINAL Unit4 MCQ - Topic order

96. Consider the following code segment.

```
ArrayList<Integer> oldList = new ArrayList();
oldList.add(100);
oldList.add(200);
oldList.add(300);
oldList.add(400);
ArrayList<Integer> newList = new ArrayList();
newList.add(oldList.remove(1));
newList.add(oldList.get(2));
System.out.println(newList);
```

What, if anything, is printed as a result of executing the code segment?

(A) `[100, 300, 400]`

(B) `[200, 300]`

(C) `[200, 400]` ✓

(D) Nothing is printed because the code segment does not compile.

Answer C

Correct. After the four calls to the `add` method, `oldList` contains `[100, 200, 300, 400]`. The call `newList.add(oldList.remove(1))` removes `200` from `oldList` and adds it to `newList`, so that `oldList` contains `[100, 300, 400]` and `newList` contains `[200]`. The call `newList.add(oldList.get(2))` adds `400` to `newList`, so that `newList` contains `[200, 400]`.

FINAL Unit4 MCQ - Topic order

97. Consider the following code segment.

```
ArrayList<Integer> numbers = new ArrayList<Integer>();
numbers.add(100);
numbers.add(200);
numbers.add(0, 300);
numbers.set(2, 400);
numbers.add(2, 500);
```

What are the contents of numbers after executing this code segment?

(A)

[100, 200, 500, 400]

(B)

[100, 200, 400, 500]

(C)

[300, 100, 400, 500]

(D) ✓

[300, 100, 500, 400]

Answer D

Correct. The first statement creates a new empty ArrayList of Integer objects. The next two statements insert 100 at index 0 and 200 at index 1. The next statement inserts 300 at index 0, which shifts 100 and 200 one position to the right. The next statement sets the element at index 2 to 400, replacing the value 200. The last statement inserts 500 at index 2, which shifts 400 one position to the right.

FINAL Unit4 MCQ - Topic order

98. Consider the following code segment.

```
ArrayList<Integer> ages = new ArrayList<Integer>();
ages.add(18);
ages.add(20);
ages.add(1, 15);
```

What are the contents of ages after executing the code segment?

(A)

[15, 18, 20]

(B) ✓

[18, 15, 20]

(C)

[18, 20, 15]

(D)

[20, 15, 18]

Answer B

Correct. The values 18 and 20 are added to the end of ages while 15 is added in between those values at index 1.

FINAL Unit4 MCQ - Topic order

99. Consider the following code segment.

```
ArrayList<Integer> nums = new ArrayList<Integer>();

nums.add(12);
nums.add(13);
nums.add(15);
nums.add(10);

if (nums.get(0) > nums.get(nums.size() - 1))
{
  int x = nums.get(0);
  nums.set(0, nums.get(nums.size() - 1));
  nums.set(nums.size() - 1, x);
}
```

Which of the following best describes the behavior of the code segment?

(A) It compares all values in nums and places nums in order from smallest to largest.

(B) It compares all values in nums and places nums in order from largest to smallest.

(C) It compares the first and last elements in nums and swaps them if the first element is larger than the last. ✓

(D) It compares the first and last elements in nums and swaps them if the first element is smaller than the last.

Answer C

Correct. The code segment checks the element at index 0 and compares that to the element at size of nums minus 1, which is the last element of nums. If the element at index 0 is greater, then it swaps the elements.

100. Consider the following code segment.

```
List<String> animals = new ArrayList<String>();

animals.add("dog");
animals.add("cat");
animals.add("snake");
animals.set(2, "lizard");
animals.add(1, "fish");
animals.remove(3);
System.out.println(animals);
```

What is printed as a result of executing the code segment?

(A) [dog, fish, cat] ✓

(B) [dog, fish, lizard]

(C) [dog, lizard, fish]

(D) The code throws an ArrayIndexOutOfBoundsException exception.

FINAL Unit4 MCQ - Topic order

101. Consider the following code segment.

```
ArrayList<String> colors = new ArrayList<String>();

colors.add("Red");
colors.add("Orange");
colors.set(1, "Yellow");
colors.add(1, "Green");
colors.set(colors.size() - 1, "Blue");
colors.remove(0);
System.out.println(colors);
```

What is printed as a result of executing the code segment?

(A) `[Red, Green]`

(B) `[Yellow, Blue]`

(C) `[Green, Blue]` ✓

(D) `[Blue, Yellow]`

Answer C

Correct. After the first two `add` method calls, the `ArrayList` contains `["Red", "Orange"]`. The method call `colors.set(1, "Yellow")` sets the `ArrayList` element at position `1` to `"Yellow"` so the `ArrayList` contains `["Red", "Yellow"]`. The method call `colors.add(1, "Green")` adds the value `"Green"` at position `1`, so the `ArrayList` contains `["Red", "Green", "Yellow"]`. The method call `colors.set(colors.size() - 1, "Blue")` updates the last element of the `ArrayList`, so the `ArrayList` contains `["Red", "Green", "Blue"]`. The `remove` method call removes the first element of the `ArrayList`, so the `ArrayList` contains `["Green", "Blue"]` when it is printed.

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102. Consider the following code segment.

```
ArrayList<String> numbers = new ArrayList<String>();
numbers.add("one");
numbers.add("two");
numbers.add(0, "three");
numbers.set(2, "four");
numbers.add("five");
numbers.remove(1);
```

Which of the following represents the contents of `numbers` after the code segment has been executed?

(A) `["one", "four", "five"]`

(B) `["three", "two", "five"]`

(C) `["three", "four", "two"]`

(D) `["three", "four", "five"]` ✓

Answer D

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103. The following incomplete statement is intended to create an `ArrayList` named `theaterClub` to store `Student` objects. Assume that the `Student` class has been properly defined.

```
ArrayList<Student> theaterClub = /* missing code */;
```

Which of the following can replace `/* missing code */` so that the statement will compile without error?

(A)

```
new Student ArrayList()
```

(B)

```
new ArrayList(Student)
```

(C) ✓

```
new ArrayList<Student>()
```

(D)

```
new ArrayList<theaterClub>()
```

Answer C

Correct. This statement correctly declares an `ArrayList` that can contain only `Student` objects. The reference type `Student` is contained in angle brackets to indicate that the `theaterClub` object is to contain elements of type `Student`.

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104. In the following code segment, `numbers` is a properly declared `ArrayList` variable that has been initialized with the following values.

```
[13, 2, -4, 0, 34, -12, -13, 18, -27]
```

```
for (int i = 0; i < numbers.size(); i++) // Line 1
{
  if (numbers.get(i) < 0)
  {
    numbers.remove(i);
  }                    // Line 6
  i--;                 // Line 7
}
```

The code segment is intended to remove all negative values from `numbers`. However, the code segment is not working as intended.

Which of the following changes can be made so that the code segment works as intended?

(A) In line 1, changing `i < numbers.size()` to `i < numbers.size() - 1`

(B) In line 7, changing `i--;` to `i++;`

(C) Removing line 7

(D) Switching lines 6 and 7 ✓

Answer D

Correct. When a negative number is removed, the remaining elements shift to the left. The value of `i` should be decreased only after an element is removed, so `i--;` should be in the body of the `if` statement.

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105. Consider the following code segment.

```
ArrayList<Integer> nums = new ArrayList<Integer>();

nums.add(3);
nums.add(2);
nums.add(1);
nums.add(0);
nums.add(0, 4);
nums.set(3, 2);
nums.remove(3);
nums.add(2, 0);
```

Which of the following represents the contents of nums after the code segment is executed?

(A)

[2, 3, 2, 1, 0]

(B)

[4, 0, 0, 1, 3]

(C)

[4, 2, 0, 2, 0]

(D) ✓

[4, 3, 0, 2, 0]

Answer D

Correct. After the call nums.add(0), the contents of nums are [3, 2, 1, 0]. The call nums.add(0, 4) inserts 4 at index 0, changing the contents to [4, 3, 2, 1, 0]. The call nums.set(3, 2) sets the element at index 3 to 2, changing the contents to [4, 3, 2, 2, 0]. The call nums.remove(3) removes the element at index 3, changing the contents to [4, 3, 2, 0]. Finally, the call nums.add(2, 0) inserts 0 at index 2, changing the contents to [4, 3, 0, 2, 0].

106. Which of the following is a reason to use an `ArrayList` instead of an array?

(A) An `ArrayList` allows faster access to its *k*th item than an array does.

(B) An `ArrayList` can store objects and an array can only store primitive types.

(C) An `ArrayList` resizes itself as necessary when items are added, but an array does not. ✓

(D) An `ArrayList` provides access to the number of items it stores, but an array does not.

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107. Consider the following statement, which is intended to create an ArrayList named al to store only elements of type Thing. Assume that the Thing class has been properly defined and includes a no-parameter constructor.

```
ArrayList<Thing> al = /* missing code */
```

Which of the following can be used to replace /* missing code */ so that the statement works as intended?

(A)

```
new Thing();
```

(B) ✓

```
new ArrayList<Thing>();
```

(C)

```
new ArrayList(Thing);
```

(D)

```
new ArrayList<>(Thing);
```

Answer B

Correct. This statement correctly declares an ArrayList that can contain only Thing objects. The reference type Thing is contained in angle brackets to indicate that the a object is to contain elements of type Thing.

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108. Consider the following statement, which is intended to create an `ArrayList` named `numbers` that can be used to store `Integer` values.

```
ArrayList<Integer> numbers = /* missing code */;
```

Which of the following can be used to replace `/* missing code */` so that the statement works as intended?

I. `new Integer()`
II. `new ArrayList<Integer>`
III. `new ArrayList<Integer>()`

(A) I and III only
(B) I and II only
(C) III only ✓
(D) II and III only

Answer C

Correct. Statement I declares an `Integer` object not an `ArrayList of Integer` objects. Statement II results in a syntax error, as the call to the `ArrayList` constructor must contain (). Statement III correctly declares an `ArrayList` object that can contain elements only of type `Integer`.

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109. Consider the following code segment.

```
ArrayList<Double> dataValues = new ArrayList<Double>();
dataValues.add(9.84);
dataValues.add(8.93);
dataValues.add(7.65);
dataValues.add(6.24);
dataValues.remove(2);
dataValues.set(2, 7.63);
```

Which of the following represents the contents of `dataValues` after executing the code segment?

(A)

[9.84, 7.63, 6.24]

(B)

[9.84, 7.63, 7.65, 6.24]

(C) ✓

[9.84, 8.93, 7.63]

(D)

[9.84, 8.93, 7.63, 6.24]

Answer C

Correct. After the four calls to the `add` method, `dataValues` contains `[9.84, 8.93, 7.65, 6.24]`. The call `dataValues.remove(2)` removes `7.65`, located at index `2`, from the list, resulting in `[9.84, 8.93, 6.24]`. The call `dataValues.set(2, 7.63)` sets the element at index `2` to `7.63`, resulting in `[9.84, 8.93, 7.63]`.

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110. Consider the following method.

```
public static void processNumbers(ArrayList<Integer> numbers)
{
   int last = numbers.size() - 1;
   numbers.add(0, numbers.get(last));
   numbers.remove(last + 1);
}
```

Which of the following best describes the preconditions necessary for the method to run without error?

(A) The `ArrayList` must be sorted from smallest to largest.

(B) The `ArrayList` must contain at least one element. ✓

(C) The `ArrayList` must contain at least two elements.

(D) The `ArrayList` must contain only positive `Integer` values.

Answer B

Correct. At least one element is needed so that numbers.size() - 1 gives a valid index.

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111. The following code segment initializes and processes an `ArrayList of String` elements. The loop is intended to delete every other element in the `ArrayList` starting with the first element.

```
ArrayList<String> nameList = new ArrayList<String>();
nameList.add("Stacey");
nameList.add("Bob");
nameList.add("Susan");
nameList.add("Pete");

for (int i = 0; i < nameList.size(); i++)
{
  /* missing code */
}
```

After the code segment is executed, `nameList` should contain `["Bob", "Pete"]`.
Which of the following can replace `/* missing code */` so that the code segment works as intended?

(A) ✓

```
nameList.remove(i);
```

(B)

```
nameList.remove(i);

i++;
```

(C)

```
nameList.set(i, "Bob");

nameList.set(i, "Pete");
```

(D)

```
nameList.set(i, "Bob");

nameList.set(i + 1, "Pete");
```

Answer A

Correct. In the first iteration (`i = 0`), the element at position 0 is removed and the remaining elements are moved up one position, reducing the size of the list to 3. In the second iteration (`i = 1`), the element at position 1 (originally at position 2) is removed and the remaining elements are moved up a position. At this point, the size of the list is 2, so at the next iteration (`i = 2`), the loop ending condition is `false` and the loop exits.

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112. Consider the following code segment.

```
ArrayList<String> oldList = new ArrayList<String>();
oldList.add("a");
oldList.add("b");
oldList.add("c");

ArrayList<String> newList = new ArrayList<String>();

/* missing code */
```

After executing this code segment, newList should contain ["c", "b", "a", "a", "b", "c"].
Which of the following can replace /* missing code */ to produce this result?

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(A)

```
for (String s : oldList)
{
  newList.add(s);
  newList.add(s);
}
```

(B) ✓

```
for (String s : oldList)
{
  newList.add(s);
  newList.add(0, s);
}
```

(C)

```
for (int j = 0; j < oldList.size(); j++)
{
  newList.add(oldList.get(0));
  newList.add(oldList.get(j));
}
```

(D)

```
for (int j = 0; j < oldList.size(); j++)
{
  newList.add(j, oldList.get(j));
  newList.add(0, oldList.get(j));
}
```

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Answer B

Correct. This code segment traverses oldList. For each iteration of the loop, it appends an element from oldList to the end of newList and inserts the same element at the beginning of newList.

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113. Consider the following method.

```
public static void transform(ArrayList<Integer> items)
{
  for (int i = items.size() - 1; i >= 0; i--)
  {
    items.add(i, items.get(i));
  }
}
```

The following code segment appears in a method in the same class as `transform`.

```
ArrayList<Integer> numbers = new ArrayList<Integer>();
numbers.add(15);
numbers.add(4);
numbers.add(19);
numbers.add(7);

transform(numbers);
```

What are the contents of `numbers` after executing the code segment?

(A)

[15, 4, 4, 19, 19, 7, 7]

(B)

[15, 4, 19, 7, 7, 19, 4, 15]

(C)

[15, 4, 19, 7, 15, 4, 19, 7]

(D) ✓

[15, 15, 4, 4, 19, 19, 7, 7]

Answer D

Correct. The loop traverses each element in the `ArrayList` in reverse. In each iteration, the loop extracts the current element and adds it before itself. So, for the first iteration, `7` is extracted and added between `19` and `7`, yielding `[15, 4, 19, 7, 7]`. In the second iteration, `19` is extracted and added before itself, yielding `[15, 4, 19, 19, 7, 7]`. The remaining iterations are similar so that each element is copied and inserted before the original.

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114. Consider the following data field and method.

```
private ArrayList list;
public void mystery(int n)
{
  for (int k = 0; k < n; k++)
  {
    Object obj = list.remove(0);
    list.add(obj);
  }
}
```

Assume that `list` has been initialized with the following `Integer` objects.

```
[12, 9, 7, 8, 4, 3, 6, 11, 1]
```

Which of the following represents the list as a result of a call to `mystery(3)`?

(A) `[12, 9, 8, 4, 3, 6, 11, 1, 7]`

(B) `[12, 9, 7, 8, 4, 6, 11, 1, 3]`

(C) `[12, 9, 7, 4, 3, 6, 11, 1, 8]`

(D) `[8, 4, 3, 6, 11, 1, 12, 9, 7]` ✓

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115. Consider the following method.

```
public void removeAll(int target, ArrayList<Integer> list)
{
  for (int index = list.size(); index >= 0; index--)  // Line 3
  {
    if (list.get(index) == target)                    // Line 5
    {
      list.remove(index);
    }
  }
}
```

The method is intended to remove all values matching a target value from its `ArrayList` parameter. However, it is not working as intended.

Which of the following changes can be made so that the method works as intended?

(A) In line 3, changing the loop initialization to `index = list.size() - 1` ✓

(B) In line 3, changing the loop condition to `index > 0`

(C) In line 3, changing the loop increment to `index++`

(D) In line 5, changing the list access expression to `list.get(target)`

Answer A

Correct. The code will throw an `IndexOutOfBoundsException` the first time through the loop because the index `list.size()` is beyond the last element in the `ArrayList`.

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116. In the following code segment, numbers is a properly declared and initialized ArrayList<Integer> variable.

```
for (int i = 0; i < numbers.size(); i++)
{
  if (numbers.get(i) < 0)
  {
    numbers.remove(i);
  }
}
```

The code segment is intended to remove all negative values from numbers. Which of the following best describes the conditions in which the code segment works as intended?

(A) The code segment works as intended for all possible contents of numbers.

(B) The code segment works as intended only when negative numbers are not in adjacent positions in numbers. ✓

(C) The code segment works as intended only when numbers is sorted.

(D) The code segment works as intended only when there are no duplicate values in numbers.

Answer B

Correct. When an element is removed from an ArrayList, the remaining elements are shifted to the left by one. This could cause elements to be skipped during a traversal. When an ArrayList has adjacent elements with negative values, the first negative element is removed and the second one is skipped, so the ArrayList will still contain elements with a negative value.

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117. The following method is a correct implementation of the insertion sort algorithm. The method correctly sorts the elements of numList so that they appear in order from least to greatest.

```
public static void insertionSort(ArrayList<Integer> numList)
{
  for (int j = 1; j < numList.size(); j++)
  {
    int temp = numList.get(j);
    int k = j;
    while (k > 0 && temp < numList.get(k - 1))
    {
      numList.set(k, numList.get(k - 1));
      k--;
    }
    numList.set(k, temp);
    /* end of outer loop */
  }
}
```

Assume that insertionSort has been called with an ArrayList parameter that has been initialized with the following Integer objects.

[60, 30, 10, 20, 50, 40]

What will the contents of numList be after three passes of the outer loop (i.e., when j == 3 at the point indicated by /* end of outer loop */)?

(A)

[10, 20, 30, 50, 60, 40]

(B) ✓

[10, 20, 30, 60, 50, 40]

(C)

[10, 30, 60, 20, 50, 40]

(D)

[20, 10, 30, 60, 50, 40]

Answer B

Correct. In each iteration of the outer loop, the method moves the element at index j to its proper position in the part of the list from index 0 to index j. After the first iteration of the outer loop, 30 is moved to index 0 so that the contents

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of numList are [30, 60, 10, 20, 50, 40]. After the second iteration of the outer loop, 10 is moved to index 0 so that the contents of numList are [10, 30, 60, 20, 50, 40]. After the third iteration of the outer loop, 20 is moved to index 1 so that the contents of numList are [10, 20, 30, 60, 50, 40].

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118. Consider the following code segment.

```
int[][] mat = {{10, 20, 30},
               {40, 50, 60}};

ArrayList<Integer> result = new ArrayList<Integer>();

/* missing code */
```

After executing this code segment, result should contain [30, 20, 10, 60, 50, 40].
Which of the following can replace /* missing code */ to produce this result?

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(A)

```
for (int j = 0; j < mat.length; j++)
{
  for (int k = 0; k <= mat[0].length; k++)
  {
     result.add(mat[j][k]);
  }
}
```

(B) ✓

```
for (int j = 0; j < mat.length; j++)
{
  for (int k = mat[0].length - 1; k >= 0; k--)
  {
    result.add(mat[j][k]);
  }
}
```

(C)

```
for (int j = mat.length - 1; j >= 0; j--)
{
  for (int k = 0; k <= mat[0].length; k++)
  {
    result.add(mat[j][k]);
  }
}
```

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(D)

```
for (int j = mat.length - 1; j >= 0; j--)
{
  for (int k = mat[0].length - 1; k >= 0; k--)
  {
    result.add(mat[j][k]);
  }
}
```

Answer B

Correct. The outer loop header traverses the rows from row 0 to row 1. The inner loop header traverses the columns from column 2 to column 0. The array elements are added to the ArrayList result as they accessed from right-to-left and from top-to-bottom.

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119. Consider the following code segment.

```
ArrayList<Integer> myList = new ArrayList<Integer>();
myList.add(75);
myList.add(20);
myList.add(99);

ArrayList<Integer> yourList = new ArrayList<Integer>();
/* missing code */
```

After the code segment is executed, yourList should contain [20, 99, 75].
Which of the following can replace /* missing code */ to produce this result?

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(A) ✓

```
int initialSize = myList.size();
for (int i = 0; i < initialSize; i++)
{
  myList.add(2, myList.remove(0));
  yourList.add(myList.get(0));
}
```

(B)

```
int initialSize = myList.size();
for (int i = 1; i <= initialSize; i++)
{
  yourList.add(myList.get(i));
}
```

(C)

```
for (Integer temp : myList)
{
  yourList.add(0, temp);
}
```

(D)

```
for (Integer temp : myList)
{
  myList.add(2, myList.remove(0));
  yourList.add(temp);
}
```

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Answer A

Correct. In each iteration, the first statement removes the first element of myList and adds it to the end. The second statement gets the first element of myList and adds it to the end of yourList. When the loop has completed execution, yourList has the values indicated.

120. In the following code segment, intList is properly declared and initialized as an ArrayList<Integer> variable.

```
int someVal = 0;

for (int i = 0; i < intList.size(); i += 2)
{
  someVal++;
}

System.out.println(someVal);
```

Which of the following best describes the value printed as a result of executing the code segment?

(A) The sum of all the elements in intList

(B) The number of elements in intList

(C) The number of even values in intList

(D) The number of elements in even-numbered indexes in intList ✓

Answer D

Correct. The loop starts at index 0 and increments by 2 each time through the loop until it reaches the size of the list. Even though it does not access any of the elements, it effectively counts the number of even values that would be valid indexes for the list.

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121. The following method is intended to remove all elements from `words` whose length is greater than `max`.

```
/** Precondition: All elements of words are non-null. */
public static void removeLongWords(ArrayList<String> words,
                                   int max)
{
  /* missing code */
}
```

Which of the following can replace `/* missing code */` so that the method works as intended?

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(A)

```
for (String w : words)
{
  if (w.length() > max)
  {
    words.remove(w);
  }
}
```

(B)

```
int i = 0;
for (String w : words)
{
  if (w.length() > max)
  {
    words.remove(w);
    i--;
  }
  i++;
}
```

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(C)

```
for (int i = 0; i < words.size(); i++)
{
  if (words.get(i).length() > max)
  {
    words.remove(i);
  }
}
```

(D) ✓

```
for (int i = words.size() - 1; i >= 0; i--)
{
  if (words.get(i).length() > max)
  {
    words.remove(i);
  }
}
```

Answer D

Correct. Removing an element while traversing `words` from the last element at index `words.size() - 1` to the first element at index `0` will result in all elements of `words` being visited, and all `String` elements with a length greater than `max` will be removed.

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122. Consider the following method, which is intended to return a list containing the elements of the parameter `myList` with all even elements removed.

```
public static ArrayList<Integer> removeEvens(ArrayList<Integer> myList)
{
 for (int i = 0; i < myList.size(); i++)
 {
   if (myList.get(i) % 2 == 0)
   {
     myList.remove(i);
   }
 }
 return myList;
}
```

Which of the following best explains why the code segment does not work as intended?

(A) The code segment causes an `IndexOutOfBoundsException` for lists with at least one even element because the indexes of all subsequent elements change by one when a list element is removed.

(B) The code segment returns a list with fewer elements than intended because it fails to consider the last element of `myList`.

(C) The code segment removes the wrong elements of `myList` because the condition in the `if` statement to test whether an element is even is incorrect.

(D) The code segment skips some elements of `myList` because the indexes of all subsequent elements change by one when a list element is removed. ✓

Answer D

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123. Consider the following method.

```
public static void mystery(List<Integer> nums)
{
  for (int k = 0; k < nums.size(); k++)
  {
    if (nums.get(k).intValue() == 0)
    {
       nums.remove(k);
    }
  }
}
```

Assume that a List<Integer> values initially contains the following Integer values.

```
[0, 0, 4, 2, 5, 0, 3, 0]
```

What will values contain as a result of executing mystery(values) ?

(A) [0, 0, 4, 2, 5, 0, 3, 0]

(B) [4, 2, 5, 3]

(C) [0, 4, 2, 5, 3] ✓

(D) The code throws an ArrayIndexOutOfBoundsException exception.

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124. In the following code segment, assume that the `ArrayList` object `numbers` has been properly declared and initialized to contain `[0, 2, 4, 5]`.

```
for (int k = numbers.size() - 1; k >= 0; k--)
{
  if (numbers.get(k) > k)
  {
    System.out.print(k + " ");
  }
}
```

What, if anything, is printed as a result of executing the code segment?

(A)

2 4 5

(B) ✓

3 2 1

(C)

5 4 2

(D) Nothing is printed because an `IndexOutOfBoundsException` is thrown.

Answer B

Correct. The code segment iterates over `numbers` from right-to-left and compares the value of each element to its index. If the value of an element is greater than its index, then the index is printed. The element at index `3`, which is `5`, is greater than `3`, so `3` is printed. The element at index `2`, which is `4`, is greater than `2`, so `2` is printed. The element at index `1`, which is `2`, is greater than `1`, so `1` is printed. The element at index `0`, which is `0`, is not greater than `0`, so `0` is not printed.

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125. In the following code segment, assume that the `ArrayList data` has been initialized to contain the `Integer` values `[4, 3, 4, 5, 3, 4]`.

```
int j = 0;
while (j < data.size() - 1)
{
  if (data.get(j) > data.get(j + 1))
  {
    System.out.print(data.get(j + 1) + " ");
  }
  j++;
}
```

What, if anything, is printed as a result of executing the code segment?

(A) ✓

3 3

(B)

4 5

(C)

4 5 4

(D) Nothing is printed because an `IndexOutOfBoundsException` occurs.

Answer A

Correct. The code segment iterates over the elements of `data`. Whenever an element is greater than the next element, the next element is printed. Because `4` is greater than the next element `(3)`, `3` is printed. Because `5` is greater than the next element `(3)`, another `3` is printed.

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126. Consider the following code segment.

```
double x = 0.0;
for (double number : numbers)
{
   x += number;
}
System.out.println(x / numbers.size());
```

Assume that numbers is an ArrayList that has been initialized with the following Double objects.

[10.0, 20.0, 30.0, 40.0, 100.0]

What is printed as a result of executing the code segment?

(A)

20.0

(B)

30.0

(C) ✓

40.0

(D)

200.0

Answer C

Correct. The code segment computes the average of the elements in numbers. It uses an enhanced for loop to compute the sum of the elements and then divides the sum by the size of the ArrayList.

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127. Consider the following method `countNegatives`, which returns the number of elements in `numList` that are less than zero.

```
public static int countNegatives(ArrayList<Integer> numList)
{
  int count = 0;
  for (int j = 0; j < numList.size(); j++)       // Line 4
  {
    if (numList.get(j) < 0)
    {
      count++;
    }
  }
  return count;
}
```

Suppose the Boolean expression `j < numList.size()` in line 4 of the method is replaced with `j <= numList.size() - 1`. Which of the following best explains the effect of this change?

(A) It has no effect on the behavior of the method. ✓

(B) It causes the method to ignore the last element in `numList`.

(C) It causes an `IndexOutOfBoundsException` to be thrown.

(D) It changes the number of times the loop executes, but all elements in `numList` will still be accessed.

Answer A

Correct. Since `j` and `numList.size()` both have integer values, the condition `j < numList.size()` is equivalent to `j <= numList.size() - 1`. The change has no effect on the behavior of the method because in both cases the `for` loop stops iterating when `j` is equal to `numList.size() - 1`.

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128. Consider the following method, which is intended to remove duplicate consecutive elements contained in nums.

```
public static void remDups(ArrayList<Integer> nums)
{
  for (int j = 0; j < nums.size() - 1; j++) // Line 3
  {
    int first = nums.get(j);
    int second = nums.get(j + 1);
    if (first == second)                  // Line 7
    {
      nums.remove(j);                     // Line 9
      j++;                                // Line 10
    }
  }
}
```

For example, if nums contains {1, 2, 2, 3, 4, 3, 5, 5, 6}, then after remDups(nums) is executed, nums should contain {1, 2, 3, 4, 3, 5, 6}.

The method does not always work as intended.

Which of the following changes can be made so that the method always works as intended?

(A) In line 3, changing the Boolean expression to j < nums.size()

(B) In line 7, changing the Boolean expression to first != second

(C) In line 9, changing the expression to nums.remove(j + 1)

(D) In line 10, changing the expression to j-- ✓

Answer D

Correct. After removing the element at index j, the loop control variable needs to be reduced by 1 to account for the removed element. Otherwise, elements in nums will be skipped.

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129. Consider the following method.

```
public static boolean mystery(ArrayList<Integer> numbers)
{
  int prev = Integer.MAX_VALUE;
  for (int num : numbers)
  {
    if (num > prev)
    {
      return false;
    }
    prev = num;
  }
  return true;
}
```

Which of the following best describes the behavior of the method?

(A) It returns false if numbers is sorted from least to greatest and returns true otherwise.

(B) It returns false if numbers is sorted from greatest to least and returns true otherwise.

(C) It returns true if numbers is sorted from least to greatest and returns false otherwise.

(D) It returns true if numbers is sorted from greatest to least and returns false otherwise. ✓

Answer D

Correct. This method compares each ArrayList element to the previous element (except for the first element, which is compared to Integer.MAX_VALUE). If the current element is greater than the previous element, it means the ArrayList is not sorted from greatest to least, and the method immediately returns false. If the code segment traverses the entire ArrayList without finding any elements that are greater than the previous element, it means the ArrayList is sorted from greatest to least and the method returns true.

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130. In the code segment shown, `myList` is an `ArrayList` of integers. The code segment is intended to remove all elements with the value `0` from `myList`.

```
int j = 0;
while (j < myList.size())
{
  if (myList.get(j) == 0)
  {
    myList.remove(j);
  }
  j++;
}
```

The code segment does not always work as intended.

For which of the following lists does the code segment **not** produce the correct result?

(A)

[0, 1, 0, 2]

(B) ✓

[1, 0, 0, 2]

(C)

[1, 2, 3, 0]

(D)

[1, 2, 3, 4]

Answer B

Correct. For this `ArrayList`, the code segment removes the element at index `1`, shifting the second instance of `0` to index `1`. The code segment then increases `j` to `2`, so the second instance of `0` is never removed.

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131. Consider the following method `findValue`, which returns `true` if `key` is found in the `ArrayList words` and returns `false` otherwise.

```
public static boolean findValue(ArrayList<String> words,
                                String key)
{
  for (int j = 0; j < words.size(); j++)      // Line 4
  {
    if (words.get(j).equals(key))
    {
      return true;
    }
  }
  return false;
}
```

Suppose the `for` loop initialization `int j = 0` in line 4 of the method is replaced with `int j = 1`. Which of the following best explains the effect of this change?

(A) It will have no effect on the behavior of the method.

(B) It will cause the method to return a different result when `key` is not in `words`.

(C) It will cause the method to return a different result when `key` is found only at the first index in `words`. ✓

(D) It will cause the method to return a different result when `key` is found only at the last index in the list.

Answer C

Correct. Changing the `for` loop to iterate starting at index `1` instead of at index `0` causes the first element in the list to be ignored. Therefore, if `key` is found only at the first index in the list, the modified method will not find the value of `key` in the list and will return a different result.

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132. In the following code segment, words is an ArrayList of two or more String objects.

```
int prev = words.get(0).length();
int count = 0;

for (int j = 1; j < words.size(); j++)
{
  int curr = words.get(j).length();
  if (curr == prev)
  {
    count += 1;
  }
  prev = curr;
}
```

Which of the following best describes the value assigned to count as a result of executing the code segment?

(A) The number of elements in words whose length is the same length as the first element

(B) The number of elements in words whose length is the same length as the element immediately preceding it ✓

(C) The number of elements in words whose value is equal to the first element

(D) The number of elements in words whose value is equal to the element immediately preceding it

Answer B

Correct. The code segment starts with prev equal to the length of the first element and compares that to curr, the length of the next element. It adds one to count each time prev and curr are the same length. It then updates prev to be equal to curr and updates curr to be equal to the next word, continuing until the end of the ArrayList.

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Consider the following correct implementation of the insertion sort algorithm. The `insertionSort` method correctly sorts the elements of `ArrayList data` into increasing order.

```
public static void insertionSort(ArrayList<Integer> data)
{
 for (int j = 1; j < data.size(); j++)
 {
   int v = data.get(j);
   int k = j;
   while (k > 0 && v < data.get(k - 1))
   {
     data.set(k, data.get(k - 1)); /* Statement 1 */
     k--;
   }
   data.set(k, v); /* Statement 2 */
   /* End of outer loop */
 }
}
```

133. Assume that `insertionSort` has been called with an `ArrayList` parameter that has been initialized with the following `Integer` objects.

`[5, 2, 4, 1, 3, 6]`

What will the contents of `data` be after three passes of the outside loop (i.e., when `j == 3` at the point indicated by `/* End of outer loop */`) ?

(A) `[1, 2, 3, 4, 5, 6]`

(B) `[1, 2, 3, 5, 4, 6]`

(C) `[1, 2, 4, 5, 3, 6]` ✓

(D) `[2, 4, 5, 1, 3, 6]`

Answer C

Correct. The insertion sort algorithm maintains a sorted portion of the `ArrayList` and an unsorted portion. In the first pass, the sorted portion contains the element at position 0 and the unsorted portion contains the elements at positions 1 through 5, inclusive. The element at position 1 is moved to its correct place in the sorted portion of the `ArrayList`, and the contents of `data` are `[2, 5, 4, 1, 3, 6]`. In the second pass, the sorted portion of the `ArrayList` contains the elements at positions 0 and 1, the element at position 2 is moved to its correct place in the sorted portion, and the contents of `data` are `[2, 4, 5, 1, 3, 6]`. In the third pass, the sorted portion of the `ArrayList` contains the elements at positions 0 through 2, inclusive, the element at position 3 is moved to its correct place in the sorted portion, and the contents of `data` at the indicated point are `[1, 2, 4, 5, 3, 6]`.

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134. Assume that `insertionSort` is called with an `ArrayList` parameter that has been initialized with the following `Integer` objects.

```
[1, 2, 3, 4, 5, 6]
```

How many times will the statements indicated by `/* Statement 1 */` and `/* Statement 2 */` execute?

(A)

Statement 1	*Statement 2*
0	0

(B) ✓

Statement 1	*Statement 2*
0	5

(C)

Statement 1	*Statement 2*
0	6

(D)

Statement 1	*Statement 2*
5	5

Answer B

Correct. Statement 1, in the `while` loop, is executed each time the first element in the unsorted portion of the `ArrayList` needs to be shifted in order to end up in its correct position in the sorted portion of the `ArrayList`. Since the `ArrayList` is initially in increasing order, Statement 1 is never executed. Statement 2 executes once for each iteration of the `for` loop. The body of the `for` loop is executed 5 times for the given `ArrayList` of 6 elements.

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135. Consider the following class definition.

```
public class Book
{
  private int pages;

  public int getPages()
  {
    return pages;
  }

  /* There may be instance variables, constructors,
     and methods that are not shown. */
}
```

The following method appears in a class other than Book. The method is intended to return the greatest number of pages found in any Book object in the ArrayList parameter bookList.

```
public static int findMaxPages(ArrayList<Book> bookList)
{
  int maxPages = Integer.MIN_VALUE;

  for (Book b : bookList)
  {
    /* missing code */
  }

  return maxPages;
}
```

Which of the following can replace /* missing code */ so the method works as intended?

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(A)

```
if (b.getPages() < maxPages)
{
  maxPages = b.getPages();
}
```

(B) ✓

```
if (b.getPages() > maxPages)
{
  maxPages = b.getPages();
}
```

(C)

```
if (booklist.get(b).getPages() < maxPages)
{
  maxPages = bookList.get(b).getPages();
}
```

(D)

```
if (bookList.get(b).getPages() > maxPages)
{
  maxPages = bookList.get(b).getPages();
}
```

Answer B

Correct. The enhanced for loop traverses bookList and the loop control variable b is assigned Book objects. The if statement compares the value returned by the call b.getPages() to the current value of maxPages. If the returned value is greater than maxPages, maxPages is updated with the new maximum value.

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136. The `removeElement` method is intended to remove all instances of `target` from the `ArrayList` object `data` passed as a parameter. The method does not work as intended for all inputs.

```
public void removeElement(ArrayList<Integer> data, int target)
{
  for (int j = 0; j < data.size(); j++)
  {
    if (data.get(j).equals(target))
    {
      data.remove(j);
    }
  }
}
```

Assume that the `ArrayList` object `scores` and the `int` variable `low_score` have been properly declared and initialized. In which of the following cases will the method call `removeElement(scores, low_score)` fail to produce the intended result?

(A) When `scores` is `[0, 2, 0, 2, 0, 6]` and `low_score` is `0`

(B) When `scores` is `[2, 4, 0, 5, 7, 0]` and `low_score` is `0`

(C) When `scores` is `[3, 4, 5, 7, 7, 2]` and `low_score` is `1`

(D) When `scores` is `[8, 8, 4, 3, 3, 6]` and `low_score` is `3` ✓

Answer D

Correct. The method does not work as intended when there are consecutive occurrences of `low_score` in `scores`. When the first instance of `3` at index `3` is removed, the elements that follow it are shifted left, resulting in `[8, 8, 4, 3, 6]`. The next element, which is also equal to `3`, moves to index `3`. But `j` is incremented to `4` after the removal, so the second instance of `3` is not removed from the list.

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137. In the code segment below, `myList` is an `ArrayList` of integers. The code segment is intended to remove all elements with the value `0` from `myList`.

```
int j = 0;
while (j < myList.size())
{
 if (myList.get(j) == 0)
 {
   myList.remove(j);
 }
 j++;
}
```

The code segment does not always work as intended. For which of the following lists does the code segment NOT produce the correct result?

(A) `{0, 1, 0, 2}`

(B) `{1, 0, 0, 2}` ✓

(C) `{1, 2, 3, 0}`

(D) `{1, 2, 3, 4}`

Answer B

Correct. For this `ArrayList`, the code segment removes the element at index `1`, shifting the second instance of `0` to index `1`. The code segment then increases `j` to `2`, so the second instance of `0` is never removed.

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138. In the following code segment, assume that the `ArrayList numList` has been properly declared and initialized to contain the `Integer` values `[1, 2, 2, 3]`. The code segment is intended to insert the `Integer` value `val` in `numList` so that `numList` will remain in ascending order. The code segment does not work as intended in all cases.

```
int index = 0;
while (val > numList.get(index))
{
 index++;
}
numList.add(index, val);
```

For which of the following values of `val` will the code segment not work as intended?

(A) `0`

(B) `2`

(C) `3`

(D) `4` ✓

Answer D

Correct. Since `val` is greater than every element in `numList`, `index` will continue to increase until it is `4`. Once `index` is `4`, an `IndexOutOfBoundsException` will occur because there is no element at index `4`.

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139. The following question refer to the following information.

Consider the following data field and method. The method `removeDups` is intended to remove all adjacent duplicate numbers from `myData`, but does not work as intended.

```
private ArrayList myData;

public void removeDups ()

{

  int k = 1;

  while (k < myData.size())

  {

    if (myData.get(k).equals(myData.get(k - 1)))

    {

      myData.remove(k);

    }

    k++;

  }

}
```

For example, if `myData` has the values `3 3 4 4 4 8 7 7 7`, after calling `removeDups`, `myData` should have the values `3 4 8 7`.

Which of the following best describes how to fix the error so that `removeDups` works as intended?

(A) The `while` condition should be `(k < myData.size() - 1)`.

(B) The `if` test should be `(myData.get(k).equals(myData.get(k + 1)))`.

(C) The body of the `if` statement should be: `myData.remove(k - 1);`

(D) There should be an `else` before the statement `k++;` ✓

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140. The following question refer to the following information.

Consider the following data field and method. The method `removeDups` is intended to remove all adjacent duplicate numbers from `myData`, but does not work as intended.

```
private ArrayList myData;

public void removeDups ()

{
  int k = 1;
  while (k < myData.size())
  {
    if (myData.get(k).equals(myData.get(k - 1)))
    {
      myData.remove(k);
    }
    k++;
  }
}
```

For example, if `myData` has the values `3 3 4 4 4 8 7 7 7`, after calling `removeDups`, `myData` should have the values `3 4 8 7`.

Assume that `myData` has the following values.

```
2 7 5 5 5 5 6 6 3 3 3
```

Which of the following represents `myData` after the incorrect `removeDups` is executed?

(A) `2 7 5 6 3`

(B) `2 7 5 6 3 3`

(C) `2 7 5 5 6 3 3` ✓

(D) `2 7 5 5 5 6 3 3`

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141. Consider the method `linearSearch`, which takes an `ArrayList` of `Integer` elements and a target `int` value as parameters and returns the index of the first appearance of the target value in the list or `-1` if the target value does not appear in the list.

```
public static int linearSearch(ArrayList<Integer> elements, int target)
{
  for (int j = 0; j < elements.size(); j++)     // Line 3
  {
    if (elements.get(j) == target)
    {
      return j;
    }
  }
  return -1;
}
```

Which of the following describes how replacing line 3 with `for (int j = (elements.size() - 1); j >= 0; j--)` will affect the behavior of `linearSearch`?

(A) The modification has no effect: the modified method will continue to return the index of the first appearance of the target value in the list, or `-1` if the target value does not appear in the list.

(B) The modified method will return the index of the last appearance of the target value in the list, or `-1` if the target value does not appear in the list. ✓

(C) The modified method will throw an `IndexOutOfBoundsException`.

(D) The modified method will return `-1` regardless of the inputs.

Answer B

Correct. Modifying the `linearSearch` method in that way causes it to search the list from right to left instead of from left to right. As a result, when the target value does appear in the list, the index of the last appearance of the target value is returned instead of the index of the first appearance.

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142. A two-dimensional array `arr` is to be created with the following contents.

```
boolean[][] arr = {{false, true, false},
                   {false, false, true}};
```

Which of the following code segments can be used to correctly create and initialize `arr` ?

(A)
```
boolean arr[][] = new boolean[2][3];
arr[0][1] = true;
arr[1][2] = true;
```
✓

(B)
```
boolean arr[][] = new boolean[2][3];
arr[1][2] = true;
arr[2][3] = true;
```

(C)
```
boolean arr[][] = new boolean[3][2];
arr[0][1] = true;
arr[1][2] = true;
```

(D)
```
boolean arr[][] = new boolean[3][2];
arr[1][0] = true;
arr[2][1] = true;
```

Answer A

Correct. This code segment creates a new array with two rows and three columns. Upon creation, each array element is assigned the default value `false`. The element in row `0` column `1` is assigned `true`, and the element in row `1` column `2` is assigned `true`.

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143. Consider the following method.

```
public static boolean checkIndices(double[][] data,
                                   int row, int col)
{
  int numRows = data.length;

  if (row < numRows)
  {
    int numCols = data[0].length;
    return col < numCols;
  }
  else
  {
    return false;
  }
}
```

Consider the following statement, which appears in a method in the same class as `checkIndices`.

```
double[][] table = new double[5][6];
```

Which of the following method calls will return the value `true`?

(A) ✓

```
checkIndices(table, 4, 5)
```

(B)

```
checkIndices(table, 4, 6)
```

(C)

```
checkIndices(table, 5, 4)
```

(D)

```
checkIndices(table, 5, 6)
```

Answer A

Correct. The value of `numRows` is `5` because the length of a two-dimensional array is the number of rows. Since the value of `row` is `4`, the `if` condition `row < numRows` is `true` and the body of the `if` is executed. The value of `numCols` is `6`

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because `data[0]` is a one-dimensional array of length `6`. Since the value of `col` is `5`, the Boolean expression `col < numCols` is `true` and the call `checkIndices(table, 4, 5)` returns `true`.

144. Consider the following code segment, which is intended to create and initialize the two-dimensional array `num` so that columns with an even index will contain only even integers and columns with an odd index will contain only odd integers.

```
int[][] num = /* missing code */;
```

Which of the following initializer lists could replace `/* missing code */` so that the code segment will work as intended?

(A) ✓

```
{{0, 1, 2}, {4, 5, 6}, {8, 3, 6}}
```

(B)

```
{{1, 2, 3}, {3, 4, 5}, {5, 6, 7}}
```

(C)

```
{{2, 1, 4}, {5, 2, 3}, {2, 7, 1}}
```

(D)

```
{{2, 4, 6}, {1, 3, 5}, {6, 4, 2}}
```

Answer A

Correct. Two-dimensional arrays are arrays of arrays. This initializer list will create a 2D array with 3 rows and 3 columns. The first row (with index 0) will be assigned the values 0, 1, and 2. The second row (with index 1) will be assigned the values 4, 5, and 6. Finally, the third row (with index 2) will be assigned the values 8, 3, and 6. Therefore, the first column (with index 0) will contain the values 0, 4, and 8. The second column (with index 1) will contain the values 1, 5, and 3. Finally, the third column (with index 2) will contain the values 2, 6, and 6. So the first and third columns (with even indices) contain only even integers and the second column (with an odd index) contains only odd integers.

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145. Consider the following code segment, which is intended to declare and initialize the two-dimensional `String` array `things`.

```
/* missing code */ = {{"spices", "garlic", "onion", "pepper"},
                      {"clothing", "hat", "scarf", "gloves"},
                      {"plants", "tree", "bush", "flower"},
                      {"vehicles", "car", "boat", "airplane"}};
```

Which of the following could replace `/* missing code */` so that `things` is properly declared?

(A)

new String[][] things

(B)

new(String[][]) things

(C)

String[] String[] things

(D) ✓

String[][] things

Answer D

Correct. A two-dimensional array is stored as arrays of arrays. The data type, `String`, followed by two pairs of square brackets, `[][]`, followed by the variable name, `things`, is the correct way to declare this two-dimensional array.

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146. A two-dimensional array `myArray` is to be created with the following contents.

```
{{0, 0, 3},
{0, 0, 0},
{7, 0, 0}}
```

Which of the following code segments can be used to correctly create and initialize `myArray` ?

I.
```
int[][] myArray = new int[3][3];
myArray[0][2] = 3;
myArray[2][0] = 7;
```

II.
```
int[][] myArray = new int[3][3];
myArray[0][2] = 7;
myArray[2][0] = 3;
```

III.
```
int[][] myArray = {{0, 0, 3}, {0, 0, 0}, {7, 0, 0}};
```

(A) I only

(B) III only

(C) I and III ✓

(D) II and III

Answer C

Correct. Code segment I creates a new array with three rows and three columns. Upon creation, each array element is assigned the default value `0`. The element in row `0` column `2` is assigned `3`, and the element in row `2` column `0` is assigned `7`. Code segment II also creates a new array with three rows and three columns, but it assigns the value `7` to the location where `3` should be assigned and vice-versa. Code segment III uses an initializer list to assign the first row `{0, 0, 3}`, assign the second row `{0, 0, 0}`, and assign the third row `{7, 0, 0}`.

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147. Consider the following code segment.

```
int[][] table = {{9, 8, 7}, {6, 5, 4}, {3, 2, 1}};
System.out.println(table[2][1]);
```

What value is printed as a result of executing the code segment?

(A) ✓
2

(B)
3

(C)
4

(D)
6

Answer A

Correct. The value 2 is stored in table[2][1]. The first square bracket indicates the row index and the second square bracket indicates the column index. Two-dimensional array indices for both rows and columns begin at 0.

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148. Assume that `arr` is a properly declared and initialized two-dimensional array of `String` objects with 3 rows and 4 columns. Which of the following statements can be used to print the four corner elements of `arr`?

(A)

```
System.out.print(arr[0, 0] + arr[0, 3] + arr[2, 0] + arr[2, 3]);
```

(B)

```
System.out.print(arr[1, 1] + arr[1, 4] + arr[3, 1] + arr[3, 4]);
```

(C)

```
System.out.print(arr[0][0] + arr[0][2] + arr[3][0] + arr[3][2]);
```

(D) ✓

```
System.out.print(arr[0][0] + arr[0][3] + arr[2][0] + arr[2][3]);
```

Answer D

Correct. For a two-dimensional array with `x` rows and `y` columns, the corners are found at row `0` and column `0`, row `0` and column `y - 1`, row `x - 1` and column `0`, and row `x - 1` and column `y - 1`.

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149. Consider the following code segment, where `letters` is a two-dimensional (2D) array that contains possible letters. The code segment is intended to print `"DIG"`.

```
String[][] letters = {{"A", "B", "C"},
                      {"D", "E", "F"},
                      {"G", "H", "I"}};
System.out.println( /* missing code */ );
```

Which of the following could replace `/* missing code */` so that the code segment works as intended?

(A) `letters[2][1] + letters[3][3] + letters[3][1]`

(B) `letters[2][0] + letters[2][2] + letters[1][0]`

(C) `letters[1][0] + letters[2][2] + letters[2][0]` ✓

(D) `letters[0][1] + letters[2][2] + letters[0][2]`

Answer C

Correct. To access elements of the 2D array, the first index is used for the row and the second index is used for the column. The missing code segment needs to access three elements from the array and concatenate them together. The string `"D"` is located in row `1` and column `0`; therefore, the first element is `letters[1][0]`. The string `"I"` is located in row `2` and column `2`; therefore, the second element is `letters[2][2]`. The string `"G"` is located in row `2` and column `0`; therefore the third element is `letters[2][0]`.

150. Consider the following code segment, where `nums` is a two-dimensional (2D) array of integers. The code segment is intended to print `"test1234"`.

```
System.out.print("test" + nums[0][0] + nums[1][0] + nums[1][1] + nums[0][1]);
```

Which of the following code segments properly declares and initializes `nums` so that the code segment works as intended?

(A) `int[][] nums = {{1, 2}, {3, 4}};`

(B) `int[][] nums = {{1, 2}, {4, 3}};`

(C) `int[][] nums = {{1, 3}, {2, 4}};`

(D) `int[][] nums = {{1, 4}, {2, 3}};` ✓

Answer D

Correct. To access elements of the two-dimensional array (2D) array `nums`, the first index is used for the row and the second index is used for the column. The value of the element at row `0` and column `0` of `nums` is `1`, the value of the element at row `1` and column `0` of `nums` is `2`, the value of the element at row `1` and column `1` of `nums` is `3`, and the value of the element at row `0` and column `1` of `nums` is `4`. Therefore, this will print the desired value.

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151. The following method is intended to return the total number of elements in the 2D array arr. The method does not work as intended.

```
public int totalElements(int[][] arr)
{
  return arr.length * arr.length;   // Line 3
}
```

Which of the following changes can be made so that the code segment works as intended?

(A) Changing line 3 to return (arr.length - 1) * (arr.length - 1);

(B) Changing line 3 to return arr.length + arr.length;

(C) Changing line 3 to return (arr.length - 1) * (arr[0].length - 1);

(D) Changing line 3 to arr.length * arr[0].length; ✓

Answer D

Correct. The expression arr.length evaluates to the number of rows in the array, and the expression arr.length[0] evaluates to the number of columns in the array. The method currently uses arr.length * arr.length, which incorrectly multiplies the number of rows by itself. The total number of elements in a 2D array can be calculated by multiplying the number of rows (arr.length) by the number of columns (arr[0].length).

FINAL Unit4 MCQ - Topic order

152. Consider the following code segment.

```
int[][] anArray = new int[10][8];

for (int j = 0; j < 8; j++)
{
 for (int k = 0; k < 10; k++)
 {
   anArray[j][k] = 5;
 }
}
```

The code segment causes an `ArrayIndexOutOfBoundsException` to be thrown. How many elements in `anArray` will be set to `5` before the exception is thrown?

(A) 8 ✓

(B) 9

(C) 64

(D) 80

Answer A

Correct. In the first iteration of the outer loop, `j` has the value `0`. The first 8 iterations of the inner loop access `anArray[0][0]` through `anArray[0][7]`. When an attempt is made to access `anArray[0][8]`, an `ArrayIndexOutOfBoundsException` is thrown since `anArray` has 8 columns, numbered 0 through 7, inclusive.

FINAL Unit4 MCQ - Topic order

153. Assume that `mat` has been declared as a 4×4 array of integers and has been initialized to contain all `1`s. Consider the following code segment.

```
int n = mat.length;
for (int j = 1; j < n; j++)
{
 for (int k = 1; k < n; k++)
 {
   mat[j][k] = mat[j - 1][k] + mat[j][k - 1];
 }
}
```

What is the value of `mat[2][2]` after the code segment has completed execution?

(A) `2`

(B) `3`

(C) `4`

(D) `6` ✓

Answer D

Correct. Every element of `mat` is modified, except those in the first row and the first column, which retain their initial values. Elements are traversed in row-major order. Each element of `mat` is replaced by the sum of the element immediately above it and the element to its left. In row `1`, `mat[1][1]` is assigned the value `mat[0][1] + mat[1][0] = 1 + 1 = 2`, then `mat[1][2]` is assigned the value `1 + 2 = 3`, and finally `mat[1][3]` is assigned the value `1 + 3 = 4`. In row `2`, `mat[2][1]` is assigned `2 + 1 = 3`, and then `mat[2][2]` is assigned `3 + 3 = 6`.

FINAL Unit4 MCQ - Topic order

154. Consider the following code segment.

```
int[][] my2Darr = new int[4][3];

for (int r = 0; r < my2Darr.length; r++)
{
  for (int c = 0; c < my2Darr[0].length; c++)
  {
    if (r == c)
    {
      my2Darr[r][c] = r;
    }
  }
}
```

Which of the following represents the contents of my2Darr after executing this code segment?

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(A)

```
{{0, 0, 0},
 {0, 0, 0},
 {0, 2, 0},
 {0, 0, 3}}
```

(B) ✓

```
{{0, 0, 0},
 {0, 1, 0},
 {0, 0, 2},
 {0, 0, 0}}
```

(C)

```
{{0, 0, 0, 0},
 {0, 0, 2, 0},
 {0, 0, 0, 3}}
```

(D)

```
{{0, 0, 0, 0},
 {0, 1, 0, 0},
 {0, 0, 2, 0}}
```

Answer B

Correct. The first line of code creates a new two-dimensional array with 4 rows and 3 columns. By using the keyword new, all elements are initialized to the default value 0. The nested for loops traverse the entire array. When the row index is equal to the column index (which occurs at my2Darr[0][0], my2Darr[1][1], and my2Darr[2][2]), the value of the array element is assigned the value of the row index.

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155. The following method is intended to return `true` if all elements in row `x` of the 2D array `nums` are positive and `false` otherwise.

```
/**
 *  Precondition: x is a valid row index of nums.
 */
public static boolean allPositiveInRow(int[][]nums, int x)
{
  boolean allPositive = /* initial value */;
  for (int y = 0; y < nums[0].length; y++)
  {
    if (nums[x][y] <= 0)
    {
      allPositive = /* updated value */;
    }
  }
  return allPositive;
}
```

Which of the following replacements for `/* initial value */` and `/* updated value */` can be used so that the method works as intended?

(A) Replace `/* initial value */` with `true` and replace `/* updated value */` with `true`.

(B) Replace `/* initial value */` with `true` and replace `/* updated value */` with `false`. ✓

(C) Replace `/* initial value */` with `false` and replace `/* updated value */` with `true`.

(D) Replace `/* initial value */` with `false` and replace `/* updated value */` with `false`.

Answer B

Correct. The `boolean` variable is initialized to `true` and is only updated to be `false` if a non-positive value is found in the row.

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156. Consider the following code segment.

```
int[][] arr2D = /* missing initialization */;
int sum = 0;

for (int j = 0; j < arr2D.length; j++)
{
  sum += arr2D[j][2];
}

System.out.println(sum);
```

Which of the following replacements for /* missing initialization */ will cause the code segment to print the value 3?

(A)

```
{{0, 0, 1},
 {0, 0, 2},
 {0, 0, 1}}
```

(B)

```
{{0, 0, 0},
 {0, 1, 2},
 {0, 0, 0}}
```

(C)

```
{{0, 0, 0},
 {0, 0, 0},
 {1, 1, 1},
 {0, 0, 0}}
```

(D) ✓

```
{{0, 0, 1, 0},
 {0, 0, 1, 0},
 {0, 0, 1, 0}}
```

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Answer D

Correct. The code segment traverses the column of arr2D with index 2 and calculates the sum of the elements in that column. For these contents of arr2D, the elements in column 2 have a sum of 3.

157. Consider the following data field and method.

```
private int[][] mat;

public int mystery(int r, int c)
{
  if (r != 0 || c != 0)
  {
    return (mat[r][c] + mystery(r - 1, c - 1));
  }
  else
  {
    return mat[r][c];
  }
}
```

Assume that `mat` is the 2-D array shown below.

	0	**1**	**2**	**3**
0	0	1	2	3
1	4	5	6	7
2	8	9	10	11
3	12	13	14	15

What value is returned as a result of the call `mystery(2, 3)`?

(A) 11
(B) 17
(C) 18
(D) No value is returned because `mystery` throws an `ArrayIndexOutOfBoundsException`. ✓

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158. Consider the following method.

```
/**
 *  Precondition: matrix has the same number of
 *                rows and columns.
 */
public static int[] operation(int[][] matrix, int r, int c)
{
  int[] result = new int[matrix.length];

  for (int j = 0; j < matrix.length; j++)
  {
    result[j] = matrix[r][j] * matrix[j][c];
  }
  return result;
}
```

The following code segment appears in another method in the same class.

```
int[][] mat = {{3, 1, 2, 4},
               {2, 0, 3, 1},
               {1, 2, 3, 3},
               {1, 1, 1, 1}};

int[] arr = operation(mat, 1, 2);
```

Which of the following represents the contents of arr after executing the code segment?

(A)

{1, 0, 6, 3}

(B)

{1, 9, 0, 4}

(C) ✓

{4, 0, 9, 1}

(D)

{4, 3, 6, 2}

Answer C

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Correct. In the first iteration of the loop, the value of result[0] is assigned the product of matrix[1][0], which is 2, and matrix[0][2], which is 2. Therefore, result[0] is assigned the value 4. In the second iteration of the loop, result[1] is assigned the product of matrix[1][1], which is 0, and matrix[1][2], which is 3. Therefore, result[1] is assigned the value 0. In the third iteration of the loop, result[2] is assigned the product of matrix[1][2], which is 3, and matrix[2][2], which is 3. Therefore, result[2] is assigned the value 9. In the final iteration of the loop, result[3] is assigned the product of matrix[1][3], which is 1, and matrix[3][2], which is 1. Therefore, result[3] is assigned the value 1.

159. Consider the following code segment.

```
int[][] board = new int[5][4];

/* code to initialize board */

int total = 0;
for (int r = 1; r < board.length; r += 2)
{
  for (int c = 0; c < board[0].length; c++)
  {
    total += board[r][c];
  }
}
```

Which of the following best describes the value of total as a result of executing the code segment?

(A) It contains the sum of elements in even-numbered rows.

(B) It contains the sum of elements in odd-numbered rows. ✓

(C) It contains the sum of elements in even-numbered columns.

(D) It contains the sum of elements in odd-numbered columns.

Answer B

Correct. Only the rows with odd numbered indexes will be traversed because the outer loop control variable is initialized to 1 and incremented by 2. All columns are traversed by the inner loop.

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160. Consider the following code segment.

```
String[][] words = {{"Hello,", "Hi,", "Hey,"},
                    {"it's", "it is", "it really is"},
                    {"nice", "great", "a pleasure"},
                    {"to", "to get to", "to finally"},
                    {"meet", "see", "catch up with"},
                    {"you", "you again", "you all"}};

for (int j = 0; j < words.length; j++)
{
  for (int k = 0; k < words[j].length; k++)
  {
    if (k == 1)
    {
      System.out.print(words[j][k] + " ");
    }
  }
}
```

What, if anything, is printed as a result of executing the code segment?

(A) Nothing is printed due to an `ArrayIndexOutOfBoundsException`.

(B)

Hello, it's nice to meet you

(C)

Hey, it really is a pleasure to finally catch up with you all

(D) ✓

Hi, it is great to get to see you again

Answer D

Correct. The outer loop traverses the rows of the array. The inner loop traverses the elements in each row (the columns). Within each of the five rows, the `if` statement selects and prints the array element in the second column (that is, the element at column index `1`).

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161. Consider the following code segment.

```
int[][] mat = new int[3][4];
for (int row = 0; row < mat.length; row++)
{
   for (int col = 0; col < mat[0].length; col++)
   {
     if (row < col)
     {
       mat[row][col] = 1;
     }
     else if (row == col)
     {
       mat[row][col] = 2;
     }
     else
     {
       mat[row][col] = 3;
     }
   }
}
```

What are the contents of mat after the code segment has been executed?

(A)
```
{{2, 1, 1},
 {3, 2, 1},
 {3, 3, 2},
 {3, 3, 3}}
```

(B)
```
{{2, 3, 3, 3},
 {1, 2, 3, 3},
 {1, 1, 2, 3}}
```

(C) ✓
```
{{2, 1, 1, 1},
 {3, 2, 1, 1},
 {3, 3, 2, 1}}
```

(D)
```
{{1, 1, 1, 1},
 {2, 2, 2, 2},
 {3, 3, 3, 3}}
```

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162. Consider the following method, which is intended to return the number of columns in the two-dimensional array `arr` for which the sum of the elements in the column is greater than the parameter `val`.

```
public int countCols(int[][] arr, int val)
{
 int count = 0;

 for (int col = 0; col < arr[0].length; col++) // Line 5
 {
   int sum = 0;
   for (int[] row : col) // Line 8
   {
     sum += row[col]; // Line 10
   }
   if (sum > val)
   {
     count++;
   }
 }
 return count;
}
```

The `countCols` method does not work as intended. Which of the following changes should be made so the method works as intended?

(A) Line 5 should be changed to `for (int col = 0; col < arr.length; col++)`

(B) Line 8 should be changed to `for (int row : col)`

(C) Line 8 should be changed to `for (int[] row : arr)` ✓

(D) Line 10 should be changed to `sum += arr[row][col];`

Answer C

Correct. Two-dimensional arrays are stored as arrays of one-dimensional arrays. Line 8 is intended to assign to `row`, a one-dimensional array of `int` values, a single row of the two-dimensional array `arr`. The original version of line 8 attempts to assign a row of `col`, but `col` is not a two-dimensional array.

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163. Consider the following method, sumRows, which is intended to traverse all the rows in the two-dimensional (2D) integer array num and print the sum of all the elements in each row.

```
public static void sumRows(int[][] num)
{
  for (int[] r : num)
  {
    int sum = 0;
    for (int j = 0; j < num.length; j++)
    {
      sum += r[j];
    }
    System.out.print(sum + " ");
  }
}
```

For example, if num contains {{3, 5}, {6, 8}}, then sumRows(num) should print the following.

8 14

The method does not always work as intended.

For which of the following two-dimensional array input values does sumRows **not** work as intended?

(A)

{{10, -18}, {48, 17}}

(B)

{{-5, 2, 0}, {4, 11, 0}}

(C) ✓

{{4, 1, 7}, {-10, -11, -12}}

(D)

{{1, 2, 3}, {4, 5, 6}, {7, 8, 9}}

Answer C

Correct. For this option, the inner for loop will iterate two times since the two-dimensional array has two rows. However, there are three elements in each row. So the sum of only the first two elements in each row is printed. The Boolean expression in the inner for loop should be j < r.length instead of j < num.length.

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164. Consider the following code segment.

```
int[][] arr = {{6, 2, 5, 7},
               {7, 6, 1, 2}};
for (int j = 0; j < arr.length; j++)
{
  for (int k = 0; k < arr[0].length; k++)
  {
    if (arr[j][k] > j + k)
    {
      System.out.println("!");
    }
  }
}
```

How many times will `"!"` be printed when the code segment is executed?

(A) 2 times

(B) 4 times

(C) 6 times ✓

(D) 8 times

Answer C

Correct. The character `"!"` is printed for each element of `arr` for which the value of the element is greater than the sum of the row and column indexes. In the first row, the sums of the row and column indexes are 0, 1, 2, and 3, and each array element is greater than the corresponding sum. In the second row, the sums of the row and column indexes are 1, 2, 3, and 4, and only the first two array elements are greater than the corresponding sums. The character `"!"` is printed 6 times.

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165. Consider the following two-dimensional array definition.

```
int[][] data = new int[5][10];
```

Consider the following code segment, where all elements in `data` have been initialized.

```
for (int j = 0; j < data.length; j++)
{
  for (int k = 0; k < data[0].length; k++)
  {
    if (j == k)
    {
      System.out.println(data[j][k]);
    }
  }
}
```

How many times is the `println` method called when the code segment is executed?

(A) 4

(B) 5 ✓

(C) 9

(D) 10

Answer B

Correct. Since the value of `data.length` is `5` (because the length of a two-dimensional array is the number of rows), the outer `for` loop executes for values of `j` from `j = 0` to `j = 4`. Since the value of `data[0].length` is `10` (because `data[0]` is an integer array of length 10), the inner `for` loop executes for values of `k` from `k = 0` to `k = 9`. The Boolean condition `j == k` in the `if` statement is true only five times, exactly when `j = k = 0`, `j = k = 1`, `j = k = 2`, `j = k = 3`, or `j = k = 4`.

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166. Consider the following code segment.

```
int[][] array2D = {{1, 2, 3, 4},
                   {5, 6, 7, 8},
                   {9, 10, 11, 12},
                   {13, 14, 15, 16}};
for (int[] i : array2D)
{
  for (int x : i)
  {
    System.out.print(x + " ");
  }
  System.out.println(" ");
}
```

How many times will the statement `System.out.print(x + " ")` be executed?

(A) 4 times

(B) 6 times

(C) 12 times

(D) 16 times ✓

Answer D

Correct. The outer loop will execute 4 times (because a two-dimensional array is equivalent to a one-dimensional array of the 4 rows). The inner loop will execute 4 times (because each row has 4 elements). The statement `System.out.print(x + " ")` will execute 16 times (because 4 times 4 equals 16).

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167. Consider the following code segment.

```
int[][] points = {{11, 12, 13, 14, 15},
                  {21, 22, 23, 24, 25},
                  {31, 32, 33, 34, 35},
                  {41, 42, 43, 44, 45}};
for (int row = 0; row < points.length; row++)
{
 for (int col = points[0].length - 1; col >= row; col--)
 {
   System.out.print(points[row][col] + " ");
 }
 System.out.println();
}
```

What is printed when this code segment is executed?

(A)
```
15 14 13 12
25 24 23
35 34
45
```

(B)
```
11 12 13 14 15
21 22 23 24
31 32 33
41 42
```

(C) ✓
```
15 14 13 12 11
25 24 23 22
35 34 33
45 44
```

(D)
```
15 14 13 12 11
25 24 23 22 21
35 34 33 32 31
45 44 43 42 41
```

Answer C

FINAL Unit4 MCQ - Topic order

168. Consider the following code segment.

```
int[][] mat = {{1, 2, 3, 4},
               {5, 6, 7, 8},
               {8, 7, 6, 5},
               {4, 3, 2, 1}};

for (int r = 0; r < mat.length; r++)
{
  /* missing code */
}
```

This code segment is intended to produce the following output.

```
1
5 6
8 7 6
4 3 2 1
```

Which of the following can be used to replace `/* missing code */` so that the code segment works as intended?

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(A)

```
for (int c = 0; c <= r; c++)
{
  System.out.print(mat[c][r] + " ");
}
System.out.println();
```

(B) ✓

```
for (int c = 0; c <= r; c++)
{
  System.out.print(mat[r][c] + " ");
}
System.out.println();
```

(C)

```
for (int c = 1; c < r; c++)
{
  System.out.print(mat[r][c] + " ");
}
System.out.println();
```

(D)

```
for (int c = 0; c <= r; c++)
{
  System.out.print(mat[r][c] + " ");
  System.out.println();
}
```

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Answer B

Correct. The outer loop will traverse from 0 to `mat.length`, which is the number of rows. For each value of the outer loop, the inner loop will traverse from 0 to `r`, the current row setting. When `r = 0`, `c` will go from 0 to 0, printing one element (`mat[0][0]`) from the first row. When `r` is 1, `c` will go from 0 to 1, printing 2 values from the second row (`mat[1][0]` and `mat[1][1]`). The traversal continues with each row printing an additional element until `r` is 3 and `c` goes from 0 to 3, printing all four values in that row. This would produce the intended output.

169. Consider the following code segment.

```
int[][] arr = {{1, 2, 3, 4},
               {5, 6, 7, 8},
               {9, 10, 11, 12}};
int sum = 0;
for (int[] x : arr)
{
 for (int y = 0; y < x.length - 1; y++)
 {
   sum += x[y];
 }
}
```

What is the value of `sum` as a result of executing the code segment?

(A) `36`

(B) `54` ✓

(C) `63`

(D) `78`

Answer B

FINAL Unit4 MCQ - Topic order

170. Consider the following code segment.

```
int[][] arr = {{1, 2, 3},
               {4, 5, 6},
               {7, 8, 9},
               {3, 2, 1}};

for (int j = 0; j < arr.length; j++)
{
  for (int k = j; k < arr[0].length; k++)
  {
    System.out.print(arr[j][k] + " ");
  }
  System.out.println();
}
```

What is printed as a result of executing the code segment?

(A)

```
1 2 3

4 5

7
```

(B) ✓

```
1 2 3

5 6

9
```

(C)

```
1 4 7

5 8

9
```

(D)

```
1 2 3

5 6

9

1
```

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Answer B

Correct. When `j` has the value `0`, `k` takes on values from `0` through `2`, inclusive, and `1 2 3` is printed. When `j` has the value `1`, `k` takes on the values `1` and `2`, and `5 6` is printed. When `j` has the value `2`, `k` takes on the value `2`, and `9` is printed. When `j` has the value `3`, the inner `for` loop body is not executed because the initial value of `k` is `3`, which is not less than the number of elements in the first row of the array.

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171. Consider the following code segment, where `num` is an integer variable.

```
int[][] arr = {{11, 13, 14, 15},
               {12, 18, 17, 26},
               {13, 21, 26, 29},
               {14, 17, 22, 28}};

for (int j = 0; j < arr.length; j++)
{
  for (int k = 0; k < arr[0].length; k++)
  {
    if (arr[j][k] == num)
    {
      System.out.print(j + k + arr[j][k] + " ");
    }
  }
}
```

What is printed when `num` has the value `14`?

(A)

14 14

(B) ✓

16 17

(C)

17 16

(D)

18 19

Answer B

Correct. The nested loops traverse the two-dimensional array in row-major order. The first occurrence of `14` found in the array is at row `0` and column `2`, so `16` is printed. The second occurrence is found at row `3` and column `0`, so `17` is printed.

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172. Consider the following code segment.

```
int[][] arr = {{10, -7, 19, 14},
               {-3, -2, -6, 11},
               {-9, 12, 10, -1}};
int ans = 1;
for (int[] row : arr)
{
  for (int value : row)
  {
    if (value > 0)
    {
      ans = ans * value;
    }
  }
}
System.out.println(ans);
```

Which of the following best describes the behavior of the code segment?

(A) It prints the product of all the elements in arr.

(B) It prints the product of all the negative elements in arr.

(C) It prints the product of all the positive elements in arr. ✓

(D) It prints nothing because of a run-time error.

Answer C

Correct. The code segment initializes ans to 1 and performs a complete traversal of arr. The code segment tests whether each element is positive by comparing it to zero. If the value of the element is positive, the code segment multiplies ans by that value.

FINAL Unit4 MCQ - Topic order

173. Consider the following method. When an element of the two-dimensional array values is accessed, the first index is used to specify the row and the second index is used to specify the column.

```
public static int enigma(int[][] values, int x, int target)
{
  int result = 0;
  for (int num : values[x])
  {
    if (num == target)
    {
      result++;
    }
  }
  return result;
}
```

Which of the following best describes the value returned by the method?

(A) The index of a column in values that contains at least one element equal to target

(B) The index of a row in values that contains at least one element equal to target

(C) The number of times that an element equal to target appears in column x of values

(D) The number of times that an element equal to target appears in row x of values ✓

Answer D

Correct. In the loop header, values[x] is row x in the two-dimensional array values. The loop traverses that row and increments the counter result each time an element is equal to target.

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174. In the following code segment, `num` is a properly declared and initialized integer variable. The code segment is intended to perform a linear search on the two-dimensional array `arr`. Each time an array element equal to `num` is found, its value should be printed. If no element equal to `num` is found, then nothing is printed. The code segment does not always work as intended.

```
int[][] arr = {{7, 3, 6, 4},
               {9, 2, 0, 5},
               {1, 4, 3, 8}};

for (int j = 0; j < arr.length - 1; j++)
{
  for (int k = 0; k < arr[0].length; k++)
  {
    if (arr[j][k] == num)
    {
      System.out.println(arr[j][k]);
    }
  }
}
```

Which of the following values of `num` can be used to show that the code segment does not always work as intended?

(A)

5

(B)

6

(C)

7

(D) ✓

8

Answer D

Correct. In the outer `for` loop header, the condition `j < arr.length - 1` causes the traversal to skip the last row of `arr`. The condition should be replaced so that the code segment traverses all rows in `arr`, for example, using `j < arr.length`. The code segment should print `8`, but because the last row is not traversed, nothing is printed. In general, any value in the last row of `arr` could be used to show that the code segment does not work as intended.

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175. In the following code segment, `board` is a properly declared and initialized 2D array of `int` values.

```
int sum = 0;
for (int r = 0; r < board.length; r++)        // Line 2
{
  for (int c = 0; c < board[0].length; c++)   // Line 4
  {
    if (c % 2 != 0)                           // Line 6
    {
      sum += board[r][c];                     // Line 8
    }
  }
}
```

The code segment is intended to compute the sum of all elements in all rows that have odd-numbered row indices. However, the code segment is not working as intended. Which of the following changes can be made so that the code segment works as intended?

(A) Changing line 2 to `for (int r = 0; r < board.length; r += 2)`

(B) Changing line 4 to `for (int c = 1; c < board[0].length; c += 2)`

(C) Changing line 6 to `if (r % 2 != 0)` ✓

(D) Changing line 8 to `sum += board[c][r];`

Answer C

Correct. The value of the outer loop control variable needs to be used to determine if the index of a row is odd so that only the elements in the appropriate rows are accessed.

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176. Assume `mat` is defined as follows.

```
int dim = 4;
int[][] mat = new int[dim][dim];
```

Consider the following code segment.

```
int sum = 0;
for (int row = 0; row < dim; row++)
{
  sum = sum + mat[row][dim - 1];
}
```

Assume that `mat` contains the following values before the code segment is executed. Note that `mat[0][3]` is `2`.

	0	**1**	**2**	**3**
0	1	1	2	2
1	1	2	2	4
2	1	3	2	6
3	1	4	2	8

What value will `sum` contain after the code segment is executed?

(A) 6
(B) 8
(C) 15
(D) 20 ✓

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177. Consider the following code segment.

```
int[][] data = {{5, 8, 6},
                {8, 2, 5},
                {9, 8, 8},
                {6, 8, 3}};

int result = data[0][0];
for (int col = 0; col < data[0].length; col++)
{
  for (int row = 0; row < data.length; row++)
  {
    if (result > data[row][col])
    {
      result = data[row][col];
    }
  }
}
```

What is the value of `result` after executing the code segment?

(A) ✓
2

(B)
3

(C)
5

(D)
9

Answer A

Correct. After initializing `result` to `5`, the code segment traverses `data` in column-major order and sets the value of `result` to the minimum value in the array, which is `2`.

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178. In the following method, the parameter `temps` is a two-dimensional array of temperatures. The method is intended to return the number of temperatures in a rectangular portion of the array that are greater than 100. The portion is defined by rows `r1` to `r2`, inclusive, and columns `c1` to `c2`, inclusive.

```
/**
 *  Preconditions: 0 <= r1 <= r2 < temps.length
 *                 0 <= c1 <= c2 < temps[0].length
 */
public static int countHighTempsInRegion(int[][] temps,
                        int r1, int r2, int c1, int c2)
{
   int count = 0;
   /* missing code */
   return count;
}
```

Which of the following can replace `/* missing code */` so that the method works as intended?

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(A)

```
for (int r = 0; r < temps.length; r++)
{
   for (int c = 0; c < temps[0].length; c++)
   {
      if (temps[r][c] > 100)
      {
         count++;
      }
   }
}
```

(B)

```
for (int r = r1; r < r2; r++)
{
   for (int c = c1; c < c2; c++)
   {
      if (temps[r][c] > 100)
      {
         count++;
      }
   }
}
```

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(C)

```
for (int r = r1; r <= r2; r++)
{
   for (int c = c1; c <= c2; c++)
   {
      if (temps[r][c] > 100)
      {
         count++;
      }
   }
}
```

(D)

```
for (int r = r1; r <= r2; r++)
{
   for (int c = c1; c <= c2; c++)
   {
      if (temps[c][r] > 100)
      {
         count++;
      }
   }
}
```

Answer C

Correct. The outer loop traverses all rows from `r1` to `r2`, inclusive. The inner loop traverses all columns from `c1` to `c2`, inclusive. The `if` statement compares the element to `100` and increments `count` as appropriate.

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179. Consider the following code segment.

```
int[][] values = {{1, 2, 3},
                  {4, 5, 6}};
int x = 0;

for (int j = 0; j < values.length; j++)
{
  for (int k = 0; k < values[0].length; k++)
  {
    if (k == 0)
    {
      values[j][k] *= 2;
    }
    x += values[j][k];
  }
}
```

What is the value of x after executing the code segment?

(A)

7

(B)

17

(C)

21

(D) ✓

26

Answer D

Correct. The nested loops traverse the two-dimensional array values. The first element of each row is doubled and then the sum of all elements is computed as 2 + 2 + 3 + 8 + 5 + 6, which evaluates to 26.

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180. Consider the following method. When an element of the two-dimensional array `values` is accessed, the first index is used to specify the row and the second index is used to specify the column.

```
public static int calculate(int[][] values, int x)
{
  int result = 0;
  for (int y = 0; y < values[0].length; y++)
  {
    if (values[x][y] > 0)
    {
      result += values[x][y];
    }
  }
  return result;
}
```

Which of the following best describes the value returned by the calculate method?

(A) A count of the positive elements in column x of values

(B) A count of the positive elements in row x of values

(C) The sum of the positive elements in column x of values

(D) The sum of the positive elements in row x of values ✓

Answer D

Correct. In the loop header, the variable y is used as a column index. The loop traverses row x of the two-dimensional array values from column 0 to column values[0].length - 1 (the last column in the array). Inside the loop, the value of each positive element is added to the variable result.

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181. Consider the following method, `countAs`, which is intended to traverse all the elements in the two-dimensional (2D) `String` array `things` and return the total number of elements that contain at least one `"a"`. The method does not always work as intended.

```
public static int countAs(String[][] things)
{
  int count = 0;
  for (int r = 0; r < things.length; r++)
  {
    for (int c = 0; c < things[r].length - 1; c++)
    {
      if (things[r][c].indexOf("a") >= 0)
      {
        count++;
      }
    }
  }
  return count;
}
```

For example, consider the following initialization.

```
String[][] order = {{"salad", "soup"},
                    {"water", "coffee"}};
```

The method call `countAs(order)` should return `2`.

For which of the following two-dimensional arrays arguments does `countAs` **not** work as intended?

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(A)

{{"lemon"}, {"lime"}}

(B)

{{"rabbit", "bird"},

{"cat", "dog"},

{"gecko", "turtle"}}

(C) ✓

{{"scarf", "gloves", "hat"},

{"shoes", "shirt", "socks"}}

(D)

{{"math", "english", "physics"},

{"golf", "baseball", "soccer"}}

Answer C

Correct. The `countAs` method contains two `for` loops. The outer `for` loop will iterate through each row in the `things` array. The inner `for` loop will iterate through all but the last element in each column. Therefore, if the last element in any column of the two-dimensional input array contains the string `"a"`, the total count will be incorrect. That is the case for this array, so the method will not work as intended. The Boolean expression in the inner `for` loop should be `c < things[r].length` instead of `c < things[r].length - 1`.

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182. Consider the following code segment.

```
int[][] mat = {{1, 2, 3, 4},
               {1, 2, 3, 4},
               {5, 6, 7, 8},
               {5, 6, 7, 8}};

int numRows = mat.length;
int numCols = mat[0].length;
int [][] changedMat = new int[numRows][numCols];
for (int row = 0; row < numRows; row++)
{
  for (int col = 0; col < numCols; col++)
  {
    changedMat[numRows – row – 1][col] = mat[row][col];
  }
}
```

Which of the following represents the contents of changedMat after executing this code segment?

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(A) ✓

{{5, 6, 7, 8},

{5, 6, 7, 8},

{1, 2, 3, 4},

{1, 2, 3, 4}}

(B)

{{1, 2, 3, 4},

{5, 6, 7, 8},

{1, 2, 3, 4},

{5, 6, 7, 8}}

(C)

{{4, 3, 2, 1},

{4, 3, 2, 1},

{8, 7, 6, 5},

{8, 7, 6, 5}}

(D)

{{1, 2, 3, 4},

{1, 2, 3, 4},

{5, 6, 7, 8},

{5, 6, 7, 8}}

Answer A

Correct. The first iteration of the inner loop copies the values of the first row (mat[0][col]) into the last row (changedMat[3][col]). The second iteration copies the values of the second row (mat[1][col]) into the second-to-last row (changedMat[2][col]). The process continues until all rows have been copied.

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183. Consider the following data field and method.

```
private int[][] mat;

public void mystery ()
{
  for (int row = 1; row < mat.length; row++)
  {
    for (int col = 0; col < mat[0].length; col++)
    {
      if (row != col)
        mat[row][col] = mat[row - 1][col];
    }
  }
}
```

Assume that `mat` contains the following values. Note that `mat[0][4]` is `2`.

```
4 1 3 4 2
1 8 7 5 3
7 4 6 9 2
3 8 1 2 4
5 6 7 0 3
```

What values does `mat` contain after a call to `mystery`?

(A)
```
4 1 3 4 2
4 8 3 4 2
4 8 6 4 2
4 8 6 2 2
4 8 6 2 3
```
✓

(B)
```
4 1 3 4 2
4 1 3 4 2
4 1 3 4 2
4 1 3 4 2
4 1 3 4 2
```

(C)
```
4 1 3 4 2
4 1 3 4 2
1 8 7 5 3
7 4 6 9 2
3 8 1 2 4
```

(D)
```
4 4 4 4 4
1 1 1 1 1
7 7 7 7 7
3 3 3 3 3
5 5 5 5 5
```

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184. Consider the following method.

```
public static int findTarget(int[] nums, int target)
{
  int answer = -1;
  for (int j = nums.length - 1; j >= nums.length / 2; j--)
  {
    if (nums[j] == target)
    {
      answer = j;
    }
  }
  return answer;
}
```

The following code segment appears in another method in the same class as findTarget.

```
int[] arr = {4, 5, 8, 3, 8, 9, 8, 1};
System.out.println(findTarget(arr, 8));
```

What is printed as a result of executing the code segment?

(A) 2

(B) 4 ✓

(C) 5

(D) 6

Answer B

Correct. The loop starts at the end of the array and works its way backward through the second half of the array. Each time the array element equals the value of target, the value of answer will be updated.

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185. Consider the following method, `inCommon`, which takes two `Integer ArrayList` parameters. The method returns `true` if the same integer value appears in both lists at least one time, and `false` otherwise.

```
public static boolean inCommon(ArrayList<Integer> a, ArrayList<Integer> b)
{
  for (int i = 0; i < a.size(); i++)
  {
    for (int j = 0; j < b.size(); j++)    // Line 5
    {
      if (a.get(i).equals(b.get(j)))
      {
        return true;
      }
    }
  }
  return false;
}
```

Which of the following best explains the impact to the `inCommon` method when line 5 is replaced by `for (int j = b.size() - 1; j > 0; j--)` ?

(A) The change has no impact on the behavior of the method.

(B) After the change, the method will never check the first element in list `b`. ✓

(C) After the change, the method will never check the last element in list `b`.

(D) After the change, the method will never check the first and the last elements in list `b`.

Answer B

Correct. When the two lists have no elements in common, or when the common element is at the last index in list `b`, the original inner `for` loop in line 5 will begin at index `0` and end at index `b.size() - 1`, thus checking every element in list `b`. The modified inner `for` loop will traverse list `b` starting with the last element at index `b.size() - 1` and end with the second element in list `b`. It will never check the element at index `0`.

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186. Consider the following method, which is intended to return `true` if `0` is found in its two-dimensional array parameter `mat` and `false` otherwise. The method does not work as intended.

```
public static boolean findZero(int[][] mat)
{
  for (int row = 0; row <= mat.length; row++)
  {
    for (int col = 0; col < mat[0].length; col++)
    {
      if (mat[row][col] == 0)
      {
        return true;
      }
    }
  }
  return false;
}
```

Which of the following values of `mat` can be used to show that the method does **not** work as intended?

(A)

```
{{30, 20},
 {10, 0}}
```

(B)

```
{{4, 3},
 {2, 1},
 {0, -1}}
```

(C)

```
{{0, 1, 2},
 {3, 4, 5},
 {6, 7, 8}}
```

(D) ✓

```
{{5, 10, 15, 20},
 {25, 30, 35, 40}}
```

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Answer D

Correct. The Boolean expression in the outer loop is `row <= mat.length` instead of `row < mat.length`. As a result, the method attempts to access a row outside the dimensions of the array when `0` is not found in the existing rows of the array. This will cause an `ArrayIndexOutOfBoundsException`. Any value of `mat` that does not contain `0` can be used to show that the method does not work as intended.

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187. Consider the following method.

```
public static int someSearch(int[] arr, int value)
{
  for (int j = 0; j < arr.length; j++)
  {
    if (arr[j] == value)
    {
      return j;
    }
  }
  return -1;
}
```

The following code segment appears in a method in the same class as `someSearch`.

```
int[] myNumbers = {1, 2, 5, 2, 6, 3, 3, 5, 3};
System.out.println(someSearch(myNumbers, 3));
```

What is printed as a result of executing the code segment?

(A)

-1

(B) ✓

5

(C)

6

(D)

8

Answer B

Correct. The `someSearch` method performs a linear search of `arr` from left to right. Although `3` appears multiple times in `arr`, the method returns the index of the first element in `arr` that matches value. In this case, that occurs when `j` is equal to `5`.

Consider the following method. This method correctly sorts the elements of array data into increasing order.

```
public static void sort(int[] data)
{
  for (int j = 0; j < data.length - 1; j++)
  {
    int m = j;
    for (int k = j + 1; k < data.length; k++)
    {
      if (data[k] < data[m])    /* Compare values */
      {
        m = k;
      }
    }
    int temp = data[m];    /* Assign to temp */
    data[m] = data[j];
    data[j] = temp;
    /* End of outer loop */
  }
}
```

188. Assume that sort is called with the array {6, 3, 2, 5, 4, 1}. Which of the following represents the contents of data after three passes of the outer loop (i.e., when j == 2 at the point indicated by /* End of outer loop */)?

(A)

{1, 2, 3, 4, 5, 6}

(B) ✓

{1, 2, 3, 5, 4, 6}

(C)

{1, 2, 3, 6, 5, 4}

(D)

{1, 3, 2, 5, 4, 6}

Answer B

Correct. The method implements a selection sort algorithm. It looks for the smallest value in the array ranging from index j to the end of the array and swaps the smallest value with the value at position j. In the first pass, 1 is the smallest element from position j = 0 to the end of the array, so it is swapped with 6, resulting in {1, 3, 2, 5, 4, 6}. In the second pass, 2 is the smallest element from position j = 1 to the end of the array, so it is swapped with 3, resulting in {1, 2, 3, 5, 4, 6}. In the third

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pass, 3 is the smallest element from position j = 2 to the end of the array, so it is swapped with itself, resulting in {1, 2, 3, 5, 4, 6}.

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189. Assume that sort is called with the array {1, 2, 3, 4, 5, 6}. How many times will the expression indicated by /* Compare values */ and the statement indicated by /* Assign to temp */ execute?

(A) ✓

/* Compare values */	/* Assign to temp */
15 times	5 times

(B)

/* Compare values */	/* Assign to temp */
15 times	6 times

(C)

/* Compare values */	/* Assign to temp */
21 times	5 times

(D)

/* Compare values */	/* Assign to temp */
21 times	6 times

Answer A

Correct. The outer loop iterates one fewer time than the length of the array. The inner loop is dependent on the outer loop, starting at the value of j + 1 and iterating to the end of the array. On the first pass through the outer loop, the inner loop will iterate five times, meaning the /* Compare values */ expression will execute 5 times. Each subsequent iteration will result in one fewer

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iteration of the inner loop. Therefore, the /* Compare values */ expression is evaluated 5 + 4 + 3 + 2 + 1, or 15, times. The /* Assign to temp */ expression is evaluated for every iteration of the outer loop. Therefore, the assignment is made 5 times.

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190. The following method is a correct implementation of the selection sort algorithm. The method sorts the elements of arr so that they are in order from least to greatest.

```
public static void selectionSort(int[] arr)
{
  for (int j = 0; j < arr.length - 1; j++)
  {
    int minIndex = j;
    for (int k = j + 1; k < arr.length; k++)
    {
      if (arr[k] < arr[minIndex])
      {
        minIndex = k;
      }
    }

    int temp = arr[j];
    arr[j] = arr[minIndex];
    arr[minIndex] = temp;
    /* end of outer loop */
  }
}
```

Assume that selectionSort has been called with an int[] argument that has been initialized with the following contents.

{40, 30, 50, 60, 10, 20}

What will the contents of arr be after three iterations of the outer loop (i.e., when j == 2 at the point indicated by /* end of outer loop */)?

(A)

{10, 20, 30, 40, 60, 50}

(B) ✓

{10, 20, 30, 60, 40, 50}

(C)

{10, 20, 50, 60, 40, 30}

(D)

{10, 30, 50, 60, 40, 20}

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Answer B

Correct. In each iteration of the outer loop, the method swaps the element at index `j` with the minimum element in the rest of the array (from `arr[j + 1]` to the end of the array). After the first iteration of the outer loop, `40` is swapped with `10` so that the contents of `arr` are `{10, 30, 50, 60, 40, 20}`. After the second iteration of the outer loop, `20` is swapped with `30` so that the contents of `arr` are `{10, 20, 50, 60, 40, 30}`. After the third iteration of the outer loop, `50` is swapped with `30` so that the contents of `arr` are `{10, 20, 30, 60, 40, 50}`.

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191. Consider the following correct implementation of the selection sort algorithm.

```
public static void selectionSort(int[] elements)
{
  for (int j = 0; j < elements.length - 1; j++)
  {
    int minIndex = j;
    for (int k = j + 1; k < elements.length; k++)
    {
      if (elements[k] < elements[minIndex])
      {
        minIndex = k;
      }
    }
    if (j != minIndex)
    {
      int temp = elements[j];
      elements[j] = elements[minIndex];
      elements[minIndex] = temp;        // Line 17
    }
  }
}
```

The following declaration and method call appear in a method in the same class as `selectionSort`.

```
int[] arr = {9, 8, 7, 6, 5};
selectionSort(arr);
```

How many times is the statement in line 17 of the method executed as a result of the call `selectionSort(arr)`?

(A) 2 ✓

(B) 3

(C) 4

(D) 5

Answer A

Correct. The statement in line 17 executes each time a value is swapped into the correct position in the array. For the given array, the values `9` and `5` are swapped. Then the values `8` and `6` are swapped. Then, since `7`, `8`, and `9` are already in the correct position, no additional swaps occur.

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192. The following method is intended to sort an array of integers into ascending order.

```
public static void sort(int[] arr)
{
  for (int j = 0; j < arr.length - 1; j++)
  {
    int minIndex = j;
    for (int k = j + 1; k < arr.length - 1; k++)
    {
      if (arr[k] < arr[minIndex])
      {
        minIndex = k;
      }
    }
    int temp = arr[minIndex];
    arr[minIndex] = arr[j];
    arr[j] = temp;
  }
}
```

This method works as intended for some, but not all, int arrays. Which of the following arrays can be used to show that the method does not work as intended?

(A)

{1, 2, 3, 4, 5}

(B) ✓

{1, 2, 3, 5, 4}

(C)

{1, 2, 4, 3, 5}

(D)

{1, 3, 2, 4, 5}

Answer B

Correct. The last element in the array is never processed by the method since the inner loop ends at arr.length - 1; thus, it will not be sorted when completed.

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193. The following method is a correct implementation of the selection sort algorithm. The method correctly sorts the elements of arr so that they appear in order from least to greatest.

```
public static void selectionSort(int[] arr)
{
  for (int j = 0; j < arr.length - 1; j++)
  {
    int minIndex = j;
    for (int k = j + 1; k < arr.length; k++)
    {
      if (arr[k] < arr[minIndex])
      {
        minIndex = k;
      }
    }
    int temp = arr[j];
    arr[j] = arr[minIndex];
    arr[minIndex] = temp;
    /* end of outer loop */
  }
}
```

Which of the following could be the contents of arr after two passes of the outer loop (i.e., when `j == 1` at the point indicated by `/* end of outer loop */`)?

(A) ✓

{10, 20, 50, 40, 30}

(B)

{10, 30, 50, 40, 20}

(C)

{20, 10, 50, 40, 30}

(D)

{20, 30, 50, 40, 10}

Answer A

Correct. After two passes in selection sort, the smallest two elements must be in the first and second positions in the sorted array.

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194. Consider the following correct implementation of the selection sort algorithm.

```
public static void selectionSort(int[] elements)
{
  for (int j = 0; j < elements.length - 1; j++)
  {
    int minIndex = j;
    for (int k = j + 1; k < elements.length; k++)
    {
      if (elements[k] < elements[minIndex])
      {
        minIndex = k;
      }
    }
    if (j != minIndex)
    {
      int temp = elements[j];
      elements[j] = elements[minIndex];
      elements[minIndex] = temp;   // line 17
    }
  }
}
```

The following declaration and method call appear in a method in the same class as selectionSort.

```
int[] arr = {30, 40, 10, 50, 20};
selectionSort(arr);
```

How many times is the statement elements[minIndex] = temp; in line 17 of the method executed as a result of the call to selectionSort?

(A) 2

(B) 3 ✓

(C) 4

(D) 5

Answer B

Correct. The statement in line 17 executes each time a value is swapped into the correct position in the array. Three swaps would occur. For the given array, the values 30 and 10 are swapped. Then the values 40 and 20 are swapped. Then, since 30 is already in the correct position, no swap occurs. Lastly, 50 and 40 are swapped.

FINAL Unit4 MCQ - Topic order

195. Consider the following method.

```
public String goAgain(String str, int index)
{
  if (index >= str.length())
    return str;

  return str + goAgain(str.substring(index), index + 1);
}
```

What is printed as a result of executing the following statement?

```
System.out.println(goAgain("today", 1));
```

(A) `todayto`

(B) `todayoday`

(C) `todayodayay` ✓

(D) `todayodaydayayy`

FINAL Unit4 MCQ - Topic order

196. Assume that methods `f` and `g` are defined as follows.

```
public int f(int x)
{
  if (x <= 0)
  {
    return 0;
  }
  else
  {
    return g(x - 1);
  }
}

public int g(int x)
{
  if (x <= 0)
  {
    return 0;
  }
  else
  {
    return (f(x - 1) + x);
  }
}
```

What value is returned as a result of the call `f(6)`?

(A) 0
(B) 6
(C) 9 ✓
(D) 12

FINAL Unit4 MCQ - Topic order

197. Consider the following method.

```
public static int mystery(int n)
{
 if (n <= 0)
 {
   return 0;
 }
 else
 {
   return n + mystery(n - 2);
 }
}
```

What value is returned as a result of the call `mystery(9)` ?

(A) `0`

(B) `16`

(C) `24`

(D) `25` ✓

Answer D

Correct. The call `mystery(9) = 9 + mystery(7) = 9 + 7 + mystery(5) = 9 + 7 + 5 + mystery(3) = 9 + 7 + 5 + 3 + mystery(1) = 9 + 7 + 5 + 3 + 1 + mystery(-1) = 9 + 7 + 5 + 3 + 1 + 0 = 25`.

FINAL Unit4 MCQ - Topic order

198. Consider the following recursive method.

```
public static int mystery(int n)
{
  if (n <= 1)
  {
    return 0;
  }
  else
  {
    return 1 + mystery(n / 2);
  }
}
```

Assuming that k is a nonnegative integer and $m = 2^k$, what value is returned as a result of the call mystery (m) ?

(A) 0

(B) k ✓

(C) $\frac{m}{2} + 1$

(D) $\frac{k}{2} + 1$

199. Consider the following method.

```
// precondition: x >= 0
public void mystery(int x)
{
  if ((x / 10) != 0)
  {
    mystery(x / 10);
  }

  System.out.print(x % 10);
}
```

Which of the following is printed as a result of the call `mystery(123456)` ?

(A) `56`

(B) `123456` ✓

(C) `654321`

(D) Many digits are printed due to infinite recursion.

FINAL Unit4 MCQ - Topic order

200. Consider the following method.

```
/** @param x an int value such that x >= 0
 */
public void mystery(int x)
{
  System.out.print(x % 10);
  if ((x / 10) != 0)
  {
    mystery(x / 10);
  }
  System.out.print(x % 10);
}
```

Which of the following is printed as a result of the call mystery (1234)?

(A) 4321

(B) 12344321

(C) 43211234 ✓

(D) Many digits are printed due to infinite recursion.

FINAL Unit4 MCQ - Topic order

201. Consider the following method.

```
public static void mystery(int[] a, int index)
{
 if (index < a.length)
 {
   mystery(a, index + 1);
 }
 if (index > 0)
 {
   System.out.print(a[index - 1]);
 }
}
```

What is printed as a result of executing the following code segment?

```
int[] array = {6, 8, 7, 9, 5};
mystery(array, 0);
```

(A) `5978`

(B) `59786` ✓

(C) `68795`

(D) Many digits are printed because of infinite recursion.

Answer B

Correct. The call `mystery(array, 0)` results in the call `mystery(array, 1)`, which results in the call `mystery(array, 2)`, etc. The call `mystery(array, 5)` is the first call in which the body of the first `if` statement is not executed. The body of the second `if` statement is executed, which prints the value of `a[4]`, which is `5`, and the method returns to the previous call to `mystery`, with `index = 4`. When the method returns, it has finished executing the body of the first `if` statement, and then it executes the body of the second `if` statement, which prints the value of `a[3]`, which is `9`, and the method returns to the previous call to `mystery`, with `index = 3`. This continues until the method gets to the method call in which `index = 0`. At this point, the body of the second `if` statement is not executed because `index` is not greater than `0`. The values `59786` have been printed, and the method exits.

FINAL Unit4 MCQ - Topic order

202. Consider the following recursive method.

```
public static String recur(int val)
{
  String dig = "" + (val % 3);

  if (val / 3 > 0)
    return dig + recur(val / 3);

  return dig;
}
```

What is printed as a result of executing the following statement?

```
System.out.println(recur(32));
```

(A) `102`
(B) `210`
(C) `1020`
(D) `2101` ✓

203. Consider the following static method.

```
private static void recur(int n)
{
  if (n != 0)
  {
    recur(n - 2);
    System.out.print(n + " ");
  }
}
```

What numbers will be printed as a result of the call `recur(7)` ?

(A) `1 3 5 7`
(B) `7 5 3 1`
(C) Many numbers will be printed because of infinite recursion.
(D) No numbers will be printed because of infinite recursion. ✓

FINAL Unit4 MCQ - Topic order

204. Consider the following recursive method.

```
private int recur(int n)
{
  if (n <= 1)
  {
    return 1;
  }
  else
  {
    return (recur(n - 2) + recur(n - 1));
  }
}
```

What value will be returned by the call `recur(4)`?

(A) 2
(B) 3
(C) 5 ✓
(D) 8

FINAL Unit4 MCQ - Topic order

205. The following recursive method is intended to return `true` if `origNum` is divisible by each digit in `numsToDivideBy`, and to return `false` otherwise. For example, `isDivisible(236, 24)` should return `true` because `236` is divisible by both `2` and `4`. The method does not always work as intended.

```
public static boolean isDivisible(int origNum, int numsToDivideBy)
{
  if (numsToDivideBy == 0)
  {
    return true;
  }
  else if (origNum % (numsToDivideBy % 10) != 0)
  {
    return false;
  }
  else
  {
    return isDivisible(origNum, numsToDivideBy / 10);
  }
}
```

Which of the following method calls can be used to demonstrate that the method does **not** work as intended?

(A)

isDivisible(120, 12)

(B)

isDivisible(135, 5)

(C) ✓

isDivisible(225, 105)

(D)

isDivisible(296, 341)

Answer C

Correct. The original call `isDivisible(225, 105)` first checks whether 225 is divisible by 5 (since `105 % 10 == 5`). Both the `if (numsToDivideBy == 0)` and `else if (origNum % (numsToDivideBy % 10) != 0)` conditions evaluate to `false`, so the call `isDivisible(225, 10)` is made. On that call, the `else if` statement attempts to evaluate `225 % (10 % 10)`, which is equal to `225 % 0`, and throws an `ArithmeticException`. The method has no check to avoid dividing by zero.

FINAL Unit4 MCQ - Topic order

206. The `bark` method below is intended to print the string `"woof"` a total of `num` times.

```
public static void bark(int num)
{
  if (num > 0)
  {
    System.out.println("woof");
    /* missing code */
  }
 }
```

Which of the following can be used to replace `/* missing code */` so that the call `bark(5)` will cause `"woof"` to be printed five times?

(A) `bark(num - 5);`

(B) `bark(num - 1);` ✓

(C) `bark(num);`

(D) `bark(num + 1);`

Answer B

Correct. This will recursively call `bark` on the value of `num - 1` until the value of the parameter is `0`. For each of the calls to `bark`, the string `"woof"` is printed (for a total of five times).

FINAL Unit4 MCQ - Topic order

207. Consider the following recursive method.

```
public static int calculateValue(int num)
{
  if (num >= 8)
  {
    return num;
  }
  else
  {
    return num + calculateValue(num + 2);
  }
}
```

What value is returned as a result of the method call `calculateValue(4)`?

(A)

28

(B) ✓

18

(C)

10

(D)

8

Answer B

Correct. In the first call to `calculateValue`, `num` is `4`, so the `else` block returns `4 + calculateValue(6)`. In the second call to `calculateValue`, `num` is `6`, so the `else` block returns `6 + calculateValue(8)`. In the third and final call to `calculateValue`, `num` is `8`, so the `if` block returns `8`. The method call returns `4 + 6 + 8`, or `18`.

FINAL Unit4 MCQ - Topic order

208. Consider the following recursive method.

```
public static int mystery(int x)
{
  if (x <= 0)
  {
    return 5;
  }
  else
  {
    return x + mystery(x - 2);
  }
}
```

What is returned as a result of the method call `mystery(5)`?

(A)

8

(B)

13

(C) ✓

14

(D)

20

Answer C

Correct. The call `mystery(5)` returns `5 + mystery(3)`. The call `mystery(3)` returns `3 + mystery(1)`. The call `mystery(1)` returns `1 + mystery(-1)`. The call `mystery(-1)` matches the base case and returns `5`. This ends the recursion and `5 + 3 + 1 + 5`, or `14`, is returned.

FINAL Unit4 MCQ - Topic order

209. Consider the following two methods, which are intended to return the same values when they are called with the same positive integer parameter `n`.

```
public static int mystery1(int n)
{
 if (n > 1)
 {
   return 5 + mystery1(n - 1);
 }
 else
 {
   return 1;
 }
}


public static int mystery2(int n)
{
 int total = 0;
 int x = 1;
 while (x < n)
 {
   total += 5;
   x++;
 }
 return total;
}
```

Which, if any, of the following changes to `mystery2` is required so that the two methods work as intended?

(A) The variable `total` should be initialized to `1`. ✓

(B) The variable `x` should be initialized to `0`.

(C) The condition in the `while` loop header should be `x < n - 1`.

(D) The condition in the `while` loop header should be `x <= n`.

Answer A

FINAL Unit4 MCQ - Topic order

210. Consider the following method.

```
/** Precondition: j <= k */
public static void mystery(int j, int k)
{
  System.out.println(j);
  if (j < k)
  {
    mystery(j + 1, k);
  }
}
```

Which of the following best describes the behavior of the `mystery` method?

(A) It repeatedly prints the value `j` due to infinite recursion.

(B) It prints the initial value of `j` a total of `k` times.

(C) It prints the integers from `j` to `k`, inclusive, in order from least to greatest. ✓

(D) It prints the integers from `j` to `k`, inclusive, in order from greatest to least.

Answer C

Correct. The method will recursively call `mystery`, increasing `j` by one until `j` and `k` are equal in value. For each of the calls to `mystery`, the value of `j` is printed.

FINAL Unit4 MCQ - Topic order

211. Consider the following method.

```
/** Precondition: num > 0 */
public static int mystery(int num)
{
  if (num / 10 == 0)
  {
    return 2;
  }
  else
  {
    return 2 + mystery(num / 10);
  }
}
```

Which of the following best describes the value returned by the method?

(A) It returns `2` more than the number of digits in `num`.

(B) It returns `2` more than the sum of the digits in `num`.

(C) It returns `2` times the number of digits in `num`. ✓

(D) It returns `2` times the sum of the digits in `num`.

Answer C

Correct. As long as `num / 10` is not equal to zero, the statement in the `else` block is executed. This statement uses `num / 10` to remove one digit from `num` and recursively calls `mystery` using that new value. Each time that happens, `2` is added to the return value until `mystery` is called with a single-digit argument. At this point, the base case is reached and `2` is returned for the last digit. Since the return value is increased by `2` for each digit in `num`, the method returns `2` times the number of digits in `num`.

FINAL Unit4 MCQ - Topic order

212. Consider the following method.

```
public static void printNums(int n)
{
  if (n < 50)
  {
    printNums(n * -2);
    System.out.print(n + " ");
  }
}
```

What is printed as a result of the call printNums(10)?

(A) ✓

-80 40 -20 10

(B)

10 -20 40 -80

(C)

40 -20 10

(D)

10 -20 40

Answer A

Correct. After calling printNums(10), n is less than 50, so printNums(-20) is called. At this point, n is still less than 50, so printNums(40) is called. At this point, n is still less than 50, so printNums(-80) is called. At this point, n is still less than 50, so printNums(160) is called. At this point, n is not less than 50, so there are no more recursive calls. Working backward from the last call, the values of n are printed in the order -80, 40, -20, and 10.

FINAL Unit4 MCQ - Topic order

213. Consider the following recursive method.

```
public static void stars(int num)
{
  if (num == 1)
  {
    return;
  }

  stars(num - 1);

  for (int i = 0; i < num; i++)
  {
    System.out.print("*");
  }
  System.out.println();
}
```

What is printed as a result of the method call stars(5)?

FINAL Unit4 MCQ - Topic order

(A)

```
*****
****
***
**
*
```

(B)

```
*****
****
***
**
```

(C)

```
*
**
***
****
*****
```

(D) ✓

```
**
***
****
*****
```

Answer D

Correct. The recursive call of the stars method occurs before any output is printed, so the method call stars(5) results in a recursive call to stars(4), then to stars(3), then to stars(2), and finally to stars(1). The call to stars(1) returns immediately

without printing any output, so the first call that produces output is stars(2), which prints a row of two stars. Then, stars(3) prints a row of three stars, stars(4) prints a row of four stars, and finally stars(5) prints a row of five stars.

214. Consider the following recursive method.

```
/** Precondition: num >= 0 */
public static int what(int num)
{
  if (num < 10)
  {
    return 1;
  }
  else
  {
    return 1 + what(num / 10);
  }
}
```

Assume that the int variable val has been declared and initialized with a value that satisfies the precondition of the method. Which of the following best describes the value, if any, returned by the call what(val)?

(A) The number of digits in val ✓

(B) The sum of the digits in val

(C) The value 1

(D) Nothing is returned. A run-time error occurs because of infinite recursion.

Answer A

Correct. Each recursive call is made with the value num / 10. The expression num / 10 uses integer division and evaluates to an integer that consists of num with the rightmost digit removed. For example, 258 / 10 is equal to 25. Each time the recursive call is made, the value returned is 1 plus the result of the recursive call. For example, what(258) is equal to 1 + what(25), and what(25) is equal to 1 + what(2), and what(2) is equal to 1. Therefore what(258) is equal to 1 + 1 + 1, or 3, which is the number of digits in 258.

FINAL Unit4 MCQ - Topic order

215. Consider the following method.

```
public static int mystery(int n)
{
  if (n < 10)
  {
    return n;
  }
  return (n % 10) + mystery(n / 10);
}
```

Assume that the `int` variable `num` is properly declared and initialized to a value greater than 0. Which of the following best describes the value returned by the call `mystery(num)`?

(A) The value of `num`

(B) The rightmost digit of `num`

(C) The sum of the digits of `num` ✓

(D) Two times the rightmost digit of `num`

Answer C

Correct. The recursive case takes the rightmost digit (`n % 10`) and adds it to the result of recursively summing the other digits.

FINAL Unit4 MCQ - Topic order

216. Consider the following method.

```
public static void mystery(int j, int k)
{
  if (j > k)
  {
    mystery(j – 2, k + 2);
  }
  System.out.print(j + " ");
}
```

What is printed as a result of the call mystery(10, 0)?

(A)

10 8 6 4

(B)

10 8 6

(C)

6 8 10

(D) ✓

4 6 8 10

Answer D

Correct. After calling mystery(10, 0), j is greater than k, so mystery(8, 2) is called. At this point, j is still greater than k, so mystery(6, 4) is called. At this point, j is still greater than k, so mystery(4, 6) is called. At this point, j is not greater than k, so there are no more recursive calls. Working backward from the last call, the values of j are printed in the order 4, 6, 8, and 10.

FINAL Unit4 MCQ - Topic order

217. Consider the following method, which is intended to return the largest value in the portion of the `int` array `data` that begins at the index `start` and goes to the end of the array.

```
 /** Precondition: 0 <= start < data.length */
public int maximum(int[] data, int start)
{
  if (start == data.length - 1)
  {
    return data[start];
  }
  /* missing statement */
  if (val > data[start])
  {
    return val;
  }
  else
  {
    return data[start];
  }
}
```

Which of the following can be used as a replacement for `/* missing statement */` so that the `maximum` method works as intended?

(A) `int val = maximum(data, start - 1);`

(B) `int val = maximum(data, start + 1);` ✓

(C) `int val = maximum(data[start - 1], start);`

(D) `int val = maximum(data[start + 1], start);`

Answer B

Correct. The variable `val` is set to the largest value among the portion of `data` that begins at the index `start + 1` and goes through the end of the array. The method then returns the larger of `data[start]` and `val`. At each subsequent index in `data`, the method returns the larger of `data[index]` and the largest of the remaining elements.

FINAL Unit4 MCQ - Topic order

218. Consider the following recursive method.

```
public static String doSomething(String str)
{
  if (str.length() < 1)
  {
    return "";
  }
  else
  {
    return str.substring(0, 1) + doSomething(str.substring(1));
  }
}
```

Assume that `myString` is a properly declared and initialized `String` object. Which of the following best describes the result of the call `doSomething(myString)`?

(A) The method call returns a `String` containing the contents of `myString` unchanged. ✓

(B) The method call returns a `String` containing the contents of `myString` with the order of the characters reversed from their order in `myString`.

(C) The method call returns a `String` containing all but the first character of `myString`.

(D) The method call returns a `String` containing only the first and second characters of `myString`.

Answer A

Correct. When there is at least one character in `str`, the method returns the first character of `str` concatenated with the value returned by the recursive call. The parameter of the recursive call is the substring of `str` starting at the second character. As a result, the original string is reconstructed recursively.

FINAL Unit4 MCQ - Topic order

219. Consider the following method, which implements a recursive binary search.

```
/** Returns an index in arr where the value x appears if x appears
*  in arr between arr[left] and arr[right], inclusive;
*  otherwise returns -1.
*  Precondition: arr is sorted in ascending order.
*                left >= 0, right < arr.length, arr.length > 0
*/
public static int bSearch(int[] arr, int left, int right, int x)
{
  if (right >= left)
  {
    int mid = (left + right) / 2;
    if (arr[mid] == x)
    {
      return mid;
    }
    else if (arr[mid] > x)
    {
      return bSearch(arr, left, mid - 1, x);
    }
    else
    {
      return bSearch(arr, mid + 1, right, x);
    }
  }
  return -1;
}
```

The following code segment appears in a method in the same class as `bSearch`.

```
int[] nums = {0, 4, 4, 5, 6, 7};
int result = bSearch(nums, 0, nums.length - 1, 4);
```

What is the value of `result` after the code segment has been executed?

(A) `1`

(B) `2` ✓

(C) `3`

(D) `4`

Answer B

Correct. When `bSearch` is first executed, `left` has the value `0`, `right` has the value `5`, and the value of `arr[mid]`, or `arr[2]`, is equal to `x`. The value `2` is returned since `arr[mid]`, or `arr[2]`, is equal to `x`. The `bSearch` method is never called recursively because `4` was found on the first call to `bSearch`.

FINAL Unit4 MCQ - Topic order

220. Consider the following method, which implements a recursive binary search.

```
/** Returns an index in myList where target appears,
 * if target appears in myList between the elements at indices
 * low and high, inclusive; otherwise returns -1.
 * Precondition: myList is sorted in ascending order.
 * low >= 0, high < myList.size(), myList.size() > 0
 */
public static int binarySearch(ArrayList<Integer> myList,
 int low, int high, int target)
{
  int mid = (high + low) / 2;
  if (target < myList.get(mid))
  {
    return binarySearch(myList, low, mid - 1, target);
  }
  else if (target > myList.get(mid))
  {
    return binarySearch(myList, mid + 1, high, target);
  }
  else if (myList.get(mid).equals(target))
  {
    return mid;
  }
  return -1;
}
```

Assume that `inputList` is an `ArrayList` of `Integer` objects that contains the following values.

```
[0, 10, 30, 40, 50, 70, 70, 70, 70]
```

What value will be returned by the call `binarySearch(inputList, 0, 8, 70)`?

(A) `-1`

(B) `5`

(C) `6` ✓

(D) `7`

Answer C

Correct. The first time `binarySearch` is executed, `low` has the value `0`, `high` has the value `8`, and `mid` is `4`. The value at index `4` is `50`, which is less than `target`. The second time `binarySearch` is executed, `low` has the value `5`, `high` has the value `8`, and `mid` is `6`. The value at index `6`, which is `70`, is equal to `target`, so `6` is returned.

FINAL Unit4 MCQ - Topic order

221. Consider the following `mergeSortHelper` method, which is part of an algorithm to recursively sort an array of integers.

```
/** Precondition: (arr.length == 0 or 0 <= from <= to <= arr.length)
* arr.length == temp.length
*/
public static void mergeSortHelper(int[] arr,
int from, int to, int[] temp)
{
 if (from < to)
 {
   int middle = (from + to) / 2;
   mergeSortHelper(arr, from, middle, temp);
   mergeSortHelper(arr, middle + 1, to, temp);
   merge(arr, from, middle, to, temp);
 }
}
```

The `merge` method is used to merge two halves of an array (`arr[from]` through `arr[middle]`, inclusive, and `arr[middle + 1]` through `arr[to]`, inclusive) when each half has already been sorted into ascending order. For example, consider the array `arr1`, which contains the values `{1, 3, 5, 7, 2, 4, 6, 8}`. The lower half of `arr1` is sorted in ascending order (elements `arr1[0]` through `arr1[3]`, or `{1, 3, 5, 7}`), as is the upper half of `arr1` (elements `arr1[4]` through `arr1[7]`, or `{2, 4, 6, 8}`). The array will contain the values `{1, 2, 3, 4, 5, 6, 7, 8}` after the method call `merge(arr1, 0, 3, 7, temp)`. The array `temp` is a temporary array declared in the calling program.

Consider the following code segment, which appears in a method in the same class as `mergeSortHelper` and `merge`.

```
int[] vals = {80, 50, 30, 20, 60, 70};
int[] temp = new int[vals.length];
mergeSortHelper(vals, 0, vals.length - 1, temp);
```

Which of the following represents the arrays merged the last time the `merge` method is executed as a result of the code segment above?

(A) `{20, 30, 50}` and `{60, 70, 80}` are merged to form `{20, 30, 50, 60, 70, 80}`.

(B) `{20, 50, 70}` and `{30, 60, 80}` are merged to form `{20, 30, 50, 60, 70, 80}`.

(C) `{20, 50, 70}` and `{30, 60, 80}` are merged to form `{20, 50, 70, 30, 60, 80}`.

(D) `{30, 50, 80}` and `{20, 60, 70}` are merged to form `{20, 30, 50, 60, 70, 80}`. ✓

Answer D

Correct. The `merge` method appears after both recursive calls in the `mergeSortHelper` method, so it is not called until the condition `from < to` evaluates to `false` and there are no more recursive calls to execute. This occurs when the two parts of the array to merge each contain one element. The original array is split into the arrays `{80}`, `{50}`, `{30}`, `{20}`, `{60}`, and `{70}`. These are then merged to form the arrays `{50, 80}`, `{30}`, `{20, 60}`, and `{70}`. These are then merged to form the arrays `{30, 50, 80}` and `{20, 60, 70}`. Lastly, these are merged to form `{20, 30, 50, 60, 70, 80}`.

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222. Consider the following instance variable and method.

```
private int[] arr;

/** Precondition: arr contains no duplicates;
 *                the elements in arr are in ascending order.
 *  @param low an int value such that 0 ≤ low ≤ arr.length
 *  @param high an int value such that low - 1 ≤ high < arr.length
 *  @param num an int value
 */
public int mystery(int low, int high, int num)
{
  int mid = (low + high) / 2;
  if (low > high)
  {
    return low;
  }
  else if (arr[mid] < num)
  {
    return mystery(mid + 1, high, num);
  }
  else if (arr[mid] > num)
  {
    return mystery(low, mid - 1, num);
  }
  else // arr[mid] == num
  {
    return mid;
  }
}
```

What is returned by the call mystery (0, arr.length − 1, num) ?

(A) The number of elements in arr that are less than num ✓

(B) The number of elements in arr that are less than or equal to num

(C) The number of elements in arr that are equal to num

(D) The index of the middle element in arr

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223. The `printRightToLeft` method is intended to print the elements in the `ArrayList words` in reverse order. For example, if `words` contains `["jelly bean", "jukebox", "jewelry"]`, the method should produce the following output.

```
jewelry
jukebox
jelly bean
```

The method is shown below.

```
public static void printRightToLeft(ArrayList<String> words)
{
  if (words.size() > 0)
  {
    System.out.println(words.get(words.size() - 1));
    /* missing code */
  }
}
```

Which of the following can be used to replace `/* missing code */` so that the `printRightToLeft` method works as intended?

(A)
```
words.remove(0);
printRightToLeft(words);
```

(B)
```
words.remove(words.size());
printRightToLeft(words);
```

(C) ✓
```
words.remove(words.size() - 1);
printRightToLeft(words);
```

(D)
```
printRightToLeft(words);
words.remove(words.size() - 1);
```

Answer C

Correct. The word at position `words.size() - 1` in `words` has already been printed, so it is removed from the `ArrayList`. The recursive call then prints the remaining elements of the `ArrayList`.

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224. The following method implements a recursive binary search algorithm.

```
public static int binarySearch(int[] elements, int low,
                               int high, int target)
{
  if (low <= high)
  {
    int mid = (high + low) / 2;
    if (target < elements[mid])
    {
      return binarySearch(elements, low, mid - 1, target);
    }
    else if (target > elements[mid])
    {
      return binarySearch(elements, mid + 1, high, target);
    }
    else if (elements[mid] == target)
    {
      return mid;
    }
  }
  return -1;
}
```

The following code segment appears in a method in the same class as binarySearch.

```
int[] numbers = {10, 10, 10, 20, 20, 30, 40, 50, 50, 60, 80};
int result = binarySearch(numbers, 0, numbers.length - 1, 10);
```

What is the value of result after executing the code segment?

(A)

-1

(B)

0

(C)

1

(D) ✓

2

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Answer D

Correct. The first time binarySearch is called, low is 0, high is 10, and mid is 5. The value at index 5 is 30, which is greater than target. The next time binarySearch is called, low is 0, high is 4, and mid is 2. The value at index 2 is 10, which is equal to target. The index 2 is returned.

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225. Consider the following method, which correctly implements a recursive binary search.

```
/**
 *  Returns an index in arr where the value x appears if x appears
 *  in arr between arr[left] and arr[right], inclusive;
 *  otherwise returns -1.
 *  Preconditions: arr is sorted in ascending order.
 *                 left >= 0, right < arr.length, arr.length > 0
 */
public static int bSearch(int[] arr, int left, int right, int x)
{
  if (left <= right)
  {
    int mid = (left + right) / 2;
    if (arr[mid] == x)
    {
      return mid;
    }
    else if (x < arr[mid])
    {
      return bSearch(arr, left, mid - 1, x);
    }
    else
    {
      return bSearch(arr, mid + 1, right, x);
    }
  }
  return -1;
}
```

The following statement appears in a method in the same class as `bSearch`. Assume that `nums` is an integer array of length 7, containing only positive integers in ascending order.

```
int result = bSearch(nums, 0, nums.length - 1, -100);
```

How many times will the `bSearch` method be called as a result of executing the statement, including the initial call?

(A) 3

(B) 4 ✓

(C) 5

(D) 7

Answer B

Correct. When `bSearch` is first executed, `left` has the value `0`, `right` has the value `6`, and `mid` has the value `3`. The target is negative and the array contains only positive values, so when `bSearch` is called again, `left` has the value `0`, `right`

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has the value `2`, and `mid` has the value `1`. When `bSearch` is called again, `left` has the value `0`, `right` has the value `0`, and `mid` has the value `0`. Then `bSearch` is called one final time, in which `left` has the value `0` and `right` has the value `-1`. The `target` is not found, so `-1` is returned. The `bSearch` method is called a total of four times.

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226. Consider the following method, which correctly implements a recursive binary search.

```
/**
 *  Returns an index in arr where the value x appears if x appears
 *  in arr between arr[left] and arr[right], inclusive;
 *  otherwise returns -1.
 *  Preconditions: arr is sorted in ascending order.
 *                 left >= 0, right < arr.length, arr.length > 0
 */
public static int bSearch(int[] arr, int left, int right, int x)
{
  if (right >= left)
  {
    int mid = (left + right) / 2;
    if (arr[mid] == x)
    {
      return mid;
    }
    else if (arr[mid] > x)
    {
      return bSearch(arr, left, mid - 1, x);
    }
    else
    {
      return bSearch(arr, mid + 1, right, x);
    }
  }
  return -1;
}
```

The following code segment appears in a method in the same class as `bSearch`.

```
int[] nums = {10, 20, 30, 40, 50};
int result = bSearch(nums, 0, nums.length - 1, 40);
```

How many times will the `bSearch` method be called as a result of executing the code segment, including the initial call?

(A) 1

(B) 2 ✓

(C) 3

(D) 4

Answer B

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Correct. When `bSearch` is first called, `left` has the value `0`, `right` has the value `4`, and `mid` has the value `2`. The value of `arr[mid]`, which is `30`, is less than `x`. When `bSearch` is called for the second time, `left` has the value `3`, `right` has the value `4`, and `mid` has the value `3`. The value of `arr[mid]`, which is `40`, is equal to `x`, and `3` is returned. The `bSearch` method is called a total of two times.

227. The following recursive method is intended to return the index of the first occurrence of goal in arr between index start and index end, inclusive. The index of the first occurrence of goal is the index closest to start where goal is found. If goal does not appear in the range of indices from start to end, the method returns -1. However, the method does not work as intended.

```
/** Precondition: 0 <= start <= end, end < arr.length */
public static int locateItem(int[] arr, int goal, int start, int end)
{
  if (start == end)                          // Line 4
  {
    return -1;
  }
  else if (arr[start] == goal)                 // Line 8
  {
    return start;                            // Line 10
  }
  else
  {
    return locateItem(arr, goal, start + 1, end);  // Line 14
  }
}
```

Which of the following changes can be made so that the method works as intended?

(A) Changing line 4 to if (start > end) ✓

(B) Changing line 8 to else if (start == goal)

(C) Changing line 10 to return arr[start];

(D) Changing line 14 to return locateItem(arr, goal, start + 1, end + 1);

Answer A

Correct. Without this change, the method returns -1 when goal appears at index end. This change allows end to be a valid return index.

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228. Consider the following method.

```
public static void find(String[] words, int s, int i)
{
  if (i != words.length)
  {
    if (i == 0 || !(words[s].equals(words[i])))
    {
      s = i;
      System.out.print(s + " ");
    }
    find(words, s, i + 1);
  }
}
```

The following code segment appears in a method in the same class as find.

```
String[] furniture = /* missing initialization */ ;
find(furniture, 0, 0);
```

Which of the following can be used to replace /* missing initialization */ so that the code segment prints 0 1 4 6 7 ?

(A)

```
{"Stool", "Chair", "Chair", "Chair",
 "Stool", "Couch", "Couch", "Table"}
```

(B)

```
{"Table", "Couch", "Couch", "Stool",
 "Chair", "Chair", "Chair", "Stool"}
```

(C) ✓

```
{"Stool", "Chair", "Chair", "Chair",
 "Couch", "Couch", "Stool", "Table"}
```

(D)

```
{"Table", "Stool", "Couch", "Couch",
 "Chair", "Chair", "Chair", "Stool"}
```

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Answer C

Correct. In the first call, the first value of s (0) is always printed. For recursive calls, each time the next string is not equal to the previous string, s is updated to the new index of the current word and is then printed. Indexes of words not equal to previous words are 1, 4, 6, and 7, which is what is printed. The method makes no recursive call when i is beyond the last valid index of the array.

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229. Consider the following method.

```
public static String[] strArrMethod(String[] arr)
{
 String[] result = new String[arr.length];
 for (int j = 0; j < arr.length; j++)
 {
   String sm = arr[j];
   for (int k = j + 1; k < arr.length; k++)
   {
     if (arr[k].length() < sm.length())
     {
       sm = arr[k]; // Line 12
     }
   }
   result[j] = sm;
 }
 return result;
}
```

Consider the following code segment.

```
String[] testTwo = {"last", "day", "of", "the", "school", "year"};
String[] resultTwo = strArrMethod(testTwo);
```

How many times is the line labeled `// Line 12` in the `strArrMethod` executed as a result of executing the code segment?

(A) 4 times ✓

(B) 5 times

(C) 6 times

(D) 15 times

Answer A

Correct. Line 12 is executed each time the variable `sm` is updated because a new smallest value is found. When `j` has the value `0`, `sm` is updated for `"day"` and `"of"`. When `j` has the value `1`, `sm` is updated for `"of"`. When `j` has the value `4`, `sm` is updated for `"year"`. When `j` has any of the values `2`, `3`, or `5`, `sm` is not updated. Line 12 is executed four times.

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230. A text file named roster.txt has the following contents. Assume that the file contains no spaces.

Cohen-Isabel
Lee-Cindy
Patel-Anisha
Sanchez-Michael

In the following code segment, a valid Scanner object named scan has been created to read from the text file.

```
File myText = new File("roster.txt");
Scanner scan = new Scanner(myText);

ArrayList<String> usernames = new ArrayList<String>();

/* missing code */
```

This code segment is intended to assign ["ICohen", "CLee", "APatel", "MSanchez"] to usernames.

Which of the following can replace /* missing code */ so that this code segment works as intended?

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(A)

```
while (scan.hasNext())
{
  String temp = scan.next().substring(0, 1);
  usernames.add(temp + scan.next());
}
```

(B)

```
while (scan.hasNext())
{
  String one = scan.next();
  String two = scan.next();
  usernames.add(two.substring(0, 1) + one);
}
```

(C) ✓

```
while (scan.hasNext())
{
  String[] temp = scan.next().split("-");
  usernames.add(temp[1].substring(0, 1) + temp[0]);
}
```

(D)

```
while (scan.hasNext())
{
  String[] temp = scan.next().split("-");
  usernames.add(temp[0].substring(0, 1) + temp[1]);
}
```

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Answer C

Correct. For every string encountered in the file, the code segment will use the split method to place the text before the hyphen in temp[0] and the text after the hyphen in temp[1]. The code segment then concatenates the first character of temp[1] and the entire value of temp[0] and adds the result to usernames.

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231. A text file named bookInfo.txt has the following contents. Assume that the file contains no spaces.

```
Persuasion,Austen_1818
Tarzan,Burroughs_1912
Frankenstein,Shelley_1818
Dracula,Stoker_1897
```

In the following code segment, a valid Scanner object named sc is created to read from the text file.

```
File bookData = new File("bookInfo.txt");
Scanner sc = new Scanner(bookData);

ArrayList<String> aList = new ArrayList<String>();
ArrayList<String> bList = new ArrayList<String>();
ArrayList<String> cList = new ArrayList<String>();

while (sc.hasNext())
{
  String temp = sc.next();
  int j = temp.indexOf(",");
  int k = temp.indexOf("_");

  aList.add(temp.substring(0, j));
  bList.add(temp.substring(j + 1, k));
  cList.add(temp.substring(k + 1));
}

System.out.println(cList.get(1));
```

What is printed as a result of executing the code segment?

(A)

1818

(B) ✓

1912

(C)

Austen

(D)

Burroughs

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Answer B

Correct. For every string encountered in the file, the code segment will use the indexOf method to determine the locations of the comma (",") and the underscore ("_"). Using the substring method, the text before the comma (a book title) is added to aList, the text between the comma and the underscore (an author name) is added to bList, and the text after the underscore (a year) is added to cList. After the code segment is executed, cList(1) contains the second year in the file, which is 1912.

232. Consider the following code segment.

```
int count = 0;
for (String word : wordList)
{
  if (word.indexOf("a") >= 0)
  {
    count++;
  }
}
System.out.println(count);
```

Assume that the `ArrayList wordList` has been initialized to contain the `String` values `["apple", "banana", "coconut", "lemon", "orange", "pear"]`.

What is printed as a result of executing the code segment?

(A)

2

(B)

3

(C) ✓

4

(D)

5

Answer C

Correct. The code segment uses an enhanced `for` loop to traverse `wordList`. The value of `count` is incremented each time a word contains the string `"a"`. There are four elements of `wordList` that contain an `"a"`, so `4` is printed.

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233. In the following code segment, arr1 and minArray are properly declared and initialized int arrays of equal length. During the traversal of the array, each element of arr1 is compared to the corresponding element of minArray located at the same index. If any element of arr1 is smaller than the corresponding element of minArray, the code segment is intended to replace that element of minArray with the corresponding element of arr1. After the code segment is executed, arr1 should remain unchanged and minArray should hold the lesser of the two elements originally found at corresponding indices in arr1 and minArray.

The code segment does not work as intended.

```
for (int c = 0; c < arr1.length; c++)  // Line 1
{                                      // Line 2
  if (arr1[c] < minArray[c])           // Line 3
  {                                    // Line 4
    arr1[c] = minArray[c];             // Line 5
  }                                    // Line 6
  else                                 // Line 7
  {                                    // Line 8
    minArray[c] = arr1[c];             // Line 9
  }                                    // Line 10
}                                      // Line 11
```

Which of the following changes can be made so that the code segment works as intended?

(A) Changing the relational operator in line 3 to >

(B) Removing lines 5 through 8 ✓

(C) Swapping the positions of line 5 and line 9

(D) Removing lines 7 through 10

Answer B

Correct. Line 5 modifies an element of arr1 if that element is smaller than the corresponding element of minArray, which is not what is intended, since arr1 should remain unchanged. Removing lines 5 through 8 will store the smaller of each pair of values in minArray.

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234. Consider the following method.

```
public static int mystery(int[] arr)
{
 int count = 0;
 int curr = arr[arr.length - 1];

 for (int value : arr)
 {
   if (value > curr)
   {
     count = count + 1;
   }
   else
   {
     count = count - 1;
   }
   curr = value;
 }

 return count;
}
```

The following code segment appears in another method of the same class.

```
int[] arr = {4, 14, 15, 3, 14, 18, 19};
System.out.println(mystery(arr));
```

What is printed as a result of executing the code segment?

(A) `-7`

(B) `-6`

(C) `3` ✓

(D) `5`

Answer C

Correct. Each element of the array is compared to the previous element. For the first element, the previous element is considered to be the last element of the array. If an array element is greater than the previous element, `count` is incremented; otherwise, `count` is decremented. The value of `count` is incremented for 5 comparisons (`14` and `4`, `15` and `14`, `14` and `3`, `18` and `14`, and `19` and `18`) and decremented for 2 comparisons (`4` and `19`, and `3` and `15`), resulting in a final value of `0 + 5 - 2 = 3`.

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235. Consider the following data field and method.

```
private int[] seq;
// precondition: seq.length > 0
public int lenIncreasing()
{
  int k = 1;

  while ((k < seq.length) && (seq[k - 1] < seq[k]))
  {
    k++;
  }

  // assertion
  return k;
}
```

Which of the following assertions is true when execution reaches the line `// assertion` in `lenIncreasing`?

(A) `(k == seq.length) && (seq[k - 1] >= seq[k])`

(B) `(k == seq.length) || (seq[k - 1] >= seq[k])` ✓

(C) `(k < seq.length) && (seq[k - 1] < seq[k])`

(D) `(k < seq.length) || (seq[k - 1] < seq[k])`

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236. Consider the following method that is intended to modify its parameter `nameList` by replacing all occurrences of `name` with `newValue`.

```
public void replace(ArrayList<String> nameList,
                    String name, String newValue)
{
  for (int j = 0; j < nameList.size(); j++)
  {
    if ( /* expression */ )
    {
      nameList.set(j, newValue);
    }
  }
}
```

Which of the following can be used to replace `/* expression */` so that `replace` will work as intended?

(A) `nameList.get(j).equals(name)` ✓

(B) `nameList.get(j) == name`

(C) `nameList[j] == name`

(D) `nameList[j].equals(name)`

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237. Consider the following code segment.

```
int[] numbers = new int[5];
numbers[0] = 2;
numbers[1] = numbers[0] + 1;
numbers[numbers[0]] = numbers[1];

for (int x = 3; x < numbers.length; x++)
{
 numbers[x] = numbers[x - 1] * 2;
}
```

Which of the following represents the contents of the array `numbers` after the code segment is executed?

(A) `{2, 3, 0, 0, 0}`

(B) `{2, 3, 1, 2, 4}`

(C) `{2, 3, 3, 6, 9}`

(D) `{2, 3, 3, 6, 12}` ✓

Answer D

Correct. After `numbers` is created, it contains `{0, 0, 0, 0, 0}`. The next three assignment statements store `2` in `numbers[0]`, `2 + 1 = 3` in `numbers[1]`, and `3` in `numbers[2]`, resulting in the contents `{2, 3, 3, 0, 0}`. The first iteration of the `for` loop assigns `3 * 2 = 6` to `numbers[3]`, and the second iteration assigns `6 * 2 = 12` to `numbers[4]`. The final contents of `numbers` after the code segment is executed is `{2, 3, 3, 6, 12}`.

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238. Consider the following method.

```
public static void methodX(int[] values)
{
 for (int j = 0; j < values.length - 1; j++)
 {
   for (int k = j + 1; k < values.length; k++)
   {
     if (values[k] < values[j])
     {
       values[j] = values[k];
     }
   }
 }
}
```

The following code segment appears in another method of the same class.

```
int[] numbers = {45, 1, 56, 10};
methodX(numbers);
```

Which of the following represents the contents of the array `numbers` after the code segment is executed?

(A) `{1, 1, 10, 10}` ✓

(B) `{1, 10, 45, 56}`

(C) `{45, 1, 56, 10}`

(D) `{45, 45, 56, 45}`

Answer A

Correct. When `j = 0` and `k = 1`, `values[k] < values[j]` and `values` is updated to `{1, 1, 56, 10}`. No changes are made to `values` when `j = 0` and `k = 2`, when `j = 0` and `k = 3`, when `j = 1` and `k = 2`, or when `j = 1` and `k = 3`. When `j = 2` and `k = 3`, `values[k] < values[j]` and `values` is updated to `{1, 1, 10, 10}`.

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239. The following method is a correct implementation of the insertion sort algorithm. The method correctly sorts the elements of `arr` so that they appear in order from least to greatest.

```
public static void insertionSort(int[] arr)
{
  for (int j = 1; j < arr.length; j++)
  {
    int temp = arr[j];
    int possibleIndex = j;
    while (possibleIndex > 0 && temp < arr[possibleIndex - 1])
    {
      arr[possibleIndex] = arr[possibleIndex - 1];
      possibleIndex--;                    // Line 10
    }
    arr[possibleIndex] = temp;
  }
}
```

The following code segment appears in a method in the same class as `insertionSort`.

```
int[] numbers = {4, 12, 4, 7, 19, 6};
insertionSort(numbers);
```

How many times is the statement in line 10 of the method executed as a result of the call to `insertionSort`?

(A) 2

(B) 3

(C) 4

(D) 5 ✓

Answer D

Correct. The statement in line 10 is executed each time an element is moved in the array. In the second iteration of the `for` loop, the `12` located at index `1` is moved to index `2`, and `4` is inserted at index `1`. The array then contains `{4, 4, 12, 7, 19, 6}` and line 10 has executed 1 time. In the third iteration of the `for` loop, the `12` located at index `2` is moved to index `3`, and `7` is inserted at index `2`. The array then contains `{4, 4, 7, 12, 19, 6}` and line 10 has executed a total of 2 times. In the fifth iteration of the `for` loop, the `19` located at index `4` is moved to index `5`, the `12` located at index `3` is moved to index `4`, the `7` located at index `2` is moved to index `3`, and `6` is inserted at index `2`. The array then contains `{4, 4, 6, 7, 12, 19}` and line 10 has executed 3 additional times, for a total of 5 times.

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240. The following method is a correct implementation of the selection sort algorithm. The method correctly sorts the elements of `data` so that they appear in order from least to greatest.

```
public static void selectionSort(int[] data)
{
  for (int j = 0; j < data.length - 1; j++)
  {
    int minIndex = j;

    for (int k = j + 1; k < data.length; k++)
    {
      if (data[k] < data[minIndex])
      {
        minIndex = k;      // Line 11
      }
    }

    if (j != minIndex)
    {
      int temp = data[j];
      data[j] = data[minIndex];
      data[minIndex] = temp;
    }
  }
}
```

The following code segment appears in a method in the same class as `selectionSort`.

```
int[] vals = {5, 10, 2, 1, 12};
selectionSort(vals);
```

How many times is the statement in line 11 of the method executed as a result of the call to `selectionSort`?

(A) 1

(B) 2

(C) 3

(D) 4 ✓

Answer D

Correct. The statement in line 11 is executed when a new smallest element is found in the array. In the first iteration of the outer `for` loop, `5` (located at index `0`) is currently the smallest element. While traversing through the rest of the array using the inner `for` loop, the code first sets `2` (located at index `2`) to the new smallest element and then sets `1` (located at index `3`) to the new smallest element. When the inner `for` loop terminates, `5` and `1` are swapped. The array now contains `{1, 10, 2, 5, 12}` and line 11 has executed 2 times. In the second iteration of the outer `for` loop, `10` (located at index `1`) is the smallest element. While traversing through the rest of the array using the inner `for` loop, the code sets `2` (located at index `2`) to

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the new smallest element. When the inner `for` loop terminates, `10` and `2` are swapped. The array now contains `{1, 2, 10, 5, 12}` and line 11 has executed 1 additional time, for a total of 3 times. In the third iteration of the outer `for` loop, `10` (located at index `2`) is currently the smallest element. While traversing through the rest of the array using the inner `for` loop, the code sets `5` (located at index `3`) to the smallest element. When the inner `for` loop terminates, `10` and `5` are swapped. The array now contains `{1, 2, 5, 10, 12}` and line 11 has executed 1 additional time, for a total of 4 times.

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241. Consider the following `mergeSortHelper` method, which is part of a correctly implemented algorithm to recursively sort an array of integers.

```
/** Preconditions: (arr.length == 0 or
 *                  0 <= from <= to <= arr.length)
 *                  arr.length == temp.length
 */
public static void mergeSortHelper(int[] arr, int from,
                                   int to, int[] temp)
{
  if (from < to)
  {
    int middle = (from + to) / 2;
    mergeSortHelper(arr, from, middle, temp);
    mergeSortHelper(arr, middle + 1, to, temp);
    merge(arr, from, middle, to, temp);
  }
}
```

The `merge` method (not shown) is used to merge two halves of an array (`arr[from]` through `arr[middle]`, inclusive, and `arr[middle + 1]` through `arr[to]`, inclusive) when each half has already been sorted into ascending order. For example, if `arr1` contains `{1, 3, 5, 7, 2, 4, 6, 8}`, then `arr1` will contain `{1, 2, 3, 4, 5, 6, 7, 8}` after the method call `merge(arr1, 0, 3, 7, temp)`.

Consider the following code segment, which appears in a method in the same class as `mergeSortHelper` and `merge`.

```
int[] numbers = {40, 10, 20, 30};
int[] temp = new int[numbers.length];
mergeSortHelper(numbers, 0, numbers.length - 1, temp);
```

How many times will the `merge` method be called as a result of running the code segment?

(A) 1

(B) 2

(C) 3 ✓

(D) 4

Answer C

Correct. The `merge` method appears after both recursive calls in the `mergeSortHelper` method, so it is not called until the condition `from < to` evaluates to `false` and there are no more recursive calls to execute. This occurs when the array parts to be merged are single elements. The `merge` method is called to merge, or sort, the elements from indices `0` to `0 (40)` and the elements from indices `1` to `1 (10)`, resulting in `{10, 40, 20, 30}`. The `merge` method is called again to merge the elements from indices `2` to `2 (20)` and the elements from indices `3` to `3 (30)`, resulting in `{10, 40, 20, 30}`. The

`merge` method is called a third time to merge the elements from indices `0` to `1` (`10`, `40`) and the elements from indices `2` to `3` (`20`, `30`), resulting in `{10, 20, 30, 40}`.

242. The following question is based on the following incomplete declaration of the class `BoundedIntArray` and its constructor definitions.

A `BoundedintArray` represents an indexed list of integers. In a `BoundedIntArray` the user can specify a size, in which case the indices range from `0` to `size - 1`. The user can also specify the lowest index, `low`, in which case the indices can range from `low` to `low + size - 1`.

```
public class BoundedIntArray
{
  private int[] myItems;   // storage for the list
  private int myLowIndex;  // lowest index

  public BoundedIntArray(int size)
  {
    myItems = new int[size];
    myLowIndex = 0;
  }

  public BoundedIntArray(int size, int low)
  {
    myItems = new int[size];
    myLowIndex = low;
  }
  // other methods not shown
}
```

Which of the following is the best reason for declaring the data fields `myItems` and `myLowIndex` to be private rather than public?

(A) This permits `BoundedIntArray` objects to be initialized and modified.

(B) This helps to prevent clients of the `BoundedIntArray` class from writing code that would need to be modified if the implementation of `BoundedIntArray` were changed. ✓

(C) This prevents compile-time errors whenever public methods are called that access the private data fields.

(D) This prevents run-time errors whenever public methods are called that access the private data fields.

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243. Consider the following class definition.

```
public class StudentInfo
{
  private String major;
  private int age;

  public String getMajor()
  { return major; }

  public int getAge()
  { return age; }

  /* There may be instance variables, constructors,
     and methods that are not shown. */
}
```

The following instance variable and method appear in another class. The averageAgeInMajor method is intended to return the average age of the StudentInfo objects in StudentList whose major is equal to theMajor, or -1 if there are no StudentInfo objects in studentList with a major that is equal to theMajor.

```
private ArrayList<StudentInfo> studentList;

public double averageAgeInMajor(String theMajor)
{
  double sum = 0.0;
  int count = 0;

  for (StudentInfo k : studentList)
  {
    /* missing code */
  }
  if (count > 0)
  {
    return sum / count;
  }
  else
  {
    return -1.0;
  }
}
```

Which of the following can be used to replace /* missing code */ so that averageAgeInMajor will work as intended?

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(A)

```
if (theMajor.equals(k.getMajor()))
{
  count++;
}
sum += k.getAge();
```

(B) ✓

```
if (theMajor.equals(k.getMajor()))
{
  sum += k.getAge();
  count++;
}
```

(C)

```
if (theMajor.equals(k.getMajor()))
{
  sum = k.getAge();
  count++;
}
```

(D)

```
if (theMajor.equals(k.getMajor()))
{
  sum += k.getAge();
}
count++;
```

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Answer B

Correct. The method uses an enhanced for loop, so k is a StudentInfo object representing an element in studentList. This code segment compares the string parameter theMajor with the string returned by k.getMajor(). If the two strings are equal, sum is incremented by the value returned by k.getAge() and count is incremented by 1.

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244. The following question is based on the following incomplete declaration of the class `BoundedIntArray` and its constructor definitions.

A `BoundedintArray` represents an indexed list of integers. In a `BoundedIntArray` the user can specify a size, in which case the indices range from `0` to `size - 1`. The user can also specify the lowest index, `low`, in which case the indices can range from `low` to `low + size - 1`.

```
public class BoundedIntArray
{
  private int[] myItems;   // storage for the list
  private int myLowIndex;  // lowest index

  public BoundedIntArray(int size)
  {
    myItems = new int[size];
    myLowIndex = 0;
  }

  public BoundedIntArray(int size, int low)
  {
    myItems = new int[size];
    myLowIndex = low;
  }
  // other methods not shown
}
```

Consider the following statements.

```
BoundedIntArray arrl = new BoundedIntArray(100, 5);
BoundedIntArray arr2 = new BoundedIntArray(100);
```

Which of the following best describes `arrl` and `arr2` after these statements?

(A) `arrl` and `arr2` both represent lists of integers indexed from `0` to `99`.

(B) `arrl` represents a list of integers indexed from `0` to `104`, and `arr2` represents a list of integers indexed from `0` to `99`.

(C) `arrl` represents a list of integers indexed from `5` to `99`, and `arr2` represents a list of integers indexed from `0` to `99`.

(D) `arrl` represents a list of integers indexed from `5` to `104`, and `arr2` represents a list of integers indexed from `0` to `99`. ✓

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245. Consider the following two methods that appear within a single class.

```
public void changeIt(int[] list, int num)
{
  list = new int[5];
  num = 0;

  for (int x = 0; x < list.length; x++)
    list[x] = 0;
}

public void start()
{
  int[] nums = {1, 2, 3, 4, 5};
  int value = 6;

  changeIt(nums, value);

  for (int k = 0; k < nums.length; k++)
    System.out.print(nums[k] + " ");

  System.out.print(value);
}
```

What is printed as a result of the call `start()`?

(A) `0 0 0 0 0 0`

(B) `0 0 0 0 0 6`

(C) `1 2 3 4 5 6` ✓

(D) `changeIt` will throw an exception.

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246. Consider the following method.

```
public void changeIt(int[] arr, int index, int newValue)
{
 arr[index] += newValue;
}
```

Which of the following code segments, if located in a method in the same class as `changeIt`, will cause the array `myArray` to contain `{0, 5, 0, 0}`?

(A)
```
int[] myArray = new int[4];
changeIt(myArray, 1, 5);
```
✓

(B)
```
int[] myArray = new int[4];
changeIt(myArray, 2, 5);
```

(C)
```
int[] myArray = new int[5];
changeIt(myArray, 1, 4);
```

(D)
```
int[] myArray = new int[5];
changeIt(myArray, 1, 5);
```

Answer A

Correct. The default initial value for `int` elements in an array is `0`. The initialization `new int[4]` instantiates an array with four elements and each element is initialized to the default initial value of `0`. This is passed to the `changeIt` method, which increments the element at index `1` by `5`, resulting in the array `{0, 5, 0, 0}`.